

Technical Supplement

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production on Argon in the NuMI Beam

Companion to the inclusive and proton-tagged analysis notes

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Abstract

This supplement contains the final configuration of the $\text{CC}\pi^\pm$ NuMI cross-section measurement (Part I), the evidence for its choices (Part II), the alternatives that were studied and not adopted (Part III) and an extended account of the physics background (Part IV).

Charged pions interact hadronically before they stop, so the range-based pion momentum saturates near 0.34 GeV/ c and carries little information about the true momentum above about 0.4 GeV/ c . Golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, recovery of the daughter energy and the proton-tagged selection were all studied, and none recovers it; a truth-level golden selection performs no better than the classifier, which places the limitation in the physics of pion interactions rather than in the estimator. The pion momentum is therefore measured in three regions. No differential $W_{\pi p}$ result is reported. The topology-level one-bin total cross section of the proton-tagged sample remains measurable, but it is not described as a $W_{\pi p}$ measurement.

The binnings of both families meet a 0.50 migration criterion, and the cross-checks of the extraction (identity closure, the D’Agostini method, ensembles of pseudo-data experiments and the consistency between configurations) are passed. Numbers shared with the analysis notes are checked for consistency automatically (`report/check_tables.py`, `xsec_analyzer/check_staleness.py`). The corrections that shaped the analysis, with the values before and after, are recorded in the change log.

1 Introduction

The inclusive analysis note and the proton-tagged note state what is measured and how. The studies that support their choices are collected here in four parts.

- **Final configuration** (Part I): the status of every result, the final binnings and the rule that selected them, the extraction configuration and uncertainty model, the validation, and the schema of the data release.
- **Evidence for the choices** (Part II): the range estimator of the pion momentum and its saturation; the binnings under the 0.68 and 0.50 migration criteria; the hadronic-mass observables, the proton-tagged selection and its additional smearing; the two-dimensional pairs; the checks of the extraction and the consistency between configurations; the control regions; the selection studies and the automated guards of the pipeline.
- **Rejected alternatives and development history** (Part III): the first binning optimisation, the θ_μ parameterisation, the lower pion threshold, the five-bin p_π response, the details of the golden-pion classifier and of the other pion-momentum estimators, and the cosmic-direction cuts. None of these is used in the submitted results; the change log gives the chronology.
- **Physics background** (Part IV): production mechanisms, nuclear effects, generator models and previous measurements.

Each section is tagged with the measurement it serves: [INCLUSIVE], [PROTON-TAGGED] or [SHARED]. Terms are defined in the glossary at the end, which is shared with the analysis notes.

Part I

Final configuration

2 Status of the results

[SHARED]

Table 1 gives the status of every inclusive product and the configurations in which it is reported, and Table 2 the same for the proton-tagged products.

Table 1: Status of the inclusive products. “yes” marks the configurations in which a product is reported.

Product	FHC	RHC	Comb.	Status
$d\sigma/dp_\mu$, $d\sigma/d\cos\theta_\mu$, $d\sigma/d\cos\theta_\pi$, $d\sigma/d\theta_{\mu\pi}$, σ in three p_π regions	yes	yes	yes	Primary result
One-bin total cross section	yes	yes	yes	Primary result
$d^2\sigma/d\cos\theta_\pi d\cos\theta_\mu$	no	no	yes	Secondary result
$d\sigma/dp_\pi$ in five bins				Not reported
$d\sigma/d\theta_\mu$				Not reported
Two-dimensional $\cos\theta_\pi \times \theta_{\mu\pi}$, $\cos\theta_\mu \times p_\mu$, $\cos\theta_\pi \times p_\pi$; FHC and RHC $\cos\theta_\pi \times \cos\theta_\mu$				Not reported

Table 2: Status of the proton-tagged products. “yes” marks the configurations in which a product is reported.

Product	FHC	RHC	Comb.	Status
One-bin total cross section	yes	yes	yes	Secondary result
$d\sigma/d\theta_p$, $d\sigma/d\theta_{\pi p}$	yes	yes	yes	Secondary result
$d^2\sigma/d\theta_p d\delta p_T$	no	no	yes	Secondary result
$d\sigma/dp_p$	yes	yes	yes	Validation-only product
$d\sigma/dW_{\text{had}}$ (two regions)	yes	yes	yes	Validation-only product
$d\sigma/d\delta p_T$, $d\sigma/dp_n$	yes	yes	yes	Validation-only product
$d\sigma/d\delta\alpha_T$, $d\sigma/d\delta\phi_T$	yes	yes	yes	Validation-only product
p_μ , p_π (two regions), $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$ on the proton-tagged sample	yes	yes	yes	Validation-only product
$d\sigma/dW_{\pi p}$				Not reported
Two-dimensional $\theta_p \times W_{\pi p}$, $\theta_{\pi p} \times W_{\pi p}$, $\theta_{\pi p} \times \delta p_T$; FHC and RHC $\theta_p \times \delta p_T$				Not reported

3 Final binnings and the binning rule

[SHARED]

Table 3 gives the bin edges of the inclusive observables. The proton-tagged edges are in Table 3 of the proton-tagged note, and those of the two two-dimensional results in their curve files (§6).

Table 3: Bin edges of every observable, from the result files. The binnings meet the 0.50 migration criterion (Sec. 8) and are the same in the three configurations.

Observable	Bins	Edges
p_μ (GeV/c)	6	0.15, 0.3, 0.425, 0.575, 0.825, 1.25, 3*
p_π (GeV/c)	3	0.175, 0.205, 0.26, 1*
$\cos\theta_\mu$	8	-1, 0.4, 0.625, 0.75, 0.825, 0.9, 0.95, 0.975, 1
$\cos\theta_\pi$	5	-1, -0.1, 0.35, 0.6, 0.8, 1
$\theta_{\mu\pi}$ (rad)	5	0, 0.6, 0.85, 1.3, 1.85, 2.6

* Numerical plotting boundary only: the upper truth bin is open-ended, and a density drawn over the conventional width represents the bin integral.

Each one-dimensional binning is selected by the following rule, established by the binning review (§8):

1. Generate candidate partitions: for every number of bins up to twelve, the maximin partition, which has the largest worst-bin migration diagonal over FHC and RHC, found by dynamic programming on a fine grid (`report/tools/maximin_binning.py`).
2. Require the event floor: at least 70 selected FHC signal events per bin at data exposure in the inclusive family, and 50 in the proton-tagged family.
3. Require every diagonal of the column-normalised migration matrix to be at least 0.50, in FHC and in RHC, on the analysis sample (§7.1); the combined configuration inherits the binning.
4. Extract the candidates that pass in full, in the three configurations: universes, covariance, unfolding and closure.
5. Compute the Wiener filter factors of each extraction and the trace of A_C , their sum, which is the number of degrees of freedom that the data determine.
6. Select the fewest bins that reach 90% of the largest trace, averaged over FHC and RHC; ties go to the larger worst diagonal.

7. Check the closure and the per-bin uncertainties of the selected binning.
8. Freeze the binning.

Four adopted binnings differ from the one the rule selects, for the reasons given in §8: $\cos\theta_\pi$ and $\theta_{\mu\pi}$ keep five bins, $\delta\alpha_T$ keeps its split at 120° , and the proton-tagged p_π keeps two regions. The 0.68 criterion of the BNB CC1 π measurement is the precedent for the migration criterion and provides a stricter comparison point; §8 gives the consequences of both. Everything in Part III, including the first binning optimisation, the five-bin p_π response and the θ_μ parameterisation, is historical and is not used in the submitted results.

A result can close while it is determined largely by the modelled migrations or by the regularisation, so the migration criterion and the Wiener filter factors are applied as separate criteria of the information a binning retains.

4 Extraction configuration and uncertainty model [SHARED]

Unfolding. Every differential result is unfolded with the Wiener-SVD method. The regularisation matrix is the second derivative of the cross-section density, built with the actual bin widths: on binnings as non-uniform as these, the uniform stencil reduces A_C to a matrix that passes the normalisation and discards the shape (§12). The additional-smearing matrix A_C of every result is released with it, and every prediction is smeared by it before it is compared with the result. The one-bin total cross sections are extracted with the Wiener filter off, $A_C \equiv 1$ (§15). The D’Agostini iterative method is a cross-check (§13).

Normalisation. Each run is normalised to its recorded exposure: the simulation of every run is scaled by the ratio of its beam-on to its simulated POT, and the beam-off samples by the per-run gate ratios (Sec. 5 of the analysis note). The flux is the summed $\nu_\mu + \bar{\nu}_\mu$ flux integrated above 60 MeV: FHC 6.81159, RHC 6.44646 and combined configuration $6.59691 \times 10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$, each configuration normalised with its own value. The 60 MeV convention includes the flux below the pion-production threshold, so it lowers the flux-averaged total relative to a threshold matched to the spectrum (Sec. 4.4 of the analysis note). The target count is $N_{\text{Ar}} = 8.710 \times 10^{29}$ argon nuclei in the fiducial volume.

Covariance. The official covariance of every result, `cov_total_plusMCS.txt`, is the framework covariance, `cov_total.txt`, plus the MCS muon-momentum-scale term. The framework covariance sums the flux (PPFX universes), cross-section (GENIE multisim and unisims), hadron re-interaction, detector-variation, POT and target-count terms and the statistical terms of the simulation, the beam-off samples and the data, each propagated through the unfolding and released separately (§6). The MCS term is the half-difference, taken as fully correlated, of the results obtained when the MCS-measured muon momenta of the pseudo-data are scaled by $\pm 5\%$ on the data side and unfolded with the nominal response. It is computed outside the framework, is zero for the results that do not use the reconstructed muon momentum, and is included in every quoted uncertainty and every summary table, among them the systematic breakdowns of Tables 22–24 (Sec. 12.3 of the analysis note). The closure χ^2 and the biases of the robustness tests are quoted with the framework covariance, since they test the extraction rather than the uncertainty of the data. Three terms lie outside the official covariance: the beamline-geometry flux term, the uncertainty of the per-run beam-off gate ratios and that of the use of the Run-1 beam-off sample for Run 2.

5 Validation

[SHARED]

The tests of the extraction go from the implementation to the model. The identity closure and the pseudo-data closure test the implementation (§14). Every extraction closes: the inclusive ones with $\chi^2 \leq 2.39/8$ and integral ratios of 0.946–1.047, the proton-tagged ones with $\chi^2 \leq 6.01/4$ (RHC $\cos \theta_\pi$) and ratios of 0.93–1.18, and the two two-dimensional results with cell ratios of 0.70–1.36. A result can close while it is determined largely by the modelled migrations or by the regularisation, so the migration criterion and the trace of A_C are applied as separate criteria (§3).

Ensembles of 100 pseudo-experiments test the size of the statistical uncertainties (§18). Seven inclusive and ten proton-tagged extractions were run, the proton angles in every configuration and the two two-dimensional results among them; the pull widths are 0.95–1.07 and 0.94–1.08, and the 68% coverages 63–73% and 62–72%.

Pseudo-data from two reweighted interaction models, GENIE multisim universe 545 and the $\Delta \rightarrow N\pi$ angular unisim, test the robustness to a model that reweighting can reach (Sec. 13.3 of the analysis note, Sec. 5.3 of the proton-tagged note). The largest bias is 1.18σ , in one sparse inclusive bin (FHC $\cos \theta_\mu$); in the proton-tagged sample the low δp_T and p_n bins move by -1.14σ and -1.16σ . The proton angles and the two two-dimensional results stay within 0.79σ .

No closure test with an independent generator has been made; the NuWro overlay available so far disagrees with standalone NuWro and is not used (Sec. 15 of the analysis note). The control regions, which test the background model, have been studied with pseudo-data only (§20).

6 Data-release schema

[SHARED]

The numerical results are in `report/data_release/` (Table 4). Each extraction is identified by a tag `<family>_<CFG>_<observable>`: the family is `incl` or `1p`, the configuration `FHC5`, `RHCFULL` or `COMB`, and the observable is the key used in `index_extractions.tsv`, for example `incl_FHC5_pmu` or `1p_COMB_thetap`. The two two-dimensional results have the tags `1p_COMB_thetap_dpt2d` and `incl_COMB_costhpi_costhmu2d`. Cross sections are in $10^{-38} \text{ cm}^2/\text{Ar}$ per unit of the variable, or per unit of each of the two variables for the two-dimensional results. Uncertainties and correlations are taken from the covariance files.

Table 4: Files of the data release in `report/data_release/`; `<tag>` is the tag of an extraction.

File	Content
<code>curves-<tag>.tsv</code>	One row per bin: <code>bin</code> , <code>low</code> , <code>high</code> , <code>width</code> , <code>unfolded_data</code> , <code>stat_err</code> , then the truth, tune and generator predictions, all already smeared by A_C . The values are densities; multiply by <code>width</code> to integrate. The two-dimensional files carry <code>inner_low</code> , <code>inner_high</code> , <code>outer_low</code> , <code>outer_high</code> and <code>area</code> in place of <code>low</code> , <code>high</code> and <code>width</code> , and order the analysis bins outer slice by outer slice, with the inner variable fastest.
<code>A_C-<tag>.tsv</code>	The additional-smearing matrix A_C : row i is the smeared bin, column j the true bin. A prediction p_j , the cross section integrated over true bin j , is compared with the result as $\text{smeared}_i = \sum_j A_C[i][j] p_j / \text{width}_i$, with the area in place of the width for the two-dimensional results.
<code>cov/<tag>/cov_total_plusMCS.txt</code>	The official covariance: the framework covariance plus the MCS momentum-scale term.
<code>cov/<tag>/cov_total.txt</code>	The framework covariance.
<code>cov/<tag>/cov-<source>.txt</code>	The components by source: flux, detector variations, cross section, re-interaction, POT, target count, simulation, beam-off and data statistics, MCS momentum scale.
<code>cov/<tag>/unfolded_signal.txt</code>	The unfolded bin integrals.
<code>total_xsec.tsv</code>	The one-bin total cross sections, with their uncertainty breakdown.
<code>index_extractions.tsv</code>	Every extraction, with its status.
<code>../status.tsv</code>	The status of every product, the single source of Tables 1 and 2 (in <code>report/</code> , next to the release).

Part II

Evidence for the choices

7 Pion momentum: the range estimator and its saturation^[INCLUSIVE]

The inclusive p_π is measured in three regions with edges $[0.175, 0.205, 0.26, \infty)$ GeV/ c , and the proton-tagged p_π in two regions, $[0.175, 0.205, \infty)$ GeV/ c . The five-bin configurations are diagnostics, not results. The golden-pion, MCS, calorimetric, regression and daughter-recovery studies of this section support the coarse range-based measurement.

7.1 Track sample and analysis sample

Migration diagonals of the pion momentum are quoted on two samples, which are not interchangeable. The track sample is every truth-matched pion track passing loose quality requirements (purity > 0.5 , track score > 0.5 , length > 5 cm), from events of any topology, 126 518 tracks in total; the golden-pion studies use it, and the estimator studies use truth-matched pion tracks of the same kind. The analysis sample is the events passing the full $CC1\mu1\pi^\pm$ selection and satisfying the signal definition, about 10 700 per beam mode; the adopted binnings, the binning studies and the hadronic-mass study use it. The analysis sample has the better response, because the selection removes the events whose pion is poorly reconstructed.

7.2 Pion-momentum bias and the conditioning of the p_π response

The p_π response is the least diagonal of any observable in the analysis. The cause is physical, and it limits what the measurement can say about the pion momentum spectrum.

7.2.1 Mass hypothesis and hadronic truncation.

The reconstructed pion momentum stored in the ntuple, `candidate_pion_mom_reco`, is `trk_range_muon_mom_v`: a range-based momentum evaluated under the muon mass hypothesis and applied to a pion track. Two distinct effects follow, and only the first is corrected.

First, the mass hypothesis biases the estimate low, because for a given range a pion carries more momentum than a muon. The size is $\sim 17\text{--}19\%$ from the CSDA range scaling, largely independent of momentum. This effect is corrected in the reco bin definitions rather than in the branch: `ccpi_ppi_bin_config_opt.txt` applies a kinetic-energy-preserving $\mu \rightarrow \pi$ mass swap, $E_\mu \rightarrow T \rightarrow E_\pi \rightarrow p_\pi$, before binning. In the lowest true bin $\langle p^{\text{reco}}/p^{\text{true}} \rangle$ goes from 0.888 with the raw branch to 0.976 with the correction.

Second, charged pions interact hadronically before coming to rest, and nothing corrects for this. The reconstructed track then ends at the interaction point rather than the stopping point, so the visible range, and any momentum inferred from it, is truncated. The effect grows rapidly with momentum and dominates at high momentum.

7.2.2 Saturation of the reconstructed momentum.

Table 5 and Figure 1 show the result: as the true momentum rises by a factor of four and a half, the reconstructed momentum barely moves, saturating at ≈ 0.34 GeV/ c . Above that the estimator carries almost no information about the energy of the pion.

Table 5: Pion momentum reconstruction versus truth, selected signal, Run-1 FHC. “raw” is the `candidate_pion_mom_reco` branch; “corr.” also applies the $\mu \rightarrow \pi$ mass swap used by the reco bin definitions. The mass correction removes the bias at threshold; nothing removes the saturation above it.

true p_π [GeV/ c]	N	$\langle p^{\text{true}} \rangle$	$\langle p^{\text{corr}} \rangle$	raw/true	corr./true
0.175–0.250	528	0.210	0.205	0.888	0.976
0.250–0.300	270	0.276	0.238	0.790	0.863
0.300–0.350	254	0.325	0.253	0.716	0.779
0.350–0.400	265	0.374	0.270	0.664	0.721
0.400–0.500	403	0.443	0.279	0.581	0.629
0.500–0.600	252	0.548	0.307	0.520	0.560
0.600–0.800	272	0.688	0.326	0.441	0.473
0.800–1.200	164	0.951	0.336	0.329	0.353

7.2.3 Effect on the five-bin response matrix.

On the analysis sample of Table 5, migrating the five-bin scheme with the corrected estimator gives a nearly lower-triangular matrix: the diagonal fraction falls from 94.3% in the first true bin to 36.5, 23.1, 10.4 and finally 6.2% in the last. For the highest true bin ($p_\pi > 0.55$ GeV/ c) only 6.2% of events are reconstructed in the corresponding reco bin while 32.6% fall into the lowest one. Every true bin above the first therefore deposits most of its events into reco bin 1, the “pile-up into the first bin” of Fig. 15.

7.2.4 Treatment in the BNB CC1 $\pi^\pm$ analysis.

The published BNB CC1 $\pi^\pm$ Xp analysis (internal note v0.91) uses an identical estimator: it too takes “the momentum of pion candidates from range”. Its treatment is summarised here and was followed by this analysis.

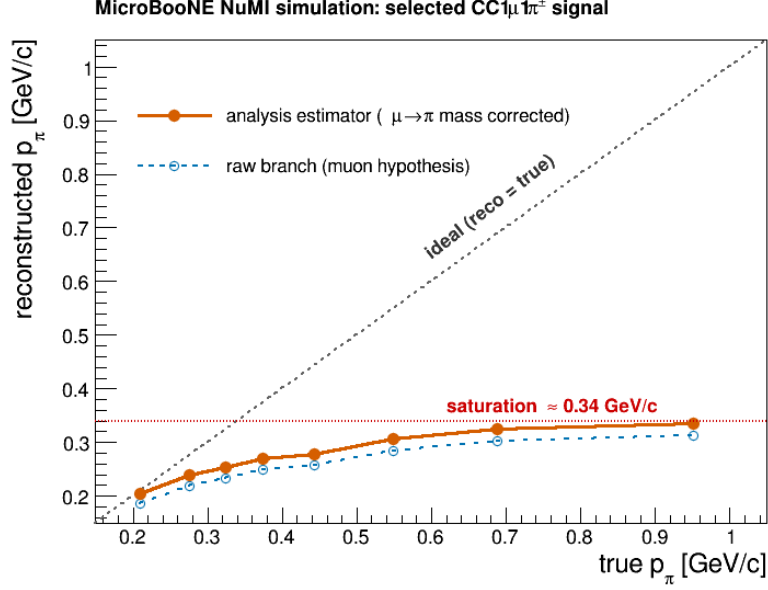


Figure 1: Mean reconstructed against mean true pion momentum, selected signal. The analysis estimator (filled, $\mu \rightarrow \pi$ mass corrected) tracks truth at threshold but flattens out near 0.34 GeV/c: a true 0.95 GeV/c pion reconstructs at the same place as a true 0.55 GeV/c one. The raw `candidate_pion_mom_reco` branch (open) sits systematically below it by the uncorrected mass-hypothesis offset. The saturation is caused by hadronic interaction truncating the track range, and no range-based estimator can recover it.

That analysis does three things this one does not. First, it defines a *golden pion*: one that comes to rest and leaves a Bragg peak, as opposed to one that scattered or interacted. Second, it trains a dedicated golden-pion BDT and applies it as an extra cut for the pion-momentum differential cross section only, leaving the total and angular results on the generic selection. The BDT separates golden from non-golden pions using track “wiggleness” (the standard deviation of the angular difference between directions of adjacent points along the fitted track, a direct handle on scattering), the number of reconstructed descendant particles, the track–shower score, and the Bragg likelihood ratio. The note quotes the cost and benefit: efficiency falls from 20.3% to 9.5% and purity is largely unchanged (51.9 $\rightarrow$ 49.5%), while the golden fraction of the selected signal rises from 36.3% to 67.7%. The same logic is applied to the muon momentum, restricting that cross section to contained muons (8.4% efficiency, 56.1% purity) so that it uses range only.

Third, and most directly comparable, the binning is chosen against an explicit criterion: each bin must hold at least 100 predicted MC events and the diagonal element of the migration (smearing) matrix must exceed 0.68. The resulting pion-momentum edges are 0.10, 0.16, 0.19, 0.22, 0.60 GeV/c with an overflow bin: fine structure only where the estimator still tracks truth, then a single wide bin from 0.22 to 0.60 and everything above 0.60 in overflow.

The five-bin scheme fails that criterion: its diagonal fractions are 94.3, 36.5, 23.1, 10.4 and 6.2%, so one of five bins passes 0.68 and the top three fail it by a wide margin. Its edges (0.175–0.550 in five bins, open above) place four boundaries inside the region where the reconstructed momentum has saturated.

7.2.5 Consequences for the unfolding.

The saturation does not bias the extracted cross section, because the response matrix contains it and the unfolding inverts it, but it makes the p_π result depend more on the regularisation than any other observable. With 6% diagonal strength in the top bin, the result above ~ 0.5 GeV/c is

effectively inferred from the response model rather than measured.

Adopted binning: three p_π regions

The adopted binning has three regions, $[0.175, 0.205]$ and $[0.205, 0.26]$ GeV/ c and everything above. Merging raises the migration diagonals from 96/35/21/10/6% (full analysis sample; 94.3/36.5/23.1/10.4/6.2% on Run 1 alone, Table 5) to 91/51/52% in FHC and 90/53/53% in RHC, which meets the 0.50 criterion of the analysis in every region (§8). Under the 0.68 criterion of the BNB measurements only two regions pass, $[0.175, 0.205]$ GeV/ c and everything above, at 91/76%. These diagonals are measured on the analysis sample (§7.1); the golden-pion study of Sec. 7.2.6 uses the looser track sample, whose diagonals are lower, and the two sets are not interchangeable.

The first boundary follows from a scan of the two-region binning on the analysis sample. The worst diagonal is always that of the upper region, and it falls monotonically as the boundary rises, so the binding constraint is the width of the first interval rather than the diagonal itself:

boundary	first-bin width	diagonal, bin 1	diagonal, bin 2
0.185	10 MeV/ c	79.7%	85.4%
0.195	20 MeV/ c	88.8%	80.0%
0.200	25 MeV/ c	90.4%	77.7%
0.205	30 MeV/c	91.0%	75.5%
0.220	45 MeV/ c	93.3%	69.1%
0.250	75 MeV/ c	95.6%	56.1%
0.300	125 MeV/ c	96.3%	39.0%

Both regions meet 0.68 anywhere between roughly 0.19 and 0.22, so the choice within that window is set by width rather than by the criterion. 0.205 is the narrowest first interval with precedent: 30 MeV/ c matches the narrowest bin used in the BNB CC1 π analysis. Going narrower trades first-bin diagonal for upper-bin diagonal, and at 10 MeV/ c the first bin itself falls to 80% on only 325 events. An exhaustive scan over all edge sets (`macros/ppi_binning_scan.C`) returns 0.205 as the best first boundary for every number of bins tried, and in the proton-tagged sample as well.

The second boundary costs 25 points on the worst diagonal. The best three-region binning, with edges 0.205 and 0.26 GeV/ c from the maximin scan of §8, reaches 91/51/52%, against 91/76% for two regions: it meets 0.50 but not 0.68. Two 30 MeV/ c intervals followed by an open region (0.205, 0.235) are worse at 91/39/62%, and degrade the upper region too, from 76% to 62%: narrowing its reconstructed range pushes events of high true momentum out of their own bin. Four regions reach only 91/39/39/42%. Above ~ 0.25 GeV/ c the reconstructed momentum barely varies, so a further boundary there separates almost nothing. The same ordering holds under a golden-pion cut (67/52/46%), so it is not an artefact of the selection.

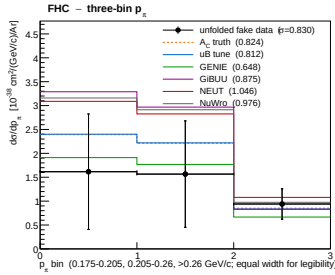
7.2.6 Closure.

The three-region scheme is carried through the full chain (universe construction, Wiener–SVD unfolding and systematics) and closes in all three configurations (Table 6). A five-bin extraction on the same sample also closes, at $\chi^2 \approx 2/5$: closure cannot expose a response that carries no shape information, whereas the migration criterion can.

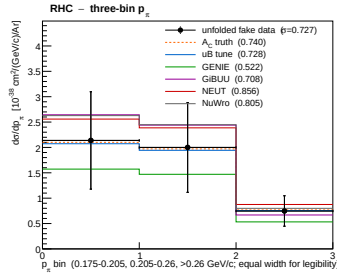
All three configurations close, with χ^2 of 1.01, 0.01 and 0.21 on three bins and unfolded-to-truth integral ratios of 1.007, 0.983 and 1.001 in FHC, RHC and combined. GENIE is the lowest prediction in every region of every configuration, as in the other observables. The released A_C for FHC is $\begin{pmatrix} 0.40 & 0.16 & 0.01 \\ 0.58 & 0.32 & 0.02 \\ -0.22 & -0.11 & 0.85 \end{pmatrix}$ (`A_C_incl_FHC5_ppi3bin.tsv`): the two near-threshold regions are mixed strongly, and together they carry one component that the data determine, so the trace is 1.57 and two filter factors exceed 0.5. The first two regions are therefore measured as a pair rather

Table 6: Three-region p_π result for all three beam configurations, $d\sigma/dp_\pi$ in $10^{-38} \text{ cm}^2/(\text{GeV}/c)/\text{Ar}$. Bin 1 is $[0.175, 0.205]$ GeV/c , bin 2 $[0.205, 0.26]$ GeV/c and bin 3 everything above. All curves are A_C -smeared, so they live in the same space as the measurement. Generated from the release (`report/tools/ppi2bin_tables.py`: closure sidecars and `curves_incl*_ppi3bin.tsv`); the uncertainty row is the total.

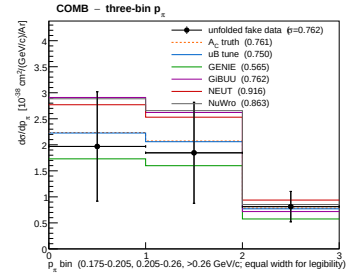
	FHC				RHC				Combined			
	bin 1	bin 2	bin 3	σ_{int}	bin 1	bin 2	bin 3	σ_{int}	bin 1	bin 2	bin 3	σ_{int}
unfolded data	1.62	1.57	0.94	0.830	2.14	2.00	0.75	0.727	1.97	1.85	0.81	0.762
uncertainty	± 1.21	± 1.11	± 0.32		± 0.96	± 0.89	± 0.30		± 1.05	± 0.97	± 0.29	
A_C truth	2.39	2.21	0.85		2.10	1.97	0.77		2.24	2.07	0.78	
uB tune	2.40	2.22	0.84		2.07	1.94	0.76		2.23	2.06	0.77	
GENIE	1.91	1.77	0.67		1.57	1.47	0.53		1.73	1.60	0.58	
GiBUU	3.29	2.97	0.83		2.64	2.44	0.67		2.91	2.62	0.72	
NEUT	3.09	2.83	1.08		2.56	2.39	0.88		2.77	2.53	0.94	
NuWro	3.16	2.91	0.97		2.63	2.45	0.80		2.88	2.66	0.85	
χ^2/ndf	1.01/3 ($p = 0.80$)				0.01/3 ($p = 1.00$)				0.21/3 ($p = 0.98$)			



(a) FHC



(b) RHC



(c) Combined

Figure 2: The three-region p_π cross section, the graphical form of Tab. 6. Unfolded pseudo-data (points, with the full systematic uncertainty) against the A_C -smeared truth, the uB tune and the four generators; every curve is A_C -smeared, so all of them live in the same space as the measurement. The regions are drawn with equal width for legibility: they are 0.030, 0.055 and 0.74 GeV/c wide, so area on these panels is not proportional to cross section, and the integrated σ is printed on each panel instead.

than separately, and a prediction is compared through the full A_C . The RHC and combined matrices mix the two regions even more (0.29/0.25 and 0.45/0.43 in RHC); the upper-region diagonal is 0.83–0.85 in all three. The total fractional uncertainty is 60% in FHC, the largest of the inclusive observables, because the regions each carry a large share of the flux normalisation with little statistical averaging. Figure 3 shows A_C for all three configurations.

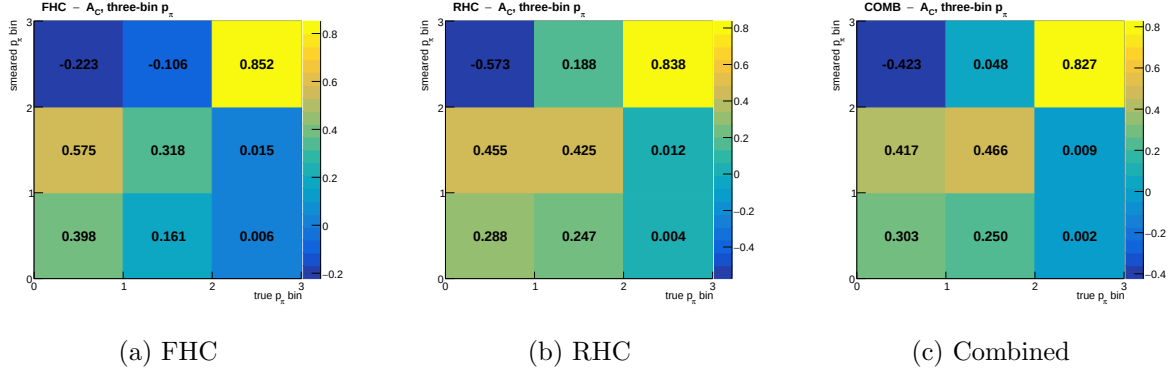


Figure 3: The additional-smearing matrix A_C for the three-region p_π scheme, in the convention $\hat{x}_i = \sum_j (A_C)_{ij} t_j$ so that a column is a true bin and a row is the smeared bin it feeds. A_C is not normalisation-preserving, so its rows do not sum to unity. The negative entries of smeared bin 3, drawn at the top, are the regularisation borrowing between the near-threshold regions and the upper one; they are why the truth curves in Fig. 2 must be A_C -smeared before being compared with the measurement rather than plotted raw.

At 91/51/52% the three regions meet the 0.50 criterion. The result remains a coarse measurement: the near-threshold regions are measured as a pair, and above ~ 0.34 GeV/ c the reconstructed momentum saturates, so the upper region is an integral over a wide range rather than a differential point. The scheme provides a result with a defensible response rather than a competitive shape measurement.

Golden-pion classifier

Following the BNB recipe, a golden-pion BDT was trained and evaluated with the macros `golden_pion_train.C` and `golden_pion_compare.C` of `macros/`. The truth label is `mc_end_p < 0.01` GeV/ c (the pion ranged out), joined to reconstruction by matching `backtracked_p` to `mc_p`, which pairs 100% of candidate tracks with no ambiguity. The inputs are the three deflection variables, the four Bragg likelihood ratios, the track–shower score and the descendant count. Training used 126 518 labelled tracks of the track sample (§7.1) from Runs 1, 2 and 4; the split is by run, and all numbers below are on Run 2, which the classifier did not see in training.

7.2.7 Classifier performance.

The ROC integral is 0.758. At 44.8% efficiency the classifier raises the golden fraction from 43.1% to 64.9%, against the BNB $36.3 \rightarrow 67.7\%$ at roughly 47% efficiency, so it reproduces the BNB benchmark.

7.2.8 Effect on the p_π migration.

Table 7 gives the migration diagonal in the five bins as the cut is tightened. The first bin passes and continues to pass; no other bin reaches 0.68 at any working point, even at 15% efficiency.

Table 7: Golden-pion BDT: efficiency, achieved golden fraction, and the p_π migration diagonal per true bin, on the held-out run of the track sample (§7.1). The BNB criterion for a usable bin is a diagonal above 0.68.

BDT cut	eff.	golden frac.	b1	b2	b3	b4	b5
none	100%	43.1%	99.2	23.7	13.2	6.4	5.5
> -0.05	57.0%	58.1%	99.1	29.4	19.7	11.2	7.1
> 0.00	44.8%	64.9%	99.1	33.1	24.5	15.4	9.1
$> +0.10$	24.6%	79.4%	99.1	42.9	36.8	29.3	15.7
$> +0.15$	15.2%	84.5%	99.2	46.9	42.2	36.1	19.5
<i>perfect (truth) golden selection</i>			98.9	36.1	24.1	14.2	8.9

7.2.9 Truth-level ceiling.

The last row of the table is to be compared with the row of matching efficiency rather than with the tightest cut. Perfect truth-level golden selection keeps 43.1% of tracks and gives 36.1, 24.1, 14.2, 8.9% in bins 2–5; the classifier at the matching 44.8% efficiency gives 33.1, 24.5, 15.4, 9.1%. They agree bin by bin. The classifier therefore already performs as well as perfect knowledge of which pions stopped, and a better golden-pion classifier cannot improve the response further. The two tightest working points exceed the truth-level row only because a hard cut on the classifier preferentially keeps long tracks, and so cuts on the reconstructed momentum itself; they are not used (§28).

7.2.10 Summary of the classifier study.

The classifier matches the published benchmark, but the p_π differential measurement above ~ 0.23 GeV/ c cannot be made to meet the BNB binning criterion with a range-based estimator, with or without a golden-pion cut. The BNB analysis meets that criterion because its resolved bins (0.10–0.16, 0.16–0.19, 0.19–0.22) all lie below the 0.175 GeV/ c threshold of this analysis, in the region where range still tracks momentum; above 0.22 it uses a single wide bin and an overflow. The options for this analysis were therefore to lower the pion threshold toward 0.10 GeV/ c and resolve the region where the estimator works (tested and not adopted, §26), to report a coarse p_π result with an open upper region (adopted, §7.2.5), or to replace the range-based estimator (§7.3). The golden-pion BDT can contribute to any of these but does not by itself solve the problem.

The limitation described in this subsection concerns attempts to recover a finer multi-bin p_π spectrum, and its numbers are for the loose truth-matched track sample. It does not apply to the adopted three-region result on the analysis sample (§7.2.5, Table 6), whose migration diagonals are 91%, 51% and 52%.

7.2.11 Scope of the p_π result.

Two routes to a finer p_π measurement were tested. The golden-pion classifier does not repair the response even at perfect truth-level selection (above), and the proton tag does not change the response appreciably (below). With merged regions the inclusive selection satisfies the binning criterion, so p_π is retained in the inclusive measurement; what it cannot support is a five-bin shape measurement. The other inclusive observables are unaffected: the angular variables and p_μ retain their shape information, and the total cross section does not use this estimator.

7.2.12 Proton-tagged response.

The second route to a sharper p_π response is the proton tag. It was measured for FHC, RHC and the combined configuration: at the same two-bin edges the proton-tagged response

is indistinguishable from the inclusive one, and marginally worse rather than better. The migration-diagonal test gives

selection	config.	N_{sig}	bin 1	bin 1 [0.175, 0.205]	bin 2 (> 0.205)	both $> 0.68?$
inclusive CC1mu1piXp	FHC	1056		91.0%	75.5%	yes
proton-tagged CC1mu1pi1p	FHC	469		88.9%	74.2%	yes
	RHC	621		89.4%	75.1%	yes
	comb.	1090		89.2%	74.7%	yes

measured on the numuMC overlays named in the corresponding `file_properties` lists, selected signal (analysis sample), with the reconstructed momentum mass-corrected as in the bin configuration. All four rows clear the criterion in both bins, and the proton-tagged values lie about two points below the inclusive ones. The three proton-tagged configurations agree to better than a percentage point, so the response is a property of the selection rather than of a particular exposure, but the proton tag does not improve it.

This is expected. The pion-momentum response is set by whether the pion ranges out or interacts hadronically before stopping, which is a property of the pion's own passage through the argon. Requiring a reconstructed proton at the vertex changes the event sample, the purity and the efficiency, but not what happens to the pion downstream, so there is no mechanism by which it would sharpen the momentum response, and the measurement confirms this.

The proton-tagged p_π extraction is therefore the pion-momentum cross section of an exclusive final state that also yields W and TKI, a Validation-only product (Table 2), not a repair of the inclusive one. The inclusive p_π in three regions also meets the adopted criterion.

Table 8: Two-bin p_π on the proton-tagged CC1mu1pi1p selection, all three beam configurations. $d\sigma/dp_\pi$ in $10^{-38} \text{ cm}^2/(\text{GeV}/c)/\text{Ar}$; bin 1 is [0.175, 0.205] GeV/c , bin 2 everything above. All curves are A_C -smeared. The proton-tagged `xsec` configs declare only the tune and the pseudo-data, so no generator rows appear. Regenerated from the release (`report/tools/ppi2bin_tables.py`); the uncertainty row is the total.

	FHC		RHC		Combined	
	bin 1	bin 2	bin 1	bin 2	bin 1	bin 2
unfolded data	1.020	0.581	0.386	0.525	0.741	0.550
uncertainty	± 0.845	± 0.251	± 0.796	± 0.222	± 0.851	± 0.220
A_C truth	1.138	0.604	1.056	0.458	1.226	0.518
uB tune	1.144	0.590	1.135	0.469	1.281	0.518
unfolded/truth	0.90	0.96	0.37	1.15	0.60	1.06
σ_{int}	0.493		0.429		0.459	
χ^2/ndf	0.02/2 ($p = 0.99$)		1.21/2 ($p = 0.55$)		0.48/2 ($p = 0.79$)	

All three extractions close, with χ^2 of 0.02, 1.21 and 0.48 on two bins. The two proton-tagged regions meet the stricter 0.68 criterion (0.74/0.75 in FHC and RHC). The proton-tagged version is extracted because it belongs to the exclusive final state, not because it is required to make p_π measurable.

7.2.13 The least-populated bin.

The RHC first bin unfolds to 0.37 times its A_C -smeared truth (0.386 ± 0.796 against 1.056) with a closure χ^2 of 1.21/2: a 0.8σ fluctuation on a bin whose uncertainty is twice its content. Independent throws move it by more than $\pm 1\sigma$. The bin holds too few events to constrain the

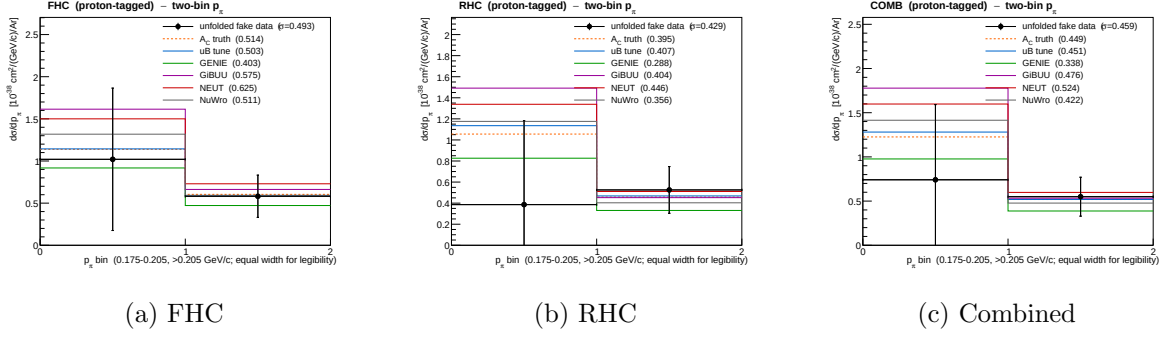


Figure 4: The two-bin p_π cross section on the proton-tagged CC1mu1pi1p selection, the graphical form of Tab. 8. The first bin is that of the inclusive result of Fig. 2, and the second is the union of the two upper inclusive regions, since three regions fail the 0.50 criterion in the proton-tagged analysis sample. The bins are drawn equal-width.

cross section, and any real-data value in it should be read with its uncertainty rather than its central value.

7.2.14 Choice of boundary for the proton-tagged sample.

The 0.205 GeV/c boundary was independently preferred by the migration-level edge scan run on the proton-tagged sample (macros/ppi_binning_scan.C, FHC; logs/binning_scan/1p_fhc.txt) as well as by the inclusive scan, so it is not simply inherited. It was not re-optimised through a separate universe, covariance and unfolding production for every candidate boundary, which would need its own universe construction per configuration (~ 1 – 2 h each). This is not expected to change the conclusion, since both bins clear the criterion with margin in all three proton-tagged configurations.

7.3 Survey of pion momentum estimators without a magnetic field

Four approaches were measured on truth-matched pion tracks. All are limited by the same mechanism: each measures what the pion did before it interacted.

estimator	$\langle \text{reco}/\text{true} \rangle$, top bin	RMS	note
range (analysis)	0.319	0.16	saturates near 0.34 GeV/c
calorimetric energy	0.383	0.31	also truncated, and $2\times$ noisier
MCS	1.011	2.1	unbiased, unusably noisy on short tracks
golden-pion selection	0.327	0.18	no gain above 0.8 GeV/c

Summary of the estimator survey. Five approaches were measured (§§28 and 29): range, calorimetry, MCS, golden-pion tagging and a multivariate regression, together with recovery of the interaction products. Each is limited by the same fact, that a charged pion which interacts before stopping destroys the information a non-magnetised detector uses to infer its momentum, and that a substantial part of the energy leaves the detector unseen. The regression is the only one that gives a material improvement, removing the $W_{\pi p}$ bias, at the cost of model dependence and of the low- W region.

One avenue has not been tried and is independent of the pion’s own reconstruction: for the proton-tagged sample, transverse kinematic imbalance constrains the pion transverse momentum from the muon and proton alone. It is limited by Fermi motion, of order 250 MeV/c, which is comparable to the present uncertainties rather than smaller, so it would help but would not change the character of the measurement. Without it, p_π is a coarse measurement in a few wide

regions at this exposure and no differential $W_{\pi p}$ result is reported, both because of physics rather than of analysis choices.

8 Binning review against the 0.68 and 0.50 migration criteria

[SHARED]

The review established the rule and the binnings of §3. It starts from the binnings produced by the first optimisation and its re-check (§24), which are called the released binnings below and in the tables.

8.1 Criteria and selection rule

Admissibility criteria. A true bin is admissible when most of its selected signal is reconstructed in the same bin: the diagonal of the column-normalised migration matrix, measured on the analysis sample (§7.1), must reach 0.68, the value used in the BNB CC1 π measurements, or 0.50, the relaxed criterion adopted for this analysis. The criterion bounds how much of a bin’s content the unfolding moves between bins, and therefore how strongly the result leans on the modelled migrations. It is applied in FHC and in RHC; the combined configuration inherits the binning. Every true bin must also hold a minimum of selected signal events, counted in FHC at data exposure: 70 in the inclusive family, the content of the sparsest released inclusive bin, and 50 in the proton-tagged family. Neither requirement says how much of the binned shape survives the regularised unfolding, and that is the third property reviewed.

Retained shape: the Wiener filter factors. In the Wiener-SVD unfolding, A_C is similar to the diagonal matrix of the Wiener filter factors $W_k = S_k^2/(S_k^2 + N_k^2)$, one for each component of the pre-whitened response. Its eigenvalues are these factors, all between zero and one, and its trace is their sum: the number of degrees of freedom that the data determine. A factor of at least 0.5 marks a component measured with a signal-to-noise ratio of at least one. For the released FHC $\cos\theta_\mu$, for example, the factors are 0.79, 0.79, 0.66, 0.35 and 0.20, a trace of 2.79 in five bins. The first optimisation and its re-check (§24.2, Appendix B of the analysis note) used the conditioning $s_{\min}/s_1$ instead. The conditioning is set by the weakest component alone, so it does not separate a binning that loses one component from one that retains nothing beyond the normalisation, and the threshold of about 0.1 used there to reject finer candidates was not applied to the released binnings, most of which fall below it: p_μ (0.006 in FHC, 0.0001 in RHC), $\theta_{\mu\pi}$ (0.007, 0.002), θ_μ (0.06, 0.005) and RHC $\cos\theta_\mu$ (0.0003). The singular values of A_C are not the filter factors either: A_C is not symmetric, and its largest singular value exceeds one.

Candidates. For each observable and each number of bins up to twelve, the partition with the largest worst-bin diagonal over FHC and RHC that meets the event floor was found by dynamic programming on a fine grid, with the edge conventions of the released configurations (`report/tools/maximin_binning.py`). The tool reproduces the unfolders’ diagonals of the released binnings to 0.01, and to 0.05 for W_{had} , two of whose edges fall between grid points. The partitions relevant to either criterion, from the coarsest that could replace the released binning to the finest passing 0.50, and for the angles also the partition at the released five bins, which separates the effect of the edges from that of the bin count, were extracted in full in FHC, RHC and combined: 141 extractions with universes, covariance, unfolding and closure (SLURM 3428213, 3428222, 3428223, 3433027 and 3433033), and the proton momentum p_p added to the proton-tagged family in two to five bins (SLURM 3433088): 153 extractions in all (`data_release/binreview_eval_2026-09-26.tsv`). They carry the tune and pseudo-data predictions only. A partition has the same diagonals in $\cos\theta_\mu$ and in θ_μ , and the θ_μ grid is finer

at forward angles, so the eight- and ten-bin θ_μ partitions, written in $\cos\theta_\mu$, are $\cos\theta_\mu$ candidates as well.

Selection rule. Among the admissible binnings of an observable, the one selected has the fewest bins that reach 90% of the largest trace of A_C , averaged over FHC and RHC; ties go to the larger worst diagonal. Bins beyond that point add components that the regularisation fills rather than the data. A choice is marked rule-sensitive when 85% or 95%, or an average that includes the combined configuration, would select a different binning. Figure 5 shows the trace against the number of bins for every candidate, Tables 9 and 10 the selected binnings, and Table 11 their edges (report/tools/binreview_tables.py, data_release/binreview_choices_2026-09-26.tsv).

8.2 Consequences and adopted binnings

Consequences of the two criteria. The released binnings fall into three groups. The seven-bin p_μ and the six-bin W_{had} fail both criteria, and the θ_p and $\theta_{\pi p}$ study binnings fall below the event floor: these change whichever criterion is adopted. The five-bin $\cos\theta_\mu$ and θ_μ and the two-bin $\delta\alpha_T$ fail 0.68 only; rebinning the muon angles to meet it gains information, while $\delta\alpha_T$ loses a second component whose filter factor is only 0.2. The remaining binnings, p_π , $\cos\theta_\pi$, $\theta_{\mu\pi}$, δp_T , $\delta\phi_T$ and p_n , pass both; of these $\cos\theta_\pi$ gains 0.45 degrees of freedom at seven bins under either criterion, and the others are at or near their information limit.

Where the two criteria select different binnings, the relaxed one adds 0.2 to 0.6 degrees of freedom per beam: p_μ (six bins against four), p_π (three against two), $\cos\theta_\mu$ (eight against seven), θ_μ (ten against eight), $\delta\phi_T$ (three against two) and $\theta_{\pi p}$ (four against three); for p_p (three against two) the gain is 0.15/0.12. $\delta\alpha_T$ and $W_{\pi p}$ gain a second bin whose component the data do not determine. A new component measured at signal-to-noise above one appears only for p_π in RHC, θ_μ and $\delta\phi_T$ in the combined configuration (where θ_μ loses one in RHC) and $\theta_{\pi p}$ in FHC and RHC. The price is diagonals down to 0.50: bins in which up to half of the selected events come from neighbouring true bins, with a correspondingly stronger dependence of the result on the modelled migrations.

Choice of criterion. The 0.68 criterion is that of the BNB measurements. Applied to every observable, it would end the exemption of p_μ , which rests on the regularisation absorbing migrations that the criterion exists to limit. Where it differs from 0.50 it forgoes at most 0.64 degrees of freedom per beam, and in all but five observable and beam combinations no component that the data determine at signal-to-noise above one; its cost falls on p_μ , which would drop from seven bins to four and from three well-measured components to two. The 0.50 criterion also ends the exemption. It keeps six p_μ bins, with two well-measured components as well, three p_π regions and eight $\cos\theta_\mu$ bins, and it accepts bins in which up to half of the selected events come from neighbouring true bins. The analysis uses the 0.50 criterion for both families. The stronger dependence on the modelled migrations that it admits is tested by the alternative-model closure and the ensembles, both repeated on the adopted binnings (Sec. 13.2 of the analysis note; Sec. 13.3 of the analysis note).

Adopted binnings. The adopted binnings are those the rule selects under 0.50 (Table 11), with four exceptions. $\cos\theta_\pi$ keeps its five bins, which pass both criteria (0.74/0.76); the seven bins the rule selects would add 0.44/0.46 degrees of freedom. $\theta_{\mu\pi}$ keeps its five bins (0.79/0.79), which retain as much as the six selected in FHC and 0.2 less in RHC. $\delta\alpha_T$ keeps its split at 120° : the rule's 122.5° raises the worst diagonal from 0.66 to 0.67 and changes the trace by less than 0.02. The proton-tagged p_π keeps two regions, because the three-region binning fails 0.50 in the proton-tagged sample. θ_μ is not reported: a partition has the same diagonals in θ_μ as in $\cos\theta_\mu$, and the cosine retains up to one degree of freedom more, because the regularisation acts on the

density per unit of the variable (§25.1). The binnings are therefore: p_μ six bins and $\cos\theta_\mu$ eight in both families; p_π three regions in the inclusive family and two in the proton-tagged one; $\cos\theta_\pi$ five and four; $\theta_{\mu\pi}$ five; W_{had} two regions split at $1.18 \text{ GeV}/c^2$; $\delta\phi_T$ three bins; δp_T , $\delta\alpha_T$ and p_n two; θ_p and $\theta_{\pi p}$ four at their maximin edges; and p_p three. No differential $W_{\pi p}$ result is reported. Table 3 and Table 3 of the proton-tagged note list the edges.

For every replaced binning the generator predictions were computed at the new edges from the event-level generator files, and the universes, covariances and result files were rebuilt (§23); the ensembles, the alternative-model closure and the D’Agostini cross-check were repeated on the adopted binnings. The control-region binnings are frozen separately (§21.5) and do not change.

8.3 Evidence for each observable

The worst diagonals of Tables 9 and 10 are measured on the analysis sample (§7.1).

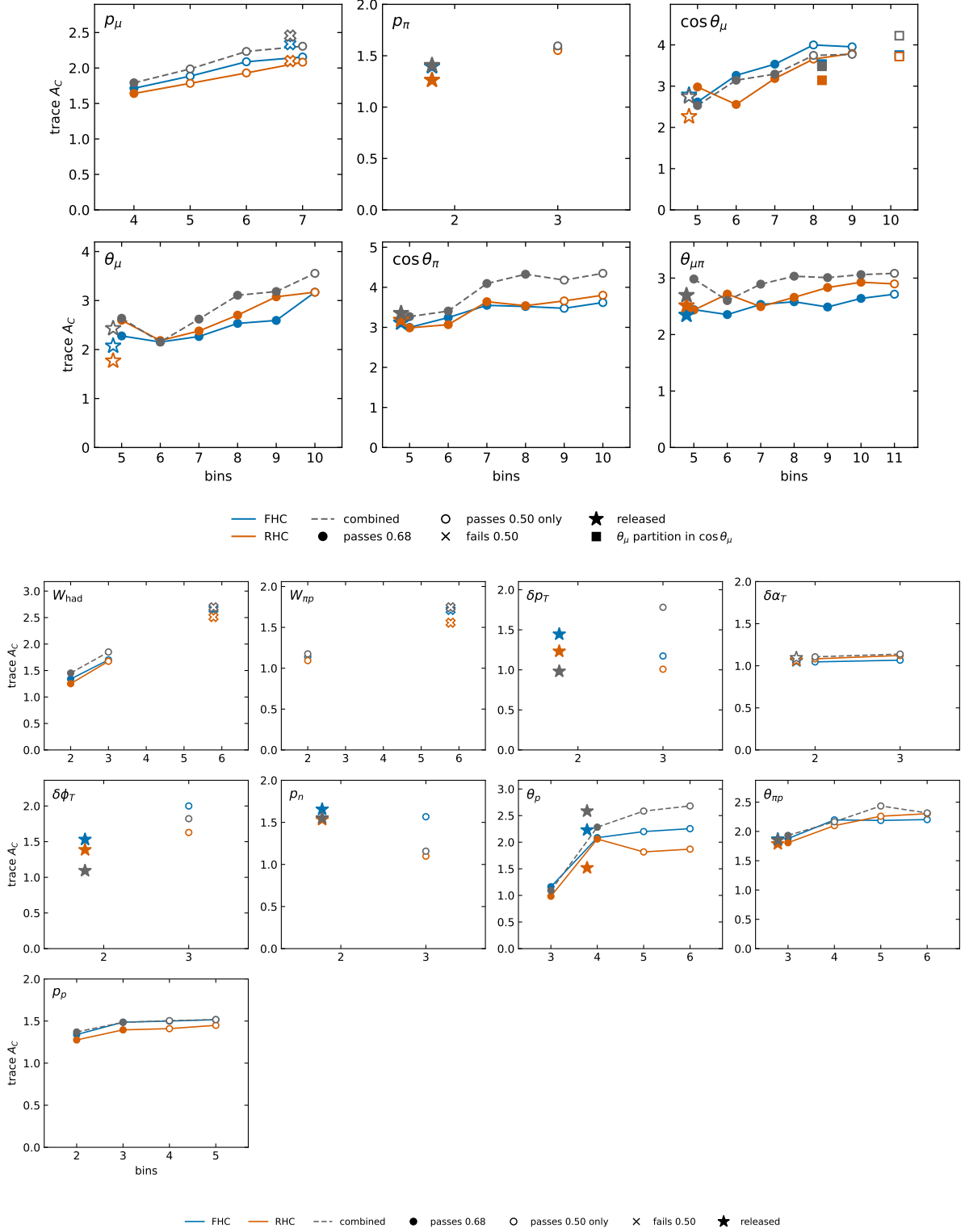


Figure 5: Degrees of freedom for signal (trace of A_C) against the number of bins, inclusive (top) and proton-tagged (bottom), FHC, RHC and combined. Lines join the maximin binnings; the star is the released binning (for θ_p and θ_{np} the study binning), drawn just left of its bin count; squares are the θ_μ partitions written in $\cos \theta_\mu$. A filled marker passes 0.68 in FHC and RHC, an open one 0.50 only, a cross neither. The three-bin W_{had} is built at an event floor of 30, below that of the family. Pseudo-data; analysis sample.

Table 9: Inclusive binnings under the two criteria, FHC, RHC and combined. For each observable: the released binning and the binning the rule of §8 selects under the 0.68 and the 0.50 criterion. “worst diag.” is the smallest column-normalised migration diagonal over the true bins; “trace A_C ” the degrees of freedom for signal; “factors ≥ 0.5 ” the number of Wiener filter factors of at least 0.5, the shape components the data determine with signal-to-noise of at least one; the last column the largest per-bin total uncertainty. [†] The choice depends on the details of the rule (§8). Fake data.

	bins	worst diag. FHC/RHC	trace A_C FHC/RHC/COMB	factors ≥ 0.5 FHC/RHC/COMB	largest bin unc. [%] FHC/RHC/COMB
p_μ					
released	7	0.33/0.31	2.3/2.1/2.5	3/3/3	58/48/48
0.68	4	0.71/0.71	1.7/1.6/1.8	2/2/2	47/44/48
0.50 [†]	6	0.57/0.59	2.1/1.9/2.2	2/2/2	47/49/43
p_π					
released	2	0.76/0.77	1.4/1.3/1.4	2/1/2	111/62/71
0.68 (= released)	2	0.76/0.77	1.4/1.3/1.4	2/1/2	111/62/71
0.50 [†]	3	0.51/0.53	1.6/1.6/1.6	2/2/2	75/45/53
$\cos \theta_\mu$					
released	5	0.67/0.68	2.8/2.3/2.7	3/2/4	52/103/49
0.68 [†]	7	0.70/0.70	3.5/3.2/3.3	4/4/4	54/61/53
0.50 [†]	8	0.66/0.66	4.0/3.7/3.7	4/4/4	62/56/49
θ_μ					
released	5	0.66/0.65	2.1/1.8/2.4	2/2/3	33/46/44
0.68 [†]	8	0.70/0.70	2.5/2.7/3.1	3/3/3	51/44/44
0.50 [†]	10	0.64/0.66	3.2/3.2/3.6	3/2/4	45/60/65
$\cos \theta_\pi$					
released	5	0.74/0.76	3.1/3.2/3.4	4/3/4	56/48/50
0.68 [†]	7	0.73/0.74	3.6/3.6/4.1	4/3/5	60/62/44
0.50 [†]	7	0.73/0.74	3.6/3.6/4.1	4/3/5	60/62/44
$\theta_{\mu\pi}$					
released	5	0.79/0.79	2.3/2.5/2.7	3/3/4	45/39/39
0.68 [†]	6	0.83/0.83	2.4/2.7/2.6	3/3/3	37/48/35
0.50 [†]	6	0.83/0.83	2.4/2.7/2.6	3/3/3	37/48/35

Table 10: Proton-tagged binnings under the two criteria, as Table 9, with an event floor of 50 selected FHC signal events per bin. “study” marks the proposed θ_p and $\theta_{\pi p}$ binnings, which are not in the released result set. * Below the event floor: the released W_{had} and $W_{\pi p}$ hold 3 and 45 events in their sparsest bin, the study θ_p and $\theta_{\pi p}$ 21 and 42. † The choice depends on the details of the rule. Fake data.

	bins	worst diag. FHC/RHC	trace A_C FHC/RHC/COMB	factors ≥ 0.5 FHC/RHC/COMB	largest bin unc. [%] FHC/RHC/COMB
W_{had}					
released*	6	0.09/0.08	2.7/2.5/2.7	4/4/4	305/162/654
0.68	2	0.71/0.72	1.3/1.3/1.5	1/1/2	68/100/91
0.50	2	0.71/0.72	1.3/1.3/1.5	1/1/2	68/100/91
$W_{\pi p}$					
released*	6	0.03/0.03	1.7/1.6/1.7	2/1/2	496/145/122
0.68	1	no binning of two or more bins is admissible			
0.50	2	0.50/0.51	1.1/1.1/1.2	1/1/1	91/136/113
δp_T					
released	2	0.74/0.72	1.4/1.2/1.0	2/1/1	121/85/24
0.68 (= released)	2	0.74/0.72	1.4/1.2/1.0	2/1/1	121/85/24
0.50 (= released)†	2	0.74/0.72	1.4/1.2/1.0	2/1/1	121/85/24
$\delta\alpha_T$					
released	2	0.66/0.66	1.1/1.1/1.1	1/1/1	45/47/46
0.68	1	no binning of two or more bins is admissible			
0.50	2	0.67/0.66	1.0/1.1/1.1	1/1/1	46/49/46
$\delta\phi_T$					
released	2	0.79/0.76	1.5/1.4/1.1	2/2/1	189/152/37
0.68 (= released)	2	0.79/0.76	1.5/1.4/1.1	2/2/1	189/152/37
0.50	3	0.62/0.60	2.0/1.6/1.8	2/2/2	188/69/48
p_n					
released	2	0.77/0.75	1.7/1.5/1.6	2/2/2	172/222/155
0.68 (= released)	2	0.77/0.75	1.7/1.5/1.6	2/2/2	172/222/155
0.50 (= released)	2	0.77/0.75	1.7/1.5/1.6	2/2/2	172/222/155
θ_p					
study*	4	0.74/0.73	2.2/1.5/2.6	2/1/3	79/91/101
0.68	4	0.69/0.68	2.1/2.1/2.3	2/2/3	41/84/43
0.50†	4	0.69/0.68	2.1/2.1/2.3	2/2/3	41/84/43
$\theta_{\pi p}$					
study*	3	0.77/0.75	1.9/1.8/1.9	2/2/2	140/187/109
0.68	3	0.75/0.74	1.9/1.8/1.9	2/2/2	97/121/99
0.50†	4	0.67/0.67	2.2/2.1/2.2	3/3/2	94/141/60
p_p					
0.68†	2	0.87/0.86	1.3/1.3/1.4	2/1/2	48/61/52
0.50†	3	0.74/0.73	1.5/1.4/1.5	2/2/2	48/58/51

Table 11: Bin edges of the binnings selected under each criterion (Tables 9 and 10); the last bin of p_μ , p_π , W_{had} , $W_{\pi p}$, δp_T and p_n is open above, as in the release. “released” where the rule keeps the released edges.

Observable	criterion	edges
p_μ (GeV/c)	0.68	0.15, 0.3, 0.525, 0.95, 3
	0.50	0.15, 0.3, 0.425, 0.575, 0.825, 1.25, 3
p_π (GeV/c)	0.68	released
	0.50	0.175, 0.205, 0.26, 1
$\cos \theta_\mu$	0.68	-1, 0.55, 0.725, 0.825, 0.9, 0.95, 0.975, 1
	0.50	-1, 0.4, 0.625, 0.75, 0.825, 0.9, 0.95, 0.975, 1
θ_μ (rad)	0.68	0, 0.16, 0.239, 0.339, 0.439, 0.578, 0.738, 0.957, 3.15
	0.50	0, 0.16, 0.239, 0.319, 0.399, 0.498, 0.618, 0.738, 0.897, 1.156, 3.15
$\cos \theta_\pi$	0.68	-1, -0.15, 0.15, 0.375, 0.6, 0.775, 0.925, 1
	0.50	-1, -0.15, 0.15, 0.375, 0.6, 0.775, 0.925, 1
$\theta_{\mu\pi}$ (rad)	0.68	0, 0.55, 0.875, 1.225, 1.625, 2.05, 2.6
	0.50	0, 0.55, 0.875, 1.225, 1.625, 2.05, 2.6
W_{had} (GeV/c ²)	0.68	0, 1.18, 2.74
	0.50	0, 1.18, 2.74
$W_{\pi p}$ (GeV/c ²)	0.68	one bin
	0.50	1.08, 1.2, 2.9
δp_T (GeV/c)	0.68	released
	0.50	released
$\delta \alpha_T$ (deg)	0.68	one bin
	0.50	0, 122.5, 180
$\delta \phi_T$ (deg)	0.68	released
	0.50	0, 22.5, 75, 180
p_n (GeV/c)	0.68	released
	0.50	released
θ_p (rad)	0.68	0, 0.44, 0.785, 1.162, π
	0.50	0, 0.44, 0.785, 1.162, π
$\theta_{\pi p}$ (rad)	0.68	0, 1.131, 1.916, π
	0.50	0, 0.911, 1.414, 1.916, π
p_p (GeV/c)	0.68	0.3, 0.42, 1.5
	0.50	0.3, 0.42, 0.6, 1.5

Inclusive observables. p_μ . The released seven bins fail both criteria. Their FHC diagonals fall from 0.80 to 0.33 with momentum, because the bins are narrower than the 30% resolution of the MCS estimator; for that reason the released p_μ binning was exempted from the criterion (Sec. 11.6 of the analysis note). Under 0.68 four bins are admissible (worst diagonal 0.71 in both beams): the trace falls from 2.3/2.1 to 1.7/1.6, and the number of components measured with signal-to-noise above one from three to two. Under 0.50 the rule selects six bins (0.57/0.59, trace 2.1/1.9). Seven bins (0.51/0.51) add 0.1 and fail 0.50 in the proton-tagged sample, which uses the same edges (0.46/0.49), whereas the six-bin edges pass there (0.53/0.55). No admissible binning keeps the third well-measured component of the released one ($W = 0.62/0.52$); the closest is 0.40/0.46, at seven bins.

p_π . The released edges (0.175 and 0.205 GeV/c, open above) are the maximin two-bin partition, and no finer binning passes 0.68: the reconstructed momentum saturates near 0.34 GeV/c (§7.2). Under 0.50 a third bin from 0.26 GeV/c passes by 0.007 (0.51/0.53). Its own filter factor is small (0.05/0.01), but the split raises the second factor from 0.55/0.40 to 0.68/0.71, so the trace rises from 1.4/1.3 to 1.6/1.6 and the largest per-bin uncertainty falls from 111% to 75% in FHC. The same edges fail 0.50 in the proton-tagged sample (0.48 in FHC).

$\cos \theta_\mu$. The released five bins fail 0.68 in the backward bin (0.67/0.68). In RHC their fifth filter factor is 0.001 and the smallest A_C row sum 0.05, so one bin is filled almost entirely by the regularisation. The maximin five bins place four bins above $\cos \theta_\mu = 0.7$, where the events are, and pass at 0.757/0.765 with a trace of 2.6/3.0 and row sums of at least 0.61. Under 0.68 the rule

selects seven bins (0.70/0.70; trace 3.5/3.2, four components above signal-to-noise one in every configuration), and under 0.50 eight (0.66/0.66; trace 4.0/3.7), beyond which the trace levels off (nine bins: 3.95/3.79). The θ_μ partitions add one bin to each without adding information: eight bins pass 0.68 (0.70/0.69) with a trace of 3.5/3.1, and ten bins pass 0.50 (0.64/0.66) with 3.75/3.72.

θ_μ . This is the same variable in another parameterisation. The released binning fails 0.68 (0.66/0.65); the rule selects eight bins under 0.68 (0.70/0.70, trace 2.5/2.7) and ten under 0.50 (0.64/0.66, trace 3.2/3.2). The same partition retains less as θ_μ than as $\cos\theta_\mu$: the eight-bin partition keeps a trace of 2.5/2.7/3.1 as θ_μ and 3.5/3.1/3.5 as $\cos\theta_\mu$, the ten-bin one 3.2/3.2/3.6 and 3.75/3.72/4.22, with the same diagonals up to the three-decimal rounding of the edges. The regulariser penalises the curvature of the cross-section density, and $d\sigma/d\theta_\mu$ and $d\sigma/d\cos\theta_\mu$ are different functions, so the parameterisation is part of the regularisation. A θ_μ result adds no information to a $\cos\theta_\mu$ result with the same partition, and retains less.

$\cos\theta_\pi$. The released five bins pass both criteria (0.74/0.76). The trace rises to 3.6/3.6 at seven bins (0.73/0.74) and no further (eight to ten bins: 3.5–3.8), so both criteria select seven bins, with a largest per-bin uncertainty of 60/62% against 56/48%. The gain, 0.45 degrees of freedom in each beam and 0.7 combined, is the one improvement of the review that does not repair a criterion failure. The seven-bin edges give 0.69/0.71 and a bin of 30 events in the proton-tagged sample, whose $\cos\theta_\pi$ keeps its own four bins.

$\theta_{\mu\pi}$. The released five bins pass both criteria (0.79/0.79), and the trace barely moves between five and eleven bins (2.3–2.7 in FHC, 2.4–2.9 in RHC); no binning has more than four filter factors above 0.2. The observable carries about two and a half degrees of freedom at this exposure however it is binned. The rule selects six bins, rule-sensitively, and any count from five to ten is equivalent within its resolution. At five bins the maximin edges (0, 0.6, 1.025, 1.5, 2.0, 2.6 rad) raise the worst diagonal to 0.86 and the smallest row sum from 0.52/0.58 to 0.73/0.62 at unchanged trace.

Proton-tagged observables. W_{had} . The released six bins fail both criteria: their FHC diagonals are 0.23, 0.45, 0.57, 0.44, 0.22 and 0.09, below 0.68 in every bin and below 0.50 in five of the six. The first bin holds three selected signal events, the per-bin uncertainty reaches 305% in FHC, 162% in RHC and 654% combined, and the single-throw closure ratio of one combined bin is 78. Under either criterion the admissible binning is two regions split at 1.18 GeV/ c^2 (0.71/0.72), with a trace of 1.3/1.3 and, in FHC and RHC, one component above signal-to-noise one. Three bins pass 0.50 only below the event floor (30 events: 0.531/0.535, per-bin uncertainty up to 325%).

$W_{\pi p}$. No two-bin partition passes 0.68. Under 0.50 a split at 1.20 GeV/ c^2 passes by 0.003, with a second filter factor of 0.29/0.27: a normalisation, not a shape. Under either criterion no differential $W_{\pi p}$ result is reported.

δp_T and p_n . The released two bins pass both criteria (0.74/0.72 and 0.77/0.75). The three-bin maximin partitions pass 0.50 but lower the trace in FHC and RHC (δp_T : 1.45/1.23 to 1.17/1.01; p_n : 1.66/1.53 to 1.57/1.10), so both criteria keep two bins. In the combined configuration the two-bin δp_T has a trace of 0.98: its shape component is not measured there.

$\delta\alpha_T$. The released two bins fail 0.68 (0.66/0.66), and so does the best two-bin partition (0.67/0.66): under 0.68 there is no $\delta\alpha_T$ shape. Under 0.50 two bins are admissible, but their second filter factor is 0.19–0.24 in every configuration and a third bin adds nothing, so $\delta\alpha_T$ contributes one well-measured number under either criterion.

$\delta\phi_T$. The released two bins pass both criteria (0.79/0.76). Under 0.50 three bins (0.62/0.60; edges 0, 22.5, 75 and 180°) raise the trace from 1.5/1.4 to 2.0/1.6, and in the combined configuration from 1.1 to 1.8; the new factor is 0.40 in FHC but 0.01 in RHC.

θ_p . The study binning (0.74/0.73) holds 43 and 21 events in its two backward bins, below the floor, and its fourth filter factor in RHC is 0.001. The maximin four bins at the floor (0.69/0.68)

have a trace of 2.1/2.1 against 2.2/1.5, and a largest per-bin uncertainty of 41/84% against 79/91%; five and six bins add nothing. Both criteria select these four bins, which pass 0.68 by 0.002 in RHC. Three bins pass more comfortably (0.77/0.75) but retain a single component (trace 1.2/1.0), so they do not serve as a fallback.

$\theta_{\pi p}$. The last bin of the study binning holds 42 events, below the floor. Under 0.68 the maximin three bins (0, 1.131, 1.916 and π rad; 0.75/0.74) meet the floor and retain the same information as the study binning (trace 1.88/1.81 against 1.87/1.79) with a smaller largest per-bin uncertainty (97/121% against 140/187%). Under 0.50 four bins (0.67/0.67) raise the trace to 2.2/2.1 and add a third component above signal-to-noise one in FHC and RHC; five or six bins add nothing.

p_p . The momentum of the leading proton, measured from its range, is added to the family. The signal requires $p_p > 0.3$ GeV/ c , which is the lower edge, and the upper bin is open. Two bins split at 0.42 GeV/ c pass 0.68 (0.87/0.86), with filter factors 0.83 and 0.51 in FHC (0.80 and 0.48 in RHC). A third bin at 0.60 GeV/ c keeps the same two components and measures the second better (0.68/0.64), at a worst diagonal of 0.74/0.73; its third factor is 0.001, and four or five bins add nothing (trace 1.49/1.40 at three bins, 1.52/1.45 at five). The rule selects the two bins under 0.68 and the three under 0.50.

Shared kinematic variables in the proton-tagged sample. The proton-tagged p_μ , p_π , $\cos\theta_\mu$ and $\theta_{\mu\pi}$ use the inclusive edges, evaluated here in the proton-tagged sample. The released p_μ gives 0.31/0.31 there and the released $\cos\theta_\mu$ 0.71/0.70. Of the selected inclusive binnings, the four-bin p_μ passes 0.68 by 0.005 in FHC (0.685/0.713), the six-bin p_μ passes 0.50 (0.53/0.55), the three-bin p_π fails 0.50 (0.48/0.51) and the five-bin maximin $\theta_{\mu\pi}$ passes at 0.84/0.86. The seven- and eight-bin $\cos\theta_\mu$ pass 0.68 (0.70/0.71) and 0.50 (0.65/0.68) there as well.

9 Hadronic mass

[PROTON-TAGGED]

9.1 The W binnings are finer than the resolution

In the six-bin scheme, $W_{\pi p}$ shows the least model tension of the proton-tagged observables. The origin is the binning, not physics.

The six bin edges were chosen for equal population, without reference to the resolution. The resolution and the bin widths, measured on selected signal in FHC Runs 1/2/4 (analysis sample), are:

	$W_{\pi p}$	W_{had}
residual RMS [GeV/ c^2]	0.142	0.283
robust half-width (16–84%)	0.101	0.177
bias (mean reco – true)	–0.111	–0.032
narrowest analysis bin	0.040	0.100
widest analysis bin	1.430	1.300
width ratio	35.7	13.0

The second and third $W_{\pi p}$ bins are 40 MeV/ c^2 wide against a resolution of 101–142 MeV/ c^2 : they are two to three times narrower than the detector can resolve. The bias alone, –111 MeV/ c^2 , is larger than three of the six bin widths. The migration diagonals are correspondingly low, 87.6/22.3/11.7/12.0/10.0/2.3%: only the first bin is measured, and the top bin, at 2.3%, is largely unmeasured.

The five-bin p_π scheme fails in the same way and for the same reason. $W_{\pi p}$ is reconstructed from the pion and proton momenta, so it inherits the saturation of the range-based pion momentum described in §7.2; its resolution cannot be better than that of its inputs.

An exhaustive edge scan on the analysis sample, with the procedure and criterion used for p_π , gives:

scheme	diagonals	verdict
current, six bins	87.6/22.3/11.7/12.0/10.0/2.3%	1 of 6 usable
best two bins (split 1.15)	83.2/77.9%	both usable
best three bins	83.2/49.7/45.9%	fails
best four bins	83.2/45.0/28.2/28.2%	fails

As with p_π , two bins is the only scheme that satisfies the criterion, and its first bin is thin (137 selected signal events, against a floor of 100). For the model comparison this means that a low generator χ^2 at six bins reflects a binning that cannot resolve the differences between generators, not agreement between them. The same test applied to W_{had} , whose six-bin upper bins are also narrower than its 177 MeV/ c^2 half-width, finds the six bins below both the 0.68 and the 0.50 criterion, and two regions split at 1.18 GeV/ c^2 as the only admissible binning (§8); that is the binning used for W_{had} .

Two-bin scheme split at 1.15 GeV/ c^2 . The boundary is more tightly constrained than for p_π , where any value between roughly 0.19 and 0.22 satisfies the criterion. Here only one choice satisfies both the diagonal criterion and the 100-event floor, and it is the scheme extracted in §9.2:

boundary [GeV/ c^2]	bin 1	bin 2	$N(\text{bin 1})$	
1.13	76.5%	88.0%	51	below the event floor
1.15	83.2%	77.9%	137	adopted
1.17	86.8%	66.8%	266	bin 2 just fails
1.19	87.6%	56.0%	468	fails
1.23	91.7%	37.6%	1114	fails

Raising the boundary increases the first-bin population at the expense of the second-bin diagonal: the two diagonals trade against each other steeply, and the window in which both exceed 0.68 is only about ± 0.01 GeV/ c^2 wide. The window is this narrow because the resolution is comparable to the full width of the peak region.

The first bin is thin, 137 selected signal events against a floor of 100, so its uncertainty is large once systematics are included; the p_π first bin, with 469 events, carries 83% (Table 8). A boundary at 1.17 doubles the first-bin population for a bin-2 diagonal of 66.8%, marginally below threshold.

The scheme is implemented in `configs/ccpi1p_Wpipr_bin_config_2bin.txt`, with the reasoning recorded in a companion `.README`. The extraction with this scheme (§9.2) meets the migration criterion, but the first bin, with about 137 selected signal events, is suppressed by the Wiener filter, which drives A_C towards rank one for statistical rather than algebraic reasons. Widening the first bin restores its population at the cost of the upper-bin diagonal, and no scanned edge satisfies both requirements at once.

The apparent $W_{\pi p}$ agreement at six bins therefore does not persist in a binning the detector can resolve. No differential $W_{\pi p}$ result is reported. The topology-level one-bin total cross section of the proton-tagged sample remains measurable, but it is not described as a $W_{\pi p}$ measurement. W_{had} carries the same limitation, since it depends on the same pion momentum.

9.2 Limits of the $W_{\pi p}$ extraction

The two-bin $W_{\pi p}$ scheme of §9.1 was run through the full extraction. It fails for a reason distinct from the six-bin failure, and the follow-up studies establish what the pion momentum can and cannot deliver.

The two-bin extraction fails on statistics. The response behaves as designed: the framework returns migration diagonals of 0.802 and 0.773, matching the truth-level scan, and the closure is acceptable at $\chi^2 = 1.17/2$ ($p = 0.56$). The extraction nonetheless carries almost no shape information, because A_C collapses. The matrices below come from the extraction pass in which the two-bin $W_{\pi p}$ scheme was run and are written with true bins as rows; the two p_π rows are the same-pass comparison, and the released inclusive p_π matrix is that of Fig. 3.

two-bin scheme	A_C	s_2/s_1
$W_{\pi p}$, first bin $N = 137$	$\begin{pmatrix} 0.018 & -0.410 \\ 0.034 & 0.892 \end{pmatrix}$	0.031
p_π proton-tagged, $N = 469$	$\begin{pmatrix} 0.498 & -0.089 \\ 0.013 & 0.854 \end{pmatrix}$	0.577
p_π inclusive, $N = 1056$	$\begin{pmatrix} 0.594 & -0.382 \\ 0.011 & 0.891 \end{pmatrix}$	0.528

The top-left entry is 0.018: A_C passes under two per cent of the first bin and takes -0.41 from the second. This is a rank-one collapse of the kind described in §25.1, but its cause is statistical rather than a defective regulariser. With 137 events the first bin has such poor signal-to-noise that the Wiener filter suppresses that mode almost entirely, and the ordering across the three schemes, $137 \rightarrow 0.031$, $469 \rightarrow 0.577$, $1056 \rightarrow 0.528$, follows population rather than anything geometric.

$W_{\pi p}$ is therefore subject to two incompatible requirements. Bins narrow enough to satisfy the response criterion leave the first bin too sparse to unfold; bins wide enough for a stable unfolding (1.19 gives 468 events, matching the working p_π case) fail the criterion at 56%.

No two- or three-bin scheme satisfies both requirements. A scan of every two- and three-bin scheme with edges between 1.12 and 1.70 GeV/ c^2 (FHC Run1+2+4, 3505 selected signal events) finds none that keeps both a migration diagonal above 0.68 and adequate statistics in every bin:

two-bin edge	N_1/N_2	diag ₁ / diag ₂	
1.15	137/3367	0.832/0.779	first bin too sparse
1.18	351/3153	0.866/0.611	
1.24	1298/2206	0.930/0.347	
1.27 (median)	1744/1760	0.951/ 0.261	upper bin unusable
1.45	2966/538	0.993/0.030	

The trend follows from the reconstruction bias. Reconstructed $W_{\pi p}$ is biased about 111 MeV low, inheriting the saturation of the range-based pion momentum (§7.2), so raising the edge far enough to balance the statistics drains the upper bin by downward migration: diag₁ climbs from 0.786 to 0.999 across the scan while diag₂ collapses from 0.927 to 0.004. There is no edge at which both bins are simultaneously populated and diagonal, and no three-bin combination does better: the third-bin diagonal never exceeds 0.347.

At this exposure $W_{\pi p}$ does not support a shape measurement unless the pion momentum improves. No differential $W_{\pi p}$ result is reported. The topology-level one-bin total cross section of the proton-tagged sample remains measurable, but it is not described as a $W_{\pi p}$ measurement. It is a one-bin extraction rather than a regularised unfolding: with a single bin there is nothing to unfold, since the response is 1×1 , A_C is trivially unity and the regularisation has no effect. It is reported through the total flux-averaged cross-section path of Sec. 14.6 of the analysis note. Generator predictions integrated over the $W_{\pi p}$ range are built by `generator_predictions/newg4/make_wpipr_1bin.C`, summing the per-bin cross sections of the six-bin flux-averaged predictions.

Six-bin $W_{\pi p}$ closure, diagnostic only. The six-bin extraction was carried through the full chain in all three configurations, detector variations included. It closes against the A_C -smeared

truth, but that closure cannot restore shape information the response does not contain (the top bin is 2% diagonal), so these numbers are a check of the machinery and not a result, and they do not appear in the result tables of the proton-tagged note. The six-bin generator predictions (GENIE 0.501, NuWro 0.646, GiBUU 0.735, NEUT $0.783 \times 10^{-38} \text{ cm}^2/\text{Ar}$, FHC) are in the release for the same diagnostic purpose.

Table 12: Six-bin $W_{\pi p}$ pseudo-data closure, diagnostic only (not reported as a differential result). σ_{int} in $10^{-38} \text{ cm}^2/\text{Ar}$; columns as in Table 8 of the proton-tagged note.

Config	σ_{int}	unf./truth	χ^2/ndf
FHC	0.466	0.97	0.92/6
RHC	0.305	0.88	2.56/6
Combined	0.359	0.85	1.12/6

The limitation comes from the pion momentum alone. Correlating the $W_{\pi p}$ residual with each input residual isolates the cause:

input	r	W residual RMS when...	
pion momentum	+0.726	all events	0.142
proton momentum	+0.012	pion momentum well reconstructed	0.061
pion $\cos \theta$	-0.029	proton momentum well reconstructed	0.130
proton $\cos \theta$	-0.032	both well reconstructed	0.038

Fixing the pion momentum alone takes the resolution from 0.142 to 0.061, a factor 2.3; fixing the proton momentum changes little, and the angles have no effect. The hadronic interaction that truncates the pion track therefore limits p_π to two bins and excludes a shape measurement of $W_{\pi p}$.

Reparameterisation to W^2 leaves the migration unchanged. A monotonic transformation with mapped edges leaves the migration matrix unchanged, and this was verified directly: splitting at $1.15 \text{ GeV}/c^2$ in W and at $1.15^2 = 1.3225 \text{ GeV}^2/c^4$ in W^2 gives diagonals of 83.2% and 77.9% in both cases, on identical events. Resolution and bin width scale together, $\sigma(W^2) = 2W\sigma(W)$, so σ/width is invariant at about 2 either way. The same argument applies to θ_μ versus $\cos \theta_\mu$ (§25.1).

10 Proton-tagged selection, binning and additional smearing

[PROTON-TAGGED]

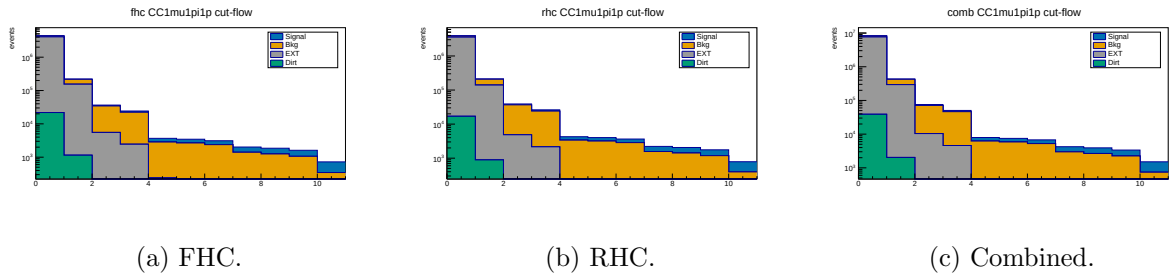


Figure 6: Proton-tagged (CC1mu1pi1p) cut-flow yields at full exposure (log scale; MC-prediction stack, Okabe-Ito: signal blue, background orange, EXT grey, dirt green). The last stage, IdentProton, is the leading-proton identification that defines the proton-tagged sample.

Table 13: Proton-tagged (CC1mu1pi1p) cut-flow yields at full exposure for FHC, RHC and combined (MC prediction; data blind). Stages None–Nonprotons are the inherited inclusive selection with the proton-tagged signal definition; the final IdentProton stage adds the leading-proton identification. Signal/Bkg from the ν_μ overlays POT-scaled per run, reprocessed with the proton-tagged selection and the full central-value weight under the safe-weight rule (`reprocess_cutflow_cf_1p.sh`); EXT from the per-run beam-off samples scaled by the run gate ratios; dirt from the GENIE NuMI dirt overlay (`macros/cutflow_yields_1p.C`). The final row agrees with the proton-tagged cut-and-count check (Table 7 of the proton-tagged note). Pred. = Signal + Bkg + EXT + Dirt. Eff. is the cumulative signal efficiency relative to the generated signal (the None stage, where all true signal is counted); Pur. = Signal/Pred.

Cut	Signal	Bkg MC	EXT	Dirt	Pred.	Eff. [%]	Pur. [%]
<i>FHC (Runs 1,2,4,5; 7.766×10^{20} POT)</i>							
None	2 888	237 145	3 256 689	19 171	3 515 893	100.0	0.08
InFiducialVol	2 424	54 365	122 860	1 005	180 654	83.9	1.3
Topological	1 917	25 340	4 299	109	31 665	66.4	6.1
MuonCandidate	1 739	17 435	1 870	73	21 117	60.2	8.2
ContainedPion	729	2 289	173	14	3 204	25.2	22.7
MuonIn3Planes	694	2 171	144	9.4	3 018	24.0	23.0
PionIn3Planes	645	1 966	113	3.9	2 728	22.3	23.6
ShowerCut	539	1 140	83	3.3	1 766	18.7	30.5
OpeningAngle	535	1 065	32	1.1	1 633	18.5	32.8
Nonprotons	484	910	23	0.7	1 417	16.7	34.1
IdentProton	331	298	6.8	0.2	637	11.5	52.0
<i>RHC (Runs 1,2,3,4a,4b,4c; 1.108×10^{21} POT)</i>							
None	3 427	288 891	3 560 975	16 958	3 870 250	100.0	0.09
InFiducialVol	2 886	67 416	139 122	889	210 312	84.2	1.4
Topological	2 318	31 870	4 767	97	39 052	67.7	5.9
MuonCandidate	2 111	21 962	2 090	65	26 227	61.6	8.0
ContainedPion	882	3 181	191	12	4 267	25.7	20.7
MuonIn3Planes	842	3 015	158	8.3	4 023	24.6	20.9
PionIn3Planes	784	2 739	121	3.4	3 648	22.9	21.5
ShowerCut	647	1 480	86	2.9	2 216	18.9	29.2
OpeningAngle	643	1 384	33	0.9	2 060	18.8	31.2
Nonprotons	573	1 162	23	0.6	1 759	16.7	32.6
IdentProton	391	388	7.1	0.2	786	11.4	49.8
<i>Combined FHC+RHC (1.885×10^{21} POT)</i>							
None	6 315	526 036	6 817 663	36 129	7 386 144	100.0	0.09
InFiducialVol	5 310	121 781	261 981	1 894	390 966	84.1	1.4
Topological	4 235	57 210	9 066	206	70 717	67.1	6.0
MuonCandidate	3 849	39 396	3 960	138	47 344	61.0	8.1
ContainedPion	1 611	5 470	364	26	7 471	25.5	21.6
MuonIn3Planes	1 536	5 186	302	18	7 041	24.3	21.8
PionIn3Planes	1 429	4 705	234	7.3	6 376	22.6	22.4
ShowerCut	1 185	2 621	169	6.2	3 982	18.8	29.8
OpeningAngle	1 178	2 448	65	2.0	3 693	18.6	31.9
Nonprotons	1 057	2 072	46	1.4	3 176	16.7	33.3
IdentProton	723	686	14	0.5	1 423	11.4	50.8

Table 14: Binning of the nine proton-tagged observables specific to the family, read from the data release (`curves_1p_*.tsv`). Every binning except the six-bin $W_{\pi p}$ meets the 0.50 migration criterion on the analysis sample with at least 50 selected FHC signal events per bin (§8); the last bin of the momenta and masses is open at truth level (the upper edge listed is the plotting boundary). $W_{\pi p}$ brackets the $\Delta(1232)$ peak; no differential $W_{\pi p}$ result is reported. The five inherited kinematic observables (p_μ , p_π , $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$) use the binnings of Table 3 of the proton-tagged note.

Observable	Definition	Bins	Released bin edges
p_p	leading proton, range	3	0.3, 0.42, 0.6, 1.5 GeV/c
$W_{\pi p}$	$M(\pi p)$, direct	6 (Not reported)	1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.9 GeV/c ²
W_{had}	calorimetric	2	0, 1.18, 2.74 GeV/c ²
δp_T	$ \vec{p}_T^\mu + \vec{p}_T^{\text{had}} $	2	0, 0.3, 2.5 GeV/c
$\delta\alpha_T$	TKI angle	2	0, 120, 180°
$\delta\phi_T$	TKI angle	3	0, 22.5, 75, 180°
p_n	inferred nucleon momentum	2	0, 0.375, 2 GeV/c
θ_p	proton polar angle	4 [†]	0, 0.44, 0.785, 1.162, π rad
$\theta_{\pi p}$	π - p opening angle	4 [†]	0, 0.911, 1.414, 1.916, π rad

[†] Maximin edges of the binning review.

10.1 Reconstruction of the proton-tagged observables from input quantities

All six observables are built from the three candidate tracks (muon, charged pion, proton) using only reconstructed quantities stored in the input ntuples: the track direction cosines $\hat{n}$, the muon-hypothesis range and multiple-scattering momenta and the proton-hypothesis range kinetic energy.

Candidate three-momenta. The muon momentum magnitude is range-based when the muon track ends inside the containment volume (Sec. 7.1 of the analysis note), and multiple-Coulomb-scattering based otherwise. The charged pion, being minimum-ionising, starts from the muon-hypothesis range momentum, converted to the pion hypothesis at fixed kinetic energy in the reconstructed-bin definition exactly as in the inclusive analysis (Sec. 8 of the analysis note); the same configuration formula is used. The proton momentum follows from its kinetic energy under the proton hypothesis, $p_p = \sqrt{T_p^2 + 2m_p T_p}$, with T_p taken from the track range (`trk_energy_proton_v`, an exact function of the track length). Each three-momentum is $\vec{p}_i = p_i \hat{n}_i$, and the energies are $E_\mu = \sqrt{p_\mu^2 + m_\mu^2}$, $E_\pi = \sqrt{p_\pi^2 + m_\pi^2}$, $E_p = T_p + m_p$ (masses $m_\mu = 105.7$, $m_\pi = 139.6$, $m_p = 938.3$ MeV).

Directly measured mass $W_{\pi p}$. The invariant mass of the pion-proton four-vector sum, using no neutrino-direction assumption:

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2}.$$

Transverse kinematic imbalance. The TKI observables treat the muon against the visible hadronic system $\vec{p}_{\text{had}} = \vec{p}_\pi + \vec{p}_p$, $E_{\text{had}} = E_\pi + E_p$. With the components transverse to the neutrino direction (the $\hat{z}$ axis of the beam frame) written $\vec{p}_T^\mu$ and $\vec{p}_T^{\text{had}}$,

$$\delta p_T = |\vec{p}_T^\mu + \vec{p}_T^{\text{had}}|, \quad \delta\alpha_T = \arccos \frac{-\vec{p}_T^\mu \cdot (\vec{p}_T^\mu + \vec{p}_T^{\text{had}})}{|\vec{p}_T^\mu| \delta p_T}, \quad \delta\phi_T = \arccos \frac{-\vec{p}_T^\mu \cdot \vec{p}_T^{\text{had}}}{|\vec{p}_T^\mu| |\vec{p}_T^{\text{had}}|},$$

both angles in degrees folded onto $[0, 180]$. For a stationary nucleon with no final-state interactions $\delta p_T \rightarrow 0$; Fermi motion broadens it and FSI drive $\delta\alpha_T \rightarrow 180^\circ$.

Calorimetric mass W_{had} and struck-nucleon momentum p_n . The calorimetric available energy is $E_{\text{cal}} = E_\mu + (E_{\text{had}} - m_p) + E_b$ with the argon removal energy $E_b = 24.78$ MeV, giving the

momentum transfer $Q^2 = -[(0, 0, E_{\text{cal}}, E_{\text{cal}}) - p^\mu]^2$ and

$$W_{\text{had}} = \sqrt{m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2}$$

(returned as 0 where reconstruction fluctuations make the argument negative). The nucleon-momentum proxy p_n combines the transverse imbalance with a longitudinal component p_L from energy-momentum balance, $p_n = \sqrt{\delta p_T^2 + p_L^2}$, using the MINERvA prescription [27] as implemented in the framework’s transverse-kinematics tool (option 1):

$$p_L = \frac{R}{2} - \frac{M_{A'}^2 + \delta p_T^2}{2R}, \quad R = M_A + p_z^\mu + p_z^{\text{had}} - E_\mu - E_{\text{had}},$$

with the target mass $M_A = 22m_n + 18m_p - 0.34381$ GeV (argon-40), the residual mass $M_{A'} = M_A - m_n + E_b$ and $E_b = 24.78$ MeV; z is the beam axis (Sec. 4.2 of the analysis note). The tool was written for a single-proton hadronic system: it receives the summed three-momentum $\vec{p}_\pi + \vec{p}_p$ and energy $E_{\text{had}} = E_\pi + E_p$ in the proton slot and forms the hadronic kinetic energy as $E_{\text{had}} - m_p$, which is why m_p appears in E_{cal} and W_{had} above. δp_T , $\delta\alpha_T$ and $\delta\phi_T$ depend only on the transverse momenta and are exact for the $\mu + (\pi p)$ system; p_L , p_n and W_{had} inherit the single-nucleon residual-mass and removal-energy assumptions of a quasi-elastic-oriented prescription and are therefore proxies, not the initial-state nucleon momentum. The generator predictions apply the identical formulas to the generator final states (Sec. 10.2), so the comparison is like for like; a truth-level closure of this prescription against a pion-production-specific definition, and a quantification of the bias from the residual-mass assumption, have not been performed (Sec. 9 of the proton-tagged note).

Two exposure conventions in the processed files. One normalisation point is specific to the proton-tagged sample. The processed ntuples use two POT conventions: the files behind the inclusive selection store the total run MC POT duplicated in every file of a multi-file run, whereas the proton-tagged files store each file’s own POT. Normalising by the individual file POT would be correct for the first set but would scale every split of a multi-file run to the full run exposure in the second, inflating the simulated background by the split multiplicity ($\sim 3\times$ for RHC) and driving the background-subtracted spectrum negative; normalising by the summed per-file POT has the mirror problem. The analysis therefore normalises each run by the sum of the distinct per-file POT values within that run, which reduces to the correct run total under either convention; FHC, with one MC file per run, is unaffected by the choice.

10.2 Additional smearing in the proton-tagged observables

Because the Wiener-SVD additional-smearing matrix A_C is not norm-preserving, each observable’s predictions are rescaled by a different factor when they are brought into the measurement space. Table 15 gives that factor, measured as the ratio of the A_C -smeared to the truth-level generator integral, for the proton-tagged observables. Its dominant behaviour is set by the response and the binning; the exact factor is prediction-dependent, and for the tested generators GENIE and GiBUU agree to a few per cent throughout.

In the released extractions the proton-tagged observables are suppressed by 0.57–0.85 (Table 15): the two-region W_{had} is the least affected (0.85), the two-bin TKI variables, $W_{\pi p}$, p_p and the proton angles sit at 0.68–0.79, and the three-bin $\delta\phi_T$ is the most affected (0.57). The suppression is not driven by sharpness alone: $W_{\pi p}$ is the sharpest distribution measured here (the ratio of its largest to smallest bin is ~ 36) yet is suppressed no more than the TKI variables, because A_C removes structure that the response cannot resolve and $W_{\pi p}$ is built from directly measured pion and proton momenta with equal-population bins that track the $\Delta(1232)$ peak.

No observable is enhanced by A_C : in the released configurations every prediction-dependent integral factor is below unity. This is an empirical property of these extractions, not a mathematical requirement on A_C , which is a regularisation operator and can suppress or enhance; an amplification would signal a mis-specified regularisation (§25.1). The suppression is applied identically to data, tune and generators, so model comparisons read off any panel are made on equal footing.

Table 15: Mean A_C row sum for the proton-tagged observables (FHC), from the released matrices `A.C_1p_FHC5_*.tsv`; equal to the ratio of the A_C -smeared to the truth-level integral for a flat prediction.

[†] Secondary result (Table 2); $W_{\pi p}$ is Not reported and the others are each a Validation-only product.

Observable	mean A_C row sum
W_{had}	0.85
$\delta\alpha_T$	0.79
p_n	0.78
$\theta_p^\dagger$	0.76
$W_{\pi p}$	0.75
p_p	0.72
δp_T	0.70
$\theta_{\pi p}^\dagger$	0.68
$\delta\phi_T$	0.57

Computed from the released matrices (`A.C_1p_FHC5_*.tsv`). The values span 0.57–0.85, and those of the inclusive FHC results 0.61–0.79, so differences in smearing cannot order the generator χ^2 across observables.

11 Two-dimensional pairs

[SHARED]

Eight two-dimensional pairs were screened and extracted in the three configurations. A two-dimensional binning is a flat list of analysis bins ordered slice by slice, so the extraction uses the same machinery, and the integrated normalisation of the two-dimensional chain agrees with the one-bin total to a median of 1.5% once each is compared with its own A_C -smeared truth. Two pairs are usable, both in the combined configuration only: $\theta_p \times \delta p_T$ ($s_{\min}/s_1 = 0.611$) and $\cos\theta_\pi \times \cos\theta_\mu$ (0.564) (Table 16). Combining the beam modes roughly doubles their conditioning ($0.43 \rightarrow 0.61$ and $0.25 \rightarrow 0.56$), through the larger sample and the shared flux rather than through any separation of ν and $\bar{\nu}$. The pairs built on $W_{\pi p}$ are not usable, as the 39% leakage between its outer bins implies.

The two usable pairs are each a Secondary result in the combined configuration only: $d^2\sigma/d\theta_p d\delta p_T$ on the proton-tagged sample, presented in the proton-tagged note, and $d^2\sigma/d\cos\theta_\pi d\cos\theta_\mu$ on the inclusive sample, presented in Sec. 14.7 of the analysis note. Their FHC and RHC extractions and the other six pairs are Not reported (Tables 1 and 2). The single-mode extractions of the two pairs are not shape measurements, because their responses cannot be inverted. Every one-dimensional result is quoted per beam mode and for the combination; the two two-dimensional results are quoted in the combined configuration only.

The pair $\theta_{\pi p} \times W_{\pi p}$ conditions at 0.537, as do the two usable pairs, yet its extraction carries no usable information: a 78% data-statistical term, per-bin uncertainties up to 276% and closure ratios down to -0.80 . Conditioning bounds how much of the response can be inverted and says nothing about how much data survives in the inverted directions, so the two must be read together. Conversely, $\cos\theta_\mu \times p_\mu$ screened best of the eight on the migration diagonal (0.70/0.68,

analysis sample) and is rank-deficient in FHC (0.002), while $\theta_p \times \delta p_T$ screened worst of the proton-tagged pairs and conditions best of all. The migration diagonal is a necessary condition, not a sufficient one; the one-dimensional binning review measures what the regularised response retains by the Wiener filter factors of A_C (§8).

Table 16: The eight two-dimensional pairs (inner $\times$ outer), ordered by the conditioning of A_C in the combined configuration. The last three columns refer to the combined configuration: the bin-averaged data-statistical term, the range of the per-bin total uncertainty and the range of the closure ratio against the A_C -smeared truth. Pseudo-data.

Pair	$s_{\min}/s_1$			combined configuration		
	FHC	RHC	COMB	data stat.	unc. [%]	closure
$\theta_p \times \delta p_T$	0.433	0.332	0.611	13.3%	29–103	0.70–1.36
$\cos \theta_\pi \times \cos \theta_\mu$	0.245	0.315	0.564	14.9%	33–64	0.91–1.26
$\theta_{\pi p} \times W_{\pi p}$	0.379	0.214	0.537	78.3%	39–276	−0.80–1.22
$\cos \theta_\pi \times \theta_{\mu\pi}$	0.037	0.133	0.171	25.5%	29–93	0.96–1.29
$\cos \theta_\mu \times p_\mu$	0.002	0.261	0.106	9.0%	33–57	0.97–1.04
$\theta_p \times W_{\pi p}$	0.012	0.152	0.112	200.4%	36–568	0.23–1.03
$\cos \theta_\pi \times p_\pi$	0.024	0.158	0.052	13.8%	32–62	0.96–1.11
$\theta_{\pi p} \times \delta p_T$	0.023	0.035	0.027	10.4%	26–52	0.89–1.15

12 Bin widths in the regularisation operator

[SHARED]

Wiener-SVD regularises with a matrix C ; this analysis uses the second-derivative form, built on the density with the actual bin widths, because of what the plain tridiagonal stencil $(1, -2, 1)$ does on these binnings. That stencil is a second derivative only on a uniform grid; these binnings are strongly non-uniform, with width ratios up to 14:1 (Table 18), so on them it is not a second derivative at all. It is near-singular, its row sums are the 10^{-6} regularisation floor, and its near-null direction is the constant vector rather than anything physical. The Wiener filter is then free to retain that direction alone, which is a rank-1 A_C that passes normalisation and discards shape.

The FHC θ_μ extraction at its pre-review binning (§25.1) is the clearest case, because its bin widths span a factor of 10.4. Unfolded with the uniform stencil, the unfolded data, the A_C -smeared truth, the MicroBooNE tune and all four generators share the same normalised shape to four decimal places (Table 17): writing the four generator predictions as the columns of an input matrix and their A_C -smeared counterparts as the columns of the output matrix, the inputs are full rank ($s_2/s_1 = 5.0 \times 10^{-2}$) while the outputs give $s_2/s_1 = 2.8 \times 10^{-6}$. Every shape degree of freedom is annihilated, each model is mapped onto one common curve and only its normalisation survives; any agreement with the data is then true by construction, and a model comparison made on such an extraction is meaningless. $\cos \theta_\mu$, the next most non-uniform binning at 14.5, is the other observable strongly affected; the remaining observables sit at width ratios of 3–6 and are affected far less.

The operator used. Both derivative regularisation matrices difference the density on the actual grid: each column is divided by its bin width and the stencil uses the true spacings between bin centres, with a one-sided first-derivative stencil at the ends. (A Dirichlet closure was rejected: it asserts the distribution vanishes at the edges, which is false here, $\cos \theta_\mu$ peaks at its upper edge.) With no widths supplied the code reduces to the uniform stencil. The widths are passed in by the unfolding application, which holds the slice definitions that the cross-section extractor does not. The effect on the conditioning of A_C , measured as s_2/s_1 of the smearing matrix itself, is given in Table 18: every inclusive observable improves and none is degraded

Table 17: FHC θ_μ unfolded with a uniform-grid second-derivative stencil: normalised bin contents. Five independent inputs, the measurement and four generator predictions plus the tune, collapse onto one shape, the signature of a rank-1 A_C .

	bin 1	bin 2	bin 3	bin 4	bin 5
Unfolded data	0.1485	0.3361	0.2899	0.1934	0.0321
A_C -smeared truth	0.1484	0.3359	0.2901	0.1933	0.0323
MicroBooNE tune	0.1484	0.3359	0.2901	0.1934	0.0322
NEUT	0.1484	0.3359	0.2901	0.1934	0.0322

except combined $\cos \theta_\mu$, marginally; in the proton-tagged family combined $W_{\pi p}$ (0.022) and $\delta\phi_T$ (0.083) would be as compromised as FHC θ_μ under the uniform stencil.

Table 18: A_C conditioning s_2/s_1 with the uniform-grid stencil (“uniform”) and with the width-aware operator (“width-aware”), measured on one pseudo-data pass at the binnings under test (the $\cos \theta_\pi$ row is a five-bin scheme other than the reported one); the width-aware values of the reported binnings follow from the released matrices. With the bin-width ratio of each observable. A value approaching zero means the additional-smearing matrix has collapsed onto a single mode and the extraction carries no shape information.

observable	width ratio	FHC		RHC		COMB
		uniform	width-aware	uniform	width-aware	uniform $\rightarrow$ width-aware
p_μ	6.3	0.34	0.90	0.31	0.94	0.19 $\rightarrow$ 0.68
$\cos \theta_\mu$	14.5	0.46	0.57	0.45	0.55	0.42 $\rightarrow$ 0.39
$\cos \theta_\pi$	3.6	0.67	0.71	0.55	0.79	0.55 $\rightarrow$ 0.74
$\theta_{\mu\pi}$	3.0	0.42	0.85	0.47	0.65	0.42 $\rightarrow$ 0.46
θ_μ	10.4	0.015	0.51	0.22	0.62	0.045 $\rightarrow$ 0.38
p_π (2 bins)	–	0.46	0.53	0.41	0.44	0.29 $\rightarrow$ 0.40
<i>proton-tagged, combined configuration</i>						
$W_{\pi p}$	–					0.022 $\rightarrow$ 0.60
W_{had}	–					0.23 $\rightarrow$ 0.86
δp_T	–					0.25 $\rightarrow$ 0.93
$\delta\alpha_T$	–					0.24 $\rightarrow$ 0.48
$\delta\phi_T$	–					0.083 $\rightarrow$ 0.85
p_n	–					0.21 $\rightarrow$ 0.88

13 The D’Agostini cross-check

[INCLUSIVE]

Seven unfolding methods, listed below, are run in parallel with the Wiener-SVD extraction. The D’Agostini comparison (Table 20 and the numbers around it) is evaluated on the released pseudo-data. The closure figures of the other methods were evaluated on a different pseudo-data throw; the comparison they support is a property of the operators and does not depend on the throw.

Table 19: Unfolding methods.

Label	Description
<code>wiener_svd</code>	Wiener-SVD (every reported result)
<code>bin_by_bin</code>	Efficiency correction only; no migration unfolding
<code>ibu</code>	D’Agostini iterative Bayesian, 4 iterations
<code>tikhonov</code>	Tikhonov-regularised least-squares, $\tau = 1.0$
<code>roo_bayes</code>	RooUnfoldBayes, 4 iterations
<code>roo_svd</code>	RooUnfoldSvd
<code>roo_invert</code>	RooUnfoldInvert (direct matrix inversion)
<code>roo_binbybin</code>	RooUnfoldBinByBin

The D’Agostini iterative-Bayesian method provides the principal quantitative cross-check of the Wiener-SVD result. Because it does not apply the Wiener-SVD additional-smearing matrix A_C , its unfolded total is free of the (non-norm-preserving) A_C suppression and is stable against the iteration count. Table 20 compares the two integrated totals for all three configurations: the D’Agostini result sits above the Wiener-SVD total by 35% (FHC), 33% (RHC) and 30% (Combined), equivalently A_C suppresses the quoted Wiener-SVD integral by 26%, 25% and 23%, and it varies by $\leq 0.4\%$ across 1, 2 and 4 iterations. The D’Agostini integral is flat across the five observables to 1.0% (FHC, 1.026–1.036), 1.1% (RHC, 0.930–0.940) and 0.5% (Combined, 0.973–0.978), and equal to the one-bin totals (1.034, 0.932, 0.975), whereas the Wiener-SVD integrals spread by 21%, 31% and 28%. The observable-to-observable spread in the Wiener-SVD integrated cross sections (Table 24 of the analysis note) is therefore the A_C smearing rather than a physical or acceptance effect.

Table 20: Wiener-SVD vs. D’Agostini (iterative Bayesian, 4 iterations) integrated cross section, averaged over the five inclusive observables (p_μ , three-region p_π , $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$; 10^{-38} cm²/Ar), evaluated on the released pseudo-data (`xsec_analyzer/dagostini_050.sh`). D’Agostini carries no A_C smearing, so it recovers a 30–35% higher total, and it is flat across observables (spread 1.0% FHC, 1.1% RHC, 0.5% combined) where the Wiener-SVD integrals spread by 21–31%, so the observable-to-observable variation of σ_{int} in Table 24 of the analysis note is the A_C smearing rather than a physical or normalisation effect. The last column is the change of the D’Agostini integral between 1 and 4 iterations.

Config	$\langle\sigma\rangle_{\text{WSVD}}$	$\langle\sigma\rangle_{\text{DAg}}$	DAg/WSVD	DAg 1→4 iter
FHC	0.764	1.031	1.35	$\leq 0.4\%$
RHC	0.702	0.934	1.33	$\leq 0.2\%$
Combined	0.751	0.975	1.30	$\leq 0.1\%$

14 Identity closure: unfolding the prediction with itself [SHARED]

The pseudo-data closure tests reported in the analysis notes compare an unfolded Poisson throw against the truth it was thrown from. They therefore test the whole chain at once, and a modest χ^2 can hide a genuine inconsistency inside the unfolding itself, because the statistical fluctuation of the throw is the dominant term.

The identity test removes that fluctuation. Wiener-SVD returns $\hat{x} = Ur$ for a measured reco vector r , and defines the additional-smearing matrix as $A_C = UR$, where R is the response (smearacceptance) matrix. Feeding an exact prediction $r = Rt$ back through the chain must therefore return

$$\hat{x} = URt = A_C t \tag{1}$$

identically, for any true vector t . Rather than pick one t , we test the matrix identity directly:

$$\|UR - A_C\|_\infty \stackrel{!}{=} 0. \quad (2)$$

This is stronger than a single-vector test and independent of the data. It is sensitive to the failure mode a pseudo-data test cannot see: the unfolding matrix and the additional-smearing matrix being built from different responses, or from responses evaluated in different universes, which would bias every extracted cross section by a fixed factor while leaving the closure χ^2 untouched.

The check is implemented inside `UnfolderNuMI` and runs on every extraction, printing a `[IDENTITY]` line. All 59 released extractions pass it: five inclusive and twelve proton-tagged observables, each in three beam configurations, the two proton angles in three configurations and the two combined two-dimensional results. The worst deviation over the 59 extractions is 6.1×10^{-5} relative, on FHC $\cos \theta_\mu$; the $\cos \theta_\mu$ extractions of the two families are the only ones above 4×10^{-6} . These are the rounding errors of the singular-value decomposition and the matrix products that follow it, not physical effects: they remain more than two orders of magnitude below the smallest systematic uncertainty in the analysis.

The deviations are larger than the 10^{-10} level obtained with regularisation matrices that are not width-aware, and the reason is arithmetic rather than physical. The width-aware operator carries factors of $1/w_j$ that span decades across a binning with a width ratio of up to 56:1, so the matrix products accumulate more floating-point error. $\cos \theta_\mu$, whose eight bins have the largest width ratio in the analysis, is the worst case. The identity itself holds exactly; only the arithmetic that verifies it is noisier. The unfolding construction is therefore exactly self-consistent, and every deviation seen in the pseudo-data closure tests is attributable to the statistical throw and to the response model, not to the unfolding implementation.

15 Total cross section and systematic breakdown [INCLUSIVE]

The measurement is the one-bin extraction of Sec. 14.6 of the analysis note on the Poisson-thrown pseudo-data (one true bin over the full $\cos \theta_\mu$ range, no overflow, Wiener filter off, $A_C \equiv 1$; `data.release/total_xsec.tsv`), with its full universe-propagated uncertainty; at unblinding it becomes the real beam-on extraction. It agrees with all four generators within 0.8σ : standalone GENIE sits $0.5\text{--}0.8\sigma$ low, while GiBUU, NEUT and NuWro lie within 0.8σ .

Table 21: Total flux-averaged cross section from the one-bin extraction (Sec. 14.6 of the analysis note; $10^{-38} \text{ cm}^2/\text{Ar}$), compared with the flux-averaged generator predictions. The measurement uncertainty is the official one-bin covariance (flux, cross section, re-interaction, detector, POT, targets and statistics; the MCS momentum-scale term is zero for the total, which uses no reconstructed muon momentum; the beamline-geometry, beam-off gate-ratio and Run-2 stand-in terms lie outside it, Sec. 12.4 of the analysis note). `data.release/total_xsec.tsv`. Values in parentheses give each generator’s deviation from the measurement in units of the measurement uncertainty. The generator predictions are integrated over the full binning with open upper bins (a closed upper edge on p_μ or p_π would discard 4–8% of each prediction).

Config	Measurement	GENIE	GiBUU	NEUT	NuWro
FHC	1.03 ± 0.40	$0.82 (+0.5\sigma)$	$1.13 (-0.2\sigma)$	$1.32 (-0.7\sigma)$	$1.25 (-0.6\sigma)$
RHC	0.93 ± 0.35	$0.67 (+0.8\sigma)$	$0.94 (-0.0\sigma)$	$1.10 (-0.5\sigma)$	$1.04 (-0.3\sigma)$
Combined	0.98 ± 0.37	$0.73 (+0.7\sigma)$	$1.02 (-0.1\sigma)$	$1.20 (-0.6\sigma)$	$1.13 (-0.4\sigma)$

The systematic breakdown by source (bin-averaged fractional uncertainty in cross-section space) is given for all three configurations in Tables 22–24. The flux (PPFX) term dominates throughout, followed by the detector variations; the data-statistical term reflects the pseudo-data

POT of each configuration. The largest detector entry is that of FHC p_π , 34.2%; elsewhere the detector term is 16.6–26.9%. The tables use the official covariance: the MCS momentum-scale row is the data-side term, computed outside the framework and non-zero only for p_μ (Sec. 12.3 of the analysis note).

Table 22: FHC ({Run 1,2,4,5}) systematic breakdown by source, bin-averaged fractional uncertainty (%), from the released systematics dumps (`logs/systdump`); “Prediction total” excludes the data statistics. The MCS momentum-scale row is the data-side term that the official covariance adds to the framework covariance.

Source	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$
Prediction total	40.6	56.7	39.6	39.0	37.1
Cross section (GENIE)	11.9	16.6	10.9	11.4	10.6
Flux (PPFX)	30.0	38.5	26.6	27.1	28.8
Detector	21.2	34.2	25.4	23.2	19.9
Reinteraction	4.2	13.3	3.5	5.1	4.1
MCS momentum scale	8.5	0.0	0.0	0.0	0.0
MC stat	3.8	6.1	5.2	4.2	2.8
EXT stat	1.1	2.3	1.7	1.2	1.0
Data stat	12.1	19.3	16.6	13.6	8.9
POT + targets	3.8	5.0	3.5	3.4	3.6
Total (incl. data stat)	42.4	60.0	43.1	41.5	38.3

Table 23: RHC ({Run 1,2,3,4a,4b,4c}) systematic breakdown by source, bin-averaged fractional uncertainty (%); same provenance as Table 22.

Source	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$
Prediction total	42.0	41.1	36.9	38.5	32.6
Cross section (GENIE)	12.6	13.7	9.7	10.8	10.7
Flux (PPFX)	29.4	27.7	24.6	25.2	25.2
Detector	23.8	24.0	23.5	25.5	16.6
Reinteraction	4.4	10.6	4.3	5.1	3.7
MCS momentum scale	10.8	0.0	0.0	0.0	0.0
MC stat	3.7	3.7	5.3	3.6	2.4
EXT stat	1.6	1.1	1.2	1.0	0.6
Data stat	13.2	12.7	18.4	12.3	8.4
POT + targets	4.0	3.9	3.4	3.5	3.5
Total (incl. data stat)	44.2	43.1	41.4	40.5	33.7

Table 24: Combined-configuration (FHC+RHC) systematic breakdown by source, bin-averaged fractional uncertainty (%); same provenance as Table 22.

Source	p_μ	p_π	$\cos \theta_\mu$	$\cos \theta_\pi$	$\theta_{\mu\pi}$
Prediction total	40.1	45.9	33.1	37.5	33.6
Cross section (GENIE)	11.7	14.9	8.4	10.7	10.1
Flux (PPFX)	27.9	31.0	22.8	25.7	25.8
Detector	23.7	26.9	21.2	23.0	18.0
Reinteraction	4.0	12.6	3.6	5.1	3.5
MCS momentum scale	8.8	0.0	0.0	0.0	0.0
MC stat	2.8	3.3	3.4	2.9	1.9
EXT stat	1.1	1.1	0.9	0.8	0.5
Data stat	9.5	11.0	11.5	9.7	6.6
POT + targets	3.7	4.2	3.1	3.5	3.4
Total (incl. data stat)	41.4	47.3	35.2	38.8	34.2

16 Presentation of the results

[INCLUSIVE]

16.1 The three-region p_π result

Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/ c : a pion of true momentum 0.95 GeV/ c reconstructs at almost the same value as

one of 0.55 GeV/c. The three regions have migration diagonals of 91%, 51% and 52% (column-normalised M , Sec. 11.2 of the analysis note) and meet the 0.50 criterion, but the upper region integrates over a very wide physical interval and is not a differential point in the usual sense. The true upper region is open-ended, $p_\pi > 0.26$ GeV/c (the true-bin definition places no upper edge on it); the 1 GeV/c that appears in the data release and in Appendix A of the analysis note is only the finite boundary adopted to assign the region a width for plotting, not a kinematic limit. The released $d\sigma/dp_\pi$ values are therefore bin-averaged densities over very different intervals (0.030, 0.055 and 0.74 GeV/c), and the panel draws the three at equal visual width, flagged on the axis, so the plotted area does not represent cross section. The result is stated as the three bin-integrated partial cross sections (Table 25); the density representation derives from them and depends on the conventional plotting width of the open region. These are the partial cross sections, $\sigma_i = (d\sigma/dp_\pi)_i \Delta p_{\pi,i}$, which do not depend on the choice of plotting boundary for the two lower regions and depend on it for the upper one only through the physical content above 1 GeV/c, which the open true bin already includes. For the open upper region the released density and the conventional width of 0.74 GeV/c form a matched representation of the extracted bin integral: a user should compare bin-integrated predictions after applying A_C , not reinterpret the density over a different finite upper boundary, since changing the boundary while holding the released density fixed would change the product:

Table 25: Three-region p_π result as bin-integrated partial cross sections (10^{-38} cm²/Ar), computed from the released `curves_incl.*_ppi3bin.tsv` as density $\times$ physical width; the uncertainty is the total uncertainty from the released covariance (the same fractional uncertainties as Tables 21 of the analysis note–23). All three regions are A_C -smeared like every other unfolded quantity, so a prediction must be smeared by the released A_C before comparison.

Region	FHC	RHC	Combined
$\sigma(0.175 < p_\pi < 0.205 \text{ GeV/c})$	0.048 ± 0.036	0.064 ± 0.029	0.059 ± 0.032
$\sigma(0.205 < p_\pi < 0.26 \text{ GeV/c})$	0.086 ± 0.061	0.110 ± 0.049	0.102 ± 0.053
$\sigma(p_\pi > 0.26 \text{ GeV/c})$	0.695 ± 0.236	0.553 ± 0.222	0.601 ± 0.216

The saturation is a physical information limit rather than a shortcoming of the range estimator, as the survey of pion momentum estimators shows (§7.3).

16.2 Interpretation of the closure χ^2 values

The closure χ^2 quoted for each extraction compares the unfolded result with the pseudo-data truth using the complete configured systematic covariance, and the resulting p -values cluster near unity (median 0.98 across all fifty-one extractions; eleven lie below 0.9 and the lowest is 0.20). This is not a sign of excellent agreement with the model. For a self-closure with correctly-sized uncertainties the p -values would be roughly uniform on $[0, 1]$, and they are not.

The reason is that the measurement and the truth it is being compared against share their systematics: both come from the same generator, the same flux and the same exposure, so the flux, cross-section and exposure terms are common-mode and largely cancel in the difference while still inflating the covariance used to judge it. Recomputing the same comparison with the statistical covariance alone raises χ^2/ndf from 0.02–0.2 to 0.5–0.8, much nearer the expected value. The table below was evaluated on a different pseudo-data throw from the released one and illustrates the effect; the released closure values are those of Tables 21 of the analysis note–23.

Extraction	full	prediction-only	statistics-only
inclusive FHC p_μ	0.14/7	0.22/7	1.01/7
inclusive comb p_μ	1.43/7	2.87/7	4.71/7
proton-tagged FHC p_μ	1.01/7	2.20/7	3.02/7
proton-tagged comb $W_{\pi p}$	0.48/6	0.77/6	4.88/6

The closure χ^2 is therefore a check that the extraction does not disagree with its input beyond the quoted band, which it passes, and not a measure of how well the band is sized. A test of the latter needs either a covariance restricted to the terms that differ between measurement and reference, or an ensemble of throws. The restricted covariance is not used for the released closure tables, whose p -values should be read with that limitation in mind; the sizing is tested instead with the ensembles of Sec. 13.2 of the analysis note and §18.

17 A_C row sums of every extraction

[SHARED]

Table 26 gives the row sums of the released additional-smearing matrices of every inclusive and proton-tagged extraction. One matrix has a negative row.

Table 26: Range of the A_C row sums of every released extraction, from the result files.

Observable	FHC	RHC	combined
<i>Inclusive</i>			
p_μ	0.45–0.98	0.35–0.72	0.45–0.73
p_π (three regions)	0.52–0.91	0.45–0.89	0.45–0.89
$\cos \theta_\mu$	0.49–0.70	0.26–0.88	0.47–0.67
$\cos \theta_\pi$	0.68–0.78	0.70–0.84	0.70–0.83
$\theta_{\mu\pi}$	0.52–1.04	0.58–1.15	0.63–1.20
<i>Proton-tagged</i>			
p_μ	0.31–1.09	0.27–0.80	0.47–0.67
p_π (two regions)	0.34–0.84	0.41–0.45	0.47–0.48
$\cos \theta_\mu$	0.49–0.98	0.38–0.88	0.50–0.93
$\cos \theta_\pi$	0.66–0.77	0.62–0.77	0.65–0.76
$\theta_{\mu\pi}$	0.37–0.87	0.31–0.75	0.35–0.77
p_p	0.62–0.77	0.60–0.74	0.62–0.77
$W_{\pi p}$	0.45–1.21	−0.14–1.26	0.32–1.24
W_{had}	0.80–0.90	0.76–0.86	0.81–0.87
δp_T	0.54–0.86	0.45–0.82	0.46–0.87
$\delta \phi_T$	0.40–0.73	0.19–0.69	0.40–0.76
$\delta \alpha_T$	0.59–0.98	0.57–0.93	0.59–0.96
p_n	0.70–0.86	0.61–0.82	0.59–0.86

Three features bear on the interpretation of the results.

The suppression is large: inclusive $\cos \theta_\mu$ has a row sum as low as 0.26 in RHC, and the proton-tagged $\delta \phi_T$ one of 0.19, a strong suppression of that output mode for a flat input. It also differs strongly between observables, which is why the integrated cross section obtained by summing a differential result varies from observable to observable while the underlying physical total cannot. The D’Agostini cross-check, whose integral is flat across observables, establishes that this spread is the smearing rather than a physical or acceptance effect.

One extraction has a negative row sum, and the pattern across configurations identifies its cause. In $W_{\pi p}$ the top row carries negative off-diagonal terms in all three configurations (summing to −0.15 FHC, −0.30 combined, −0.61 RHC against diagonals of 0.73, 0.62 and 0.47); only RHC tips negative overall (row sum −0.14), because its diagonal element is the smallest of the three. The first $\cos \theta_\mu$ bin of the proton-tagged RHC sample has a positive row sum with the eight adopted bins.

This is therefore not an RHC-specific defect but the regularisation pushing back against the least-determined bin of the observable: for $W_{\pi p}$, the same top bin that is only 2% diagonal in the six-bin response and is the reason no differential $W_{\pi p}$ result is reported. A negative row is not forbidden, since A_C is a regularisation operator rather than a probability matrix, but

a prediction smeared through one is being told that raising the truth in that bin lowers the measured content, which is a statement about the regularisation and not about the detector. That row should be used with corresponding caution.

18 Ensembles: reference construction and full-covariance pulls

[SHARED]

Ensemble coverage and the size of the uncertainties. The closure test above uses a single throw. It can show that the extraction reproduces its input, but a single residual has no width, so it cannot say whether the quoted uncertainty is correctly sized. To test that, one hundred statistically independent Poisson realisations of the data of each configuration (six extractions: p_μ and $\cos\theta_\mu$ in FHC, $\cos\theta_\mu$ in RHC, the three-region p_π in the combined configuration, the FHC and RHC one-bin totals; 100 members each) were each put through the whole chain (thrown, universes rebuilt from that throw, unfolded), and the per-bin pull

$$p_i = \frac{\hat{x}_i - (A_C x_{CV})_i}{\sigma_i}$$

formed against the fixed central-value truth smeared by that member’s own A_C . (The per-throw pseudo-data truth is not a valid reference: it follows the Poisson draw, so numerator and reference fluctuate together and the width comes out too small.)

The throws follow the weight rule of Eq. 4 of the analysis note, and the extractions are chosen to bracket the conditioning and the configurations: the well-behaved p_μ ; $\cos\theta_\mu$, whose A_C row sums fall as low as 0.49 in FHC (0.26 in RHC) so that the regularisation removes up to 51% (74%) of a bin’s normalisation; the three-region p_π of the combined configuration; and the one-bin totals, which have no regularisation at all. The four differential ensembles use the released binnings (members built from staged configurations identical to the released ones). If the uncertainties were mis-sized anywhere, it would show in $\cos\theta_\mu$; if the extraction carried a bias independent of the regularisation, it would show in the one-bin totals.

Reference covariance	pull mean	pull width	68% / 95% coverage
<i>statistical only</i>			
$d\sigma/dp_\mu$, FHC	−0.11	1.07	64.2% / 93.5%
$d\sigma/d\cos\theta_\mu$, FHC	−0.06	0.97	71.8% / 94.5%
$d\sigma/d\cos\theta_\mu$, RHC	−0.05	1.05	63.2% / 94.0%
$d\sigma/dp_\pi$ (three regions), combined	−0.11	0.95	72.7% / 95.7%
one-bin total, FHC	−0.10	0.97	67.0% / 95.0%
one-bin total, RHC	−0.21	0.99	64.0% / 94.0%
<i>full</i>			
$d\sigma/dp_\mu$, FHC	−0.03	0.32	99.5% / 100.0%
$d\sigma/d\cos\theta_\mu$, FHC	−0.02	0.37	98.5% / 100.0%
$d\sigma/d\cos\theta_\mu$, RHC	−0.01	0.46	96.1% / 100.0%
$d\sigma/dp_\pi$ (three regions), combined	−0.02	0.22	100.0% / 100.0%
one-bin total, FHC	−0.01	0.11	100.0% / 100.0%
one-bin total, RHC	−0.02	0.11	100.0% / 100.0%
expected	0	1	68% / 95%

Against the statistical covariance all six widths are consistent with unity (0.95–1.07, each determined to about $\pm 7\%$, $1/\sqrt{2 \times 100}$, more for the multi-bin extractions, whose bins are correlated), and the interval coverage is correct within the same statistics. The statistical uncertainties are therefore correctly sized, and the strongly suppressed observable behaves no differently from the well-conditioned one. Of the two, the $\cos\theta_\mu$ result is the more informative:

p_μ was expected to pass, whereas it was not known in advance whether the severe A_C suppression in $\cos\theta_\mu$ would distort its uncertainties.

Against the full covariance the pull width is 0.22–0.46 for the differential results and 0.11 for the totals, and coverage is nearly 100%: the quoted uncertainty is two to five times the throw-to-throw spread for the differential results and about nine times for the totals, because the ensemble varies only the data statistics while the denominator carries flux, detector and cross-section terms that are identical in every member. This is the same effect as the closure p -values clustered near unity, measured here directly.

The ensemble means sit below the reference in all six cases, by 0.4–1.1% (0.6–2.0 σ on the mean; largest in significance for the RHC total, -0.81% at 2.0 σ , and in size for FHC $\cos\theta_\mu$, -1.14% at 1.3 σ), with pull means of -0.05 to -0.11 and -0.21 for the RHC total (Table above; `data.release/ensemble.2026-09-26.tsv`). The extractions of one configuration share the same throws and are correlated, so this is a few measurements of a small negative offset, each consistent with zero at the present ensemble size. The throw itself is unbiased: the mean selected count over the 32 members of an FHC ensemble built with the same throw rule is 1578.7 against the neutrino-MC expectation of 1580.6 (ratio 0.999), so whatever offset remains is in the extraction, not in the pseudo-data. The offset is at most two standard deviations of the ensemble mean and is largest where the regularisation is strongest, but the one-bin totals, which have no regularisation, also sit low, so the excess in $\cos\theta_\mu$ over the common $\approx 1\%$ is the only part that can be attributed to A_C . No correction is applied and no uncertainty is assigned; a larger ensemble would settle whether the common offset is real.

A truncated read of the pseudo-data occurs when two ensembles sharing a throw number write the same throw file at the same time and a member opens it mid-write, seeing part of the tree. Ten of the 600 members were first built from such a read. Four RHC-total members (Run 2 read at 47–72%) and one FHC $\cos\theta_\mu$ member (43%) were affected materially; they widened the RHC-total pull width to 1.28. Every member was checked for this (the number of events univmake read against the entries on disk; 2791 of 2800 inputs agree exactly, and the five remaining one-entry differences are one out-of-range truth event in the one-bin binning, which does not enter the reco-level data), and the affected members were rebuilt. Each throw is created once and never replaced (`macros/throw.guard.h`; change log). The four differential ensembles of the table were built this way, and the entry-count scan of their 2000 inputs and of the proton-tagged W_{had} ensemble finds no truncated read; the members short by one entry are the boundary case of an event with $\cos\theta_\mu = 1.0$ exactly (change log).

The study covers the inclusive sample in all three configurations. The two-dimensional results and the proton angles have their own 100-member ensembles: the inclusive $\cos\theta_\pi \times \cos\theta_\mu$ (Sec. 14.7 of the analysis note), and on the proton-tagged sample θ_p and $\theta_{\pi p}$ in each configuration and $\theta_p \times \delta p_T$, with statistical pull widths of 0.94–1.04 (`data.release/ensemble.v1.2026-09-28.tsv`). The pulls of a two-dimensional result are read from its all-slice closure file, which holds every analysis bin in the order of the covariance. The proton-tagged family, where the statistics are sparser, has in addition three ensembles of validation-only products (W_{had} in its two regions and the two-bin δp_T in FHC, the two-bin p_n combined): statistical pull widths 0.94–1.08, 68% coverage 62–72% and integral offsets within 0.7% (Sec. 5.2 of the proton-tagged note, `data.release/ensemble.1p.2026-09-26.tsv`).

19 Consistency between configurations and with the sibling ν_e measurement [INCLUSIVE]

19.1 Consistency with the sibling ν_e CC 1π measurement

This analysis is one of a pair of MicroBooNE NuMI CC single-charged-pion measurements; the companion ν_e CC 1π analysis shares the same cross-section-extraction framework, beam and

reconstruction. A cross-check against its internal note confirms consistency on the essential conventions and identifies the differences that a combined NuMI CC π programme should reconcile.

Consistent by construction. Both use the Wiener-SVD extraction framework; the updated NuMI flux with the fixed geometry bug and Geant v4.10.4; the same signal topology (exactly one $\pi^\pm$, zero π^0 or heavier mesons, any number of protons, charged current), and the same systematic suite, PPF π flux hadronic universes (the 12 beamline-geometry unisims the ν_e analysis carries are evaluated but not in this analysis’s official covariance, Sec. 12.4 of the analysis note), the 600-universe GENIE cross-section multisim with unisims, the Geant4 reinteraction reweighting, the standard detector-variation set, and 2% / 1% / 100% POT / target / dirt normalisation uncertainties. The FHC beam-on data POT per run agrees exactly (Run 2 1.268, Run 4 2.075×10^{20}).

Differences reconciled or flagged. (i) **POT accounting:** the ν_e note quotes Run 1 FHC = 3.283×10^{20} POT (9 846 635 triggers), the figure that includes the open-trigger stream. The beam-on sample processed here is the non-open-trigger dataset, 2.192×10^{20} POT with 5 748 692 triggers, confirmed against the beam bookkeeping; the total FHC exposure is 7.766×10^{20} over Runs 1, 2, 4 and 5. (ii) **Run 4:** the Run4 overlay is identical to Run4c+Run4d combined (same 859,368 events and 2.834×10^{21} POT); only one may be used to avoid double counting. (iii) **Pion threshold:** ν_e uses $KE_\pi > 40$ MeV ($p_\pi > 0.113$ GeV/ c) versus our $p_\pi > 0.175$ GeV/ c (Table 8 of the analysis note). (iv) **Opening-angle cut:** ν_e excludes $\theta_{e\pi} > 170^\circ$ (a shower-specific back-to-back misreconstruction), whereas our muon-track topology uses $\theta_{\mu\pi} < 2.6$ rad (149°). (v) **Regularisation:** the ν_e analysis smooths on the first derivative, this analysis on the second (§12). (vi) **Scope:** both analyses cover Runs 1–5 in FHC and RHC; the present measurement reports FHC, RHC and the combined configuration.

19.2 The combined configuration and its inputs

The integrated cross section extracted from a given configuration is not a configuration-independent number: it depends on the additional-smearing matrix A_C , which differs per configuration and per observable, so comparisons across configurations are made observable by observable and against the pseudo-data truth rather than between measurements. Two tests compare the combined configuration with its inputs.

The combined truth against the exposure-weighted average of its inputs. A configuration spanning both horn currents should predict a cross section between those of its two inputs, equal to their average weighted by the fluence of each mode (0.425 FHC, 0.575 RHC). The realised truths of the one-bin totals (Table 28 of the analysis note) give a ratio of the combined truth to that average of 1.000 for the inclusive family and 1.001 for the proton-tagged family. Observable by observable, the A_C -smeared truth integrals give 1.006 (0.981–1.030) and 1.046 (0.984–1.187), the spread being the configuration dependence of A_C rather than of the truth. The test is sensitive to the treatment of the detector variations: the two per-mode detector-variation sets of the combined configuration must be pooled as one sample scaled once to the data exposure, since scaling each set to the full exposure and summing would double the detector covariance, over-regularise the combined extraction and push its truth ratio to 0.91 with a worst observable at 0.77. With the pooled treatment the combined detector term lies between or close to those of its inputs (analysis note, Tables 21–23: 23.7% against 21.2% in FHC and 23.8% in RHC for p_μ).

The RHC deficit is in the input, not the extraction. The one-bin totals give an RHC/FHC ratio of the injected tune of 0.901 (inclusive) and 0.878 (proton-tagged), and the extracted one-bin ratios are 0.901 and 0.873 (Table 28 of the analysis note). Normalisation is not implicated: the exposures sum exactly ($7.766 \times 10^{20} + 1.1082 \times 10^{21} = 1.8848 \times 10^{21}$ POT) and the per-configuration flux values reproduce the POT-weighted combination to 0.01%. The

deficit is therefore a property of the $\bar{\nu}_\mu$ -dominant sample, and the number to be compared with data at unblinding is the tune ratio.

External reproduction of the released curves. The transformation $(d\sigma/dx)_i = w_i^{-1} \sum_j (A_C)_{ij} p_j$ applied to the released truth-level predictions reproduces every released generator curve to a worst relative per-bin deviation of 4×10^{-8} , the precision of the text files (96 curves). This check is part of the release procedure because an internal closure cannot make it: the unfolded data and the truth can both be right while the model comparison is wrong, and only an external user’s arithmetic exposes it.

20 Control regions: definitions and roles

[INCLUSIVE]

The four reconstruction-level control regions of the inclusive $\text{CC } 1\mu 1\pi Xp$ selection are $\text{CC}0\pi$, multi- π , π^0 and large muon-pion opening angle (called “cosmic” below for brevity: it is cosmic-enriched, 34–41% beam-off, not cosmic-pure); Sec. 10.1 of the analysis note gives their definitions. Throughout, “beam-off”, “EXT” and “cosmic data” name the same data-driven component, the beam-off trigger sample scaled by gate counts; “signal contamination” means the inclusive signal unless the proton-tagged selection is named. Each region is anchored by the inversion of one nominal signal requirement; requirements that become undefined once that inversion is made, or that would sculpt the targeted control sample, are dropped as listed in Table 27. Each region is disjoint from the signal region at reconstruction level (demonstrated event by event, §21.1); their expected signal contaminations are 8–24%. The selections, the run-dependent normalisation, the beam-off scaling, the displayed observables and binning, the covariance construction, the goodness-of-fit tests and the responses to a discrepancy are fixed in §21.5 before beam-on data in the regions are examined. The data are compared with the complete prediction (neutrino MC + beam-off + dirt) separately in FHC and RHC, with the combined configuration as a cross-check. The proton-tagged control regions are discussed in §21.6.

Role of each region. The transfer factors of §21.3 set the physics role of each region.

- **CC0 π** is the high-statistics validation of the dominant neutrino background class: $\text{CC}0\pi$ events with a proton or other track taken as the pion are 215 of the 560 FHC neutrino-background events (38%). The region has 72% target purity (65% of the full stack) and 0.6% raw statistical precision. It validates the normalisation and kinematics of the underlying $\text{CC}0\pi$ population, not the proton-to-pion misidentification probability that transfers that population into the signal region: the transfer carries a 31–34% uncertainty dominated by the detector variations. It is therefore a validation test rather than a per-cent-level constraint, and it is conditionally eligible for a fitted normalisation correction under §21.5. The direct test of the misidentification probability would be an inverted pion-PID region, which has not been defined.
- **π^0** tests the shower veto and the $\text{CC}+\text{NC } \pi^0$ background. Its target purity is 50.1% (FHC) and 51.0% (RHC) of the neutrino MC, its transfer factor 0.02 with a 17–19% uncertainty set by the Monte-Carlo-statistical floor of the detector-variation samples. The region is diagnostic only; a fit of the $\text{CC}+\text{NC } \pi^0$ normalisation is considered only after the pre-fit result is recorded, together with the fit model, the leverage of the fixed components, the transfer uncertainty and its floor, and the effect on the signal-region prediction.
- **Multi- π** is diagnostic only. Its transfer factor is $\simeq 2.5$ (the signal region holds more of the target class than the region does) and its target purity 47%; it is not used to fit or constrain the signal-region normalisation, but it exposes gross modelling failures and shape discrepancies.

- **Cosmic** validates the beam-off normalisation and shape to $\sim 50\%$ once the prediction uncertainties of its neutrino component are included (17% from data statistics alone). This precision is sufficient because beam-off is 1.9% of the selected signal-region prediction: a 50% uncertainty on it is an effect of order 1% on the selected sample. It is not eligible for a fitted correction.

About 30% of the FHC neutrino background has no region of its own: NC and ν_e CC without a π^0 , out-of-fiducial-volume events and CC1 $\pi^\pm$ outside the measured phase space (§21.3). These components keep their model uncertainties unchanged; the CC0 π and π^0 distributions carry indirect sensitivity to the first two through the shared reconstruction, and each is small enough individually that no dedicated region is justified at this exposure.

Sensitivity of the test and its blind direction. The pass/fail statistic is a global normalisation χ^2 over the eight (region, mode) totals with the full pre-fit covariance (§21.5). It detects a discrepancy confined to one region beyond 12% of the CC0 π or π^0 total and beyond 50–62% of the multi- π or cosmic total, an FHC/RHC inconsistency, and, through the normalisation-marginalised shape tests, a shape difference. A common normalisation shift of all regions fails it only beyond 58%, because the correlated flux, cross-section and detector terms absorb it. That blind direction is acceptable because it is the direction the signal-region measurement already carries as its flux and cross-section uncertainty: the control regions test the background model region by region, not the overall normalisation.

Scope of the pseudo-data studies. The control-region machinery has been exercised on Poisson-thrown central-value pseudo-data. Those studies validate the selection logic, the per-run exposure scaling, the central-value weighting and, in the full-stack variant of §21.2, the exact bookkeeping path that beam-on data will follow. They do not validate the background model: the pseudo-data are drawn from the same simulation that supplies the prediction, so agreement is guaranteed up to statistics. The background model is the largest untested assumption in the analysis, and the control regions exist to test it. The control-region comparison is made at reconstruction level, so it does not depend on the unfolding or on a closure test with an independent generator.

Table 27: Control-region definitions in terms of the selection stages. The common preselection *pre* is software trigger, reconstructed vertex in the fiducial volume, topological score > 0.67 and a qualifying muon candidate (Cuts 1–4 of Sec. 8 of the analysis note); *shower* is the shower veto (Cut 8), *final* the non-proton multiplicity cut (Cut 10), *contained/gap* the pion containment and three-plane requirements (Cuts 5–7), *opening* the $\theta_{\mu\pi} < 2.6$ rad cut (Cut 9), and n_π the number of qualifying contained pion candidates. The signal region is *pre*, *contained*, *gap*, *shower*, *opening* and *final* with $n_\pi = 1$. “Dropped” lists nominal requirements not applied in the region; “target” the signal-region background the region is meant to constrain. Each region is stored as a flag alongside the signal-region flag when the events are processed.

Region	Definition (branch level)	Dropped vs. signal	Target background
CC0 π	<i>pre</i> , <i>shower</i> , <i>final</i> ; $n_\pi = 0$	contained, gap, opening (no pion to define them)	CC0 π events with a proton or other track taken as the pion; pion absorption feed-in
multi- π	<i>pre</i> , <i>final</i> ; $n_\pi \geq 2$	shower, contained/-gap of the extra pion, opening	multi-pion events with one pion lost
π^0	<i>pre</i> , shower veto failed; $n_{\text{shower}} \geq 1$	contained, gap, opening, final	CC/NC π^0 with both photons missed; shower-veto efficiency
cosmic	<i>pre</i> , <i>contained</i> , <i>gap</i> , <i>shower</i> , <i>final</i> ; opening-angle cut inverted	opening only	residual beam-off (cosmic) in the signal region; its normalisation

The π^0 region is the furthest from a single inversion: besides the pion requirements, which have no meaning once no pion candidate is demanded, it also drops the final multiplicity cut. The cut is dropped to avoid a plausible topology-dependent sculpting (photon showers reconstructed as additional track-like primaries). The comparison of §21.4 finds no material change in target purity, transfer uncertainty or the frozen leading-shower-energy shape. The no-cut definition is retained because it provides 11–14% more events, not because the study demonstrated a material bias from the cut.

Expected yields and precision. Table 13 of the analysis note gives the expected composition per region and beam. The raw statistical precision of the data samples is 0.6% (CC0 π), 0.8% (π^0) and 6% (multi- π) for the combined configuration. The neutrino-MC prediction in each region carries the flux, cross-section and re-interaction uncertainties of Table 29, including the detector variations. The regions carry a few-per-cent sensitivity to the signal normalisation in principle (the CC0 π region holds 987 signal events, 9% of its neutrino MC, so a 30% signal excess would appear as a 2.8% excess), which is inside the 9–24% detector-variation uncertainty on the region prediction and so invisible to every pull of §21.5. This keeps the regions blind to the signal, and the response tree holds the signal at its central value in any case.

21 Control-region machinery

[INCLUSIVE]

21.1 Region overlaps

The four regions are disjoint from the signal region by construction: CC0 π and multi- π require $n_\pi = 0$ and $n_\pi \geq 2$ against the signal region’s $n_\pi = 1$; the π^0 region requires the shower veto to fail; the cosmic region requires the opening-angle cut to fail. An event-by-event check on the files instrumented with the control-region flags confirms this: the number of events carrying both the signal-region flag and any control-region flag is exactly zero in the Monte Carlo and in the pseudo-data of every configuration. The regions are not mutually exclusive: multi- π does not apply the shower veto, so events with two pion candidates and a failed shower veto belong to

both multi- π and π^0 (47 of the 124 FHC multi- π MC events, 68 of 159 RHC); all other pairs are disjoint. Any simultaneous fit of the regions must account for that overlap; the per-region comparisons of §21.5 do not need to.

21.2 Full-stack pseudo-data test

Table 14 of the analysis note compares the neutrino-MC throw with the neutrino MC alone. To exercise the exact path that beam-on data will take, the same comparison was run with pseudo-data consisting of the neutrino-MC throw plus an independent Poisson throw of the beam-off and dirt expectations, compared with the complete prediction. The data/prediction ratios are 0.996, 1.047, 0.992, 1.000 (FHC; CC0 π , multi- π , π^0 , cosmic), 0.992, 1.122, 1.012, 1.080 (RHC) and 0.995, 1.079, 1.006, 1.027 (combined), each within its statistical uncertainty (0.6–10%). This tests bookkeeping, not physics. Figures 7 and 8 show, for every region and both beam modes, the frozen distributions exactly as they will be drawn for the beam-on comparison, with full-stack pseudo-data in place of the beam-on points.

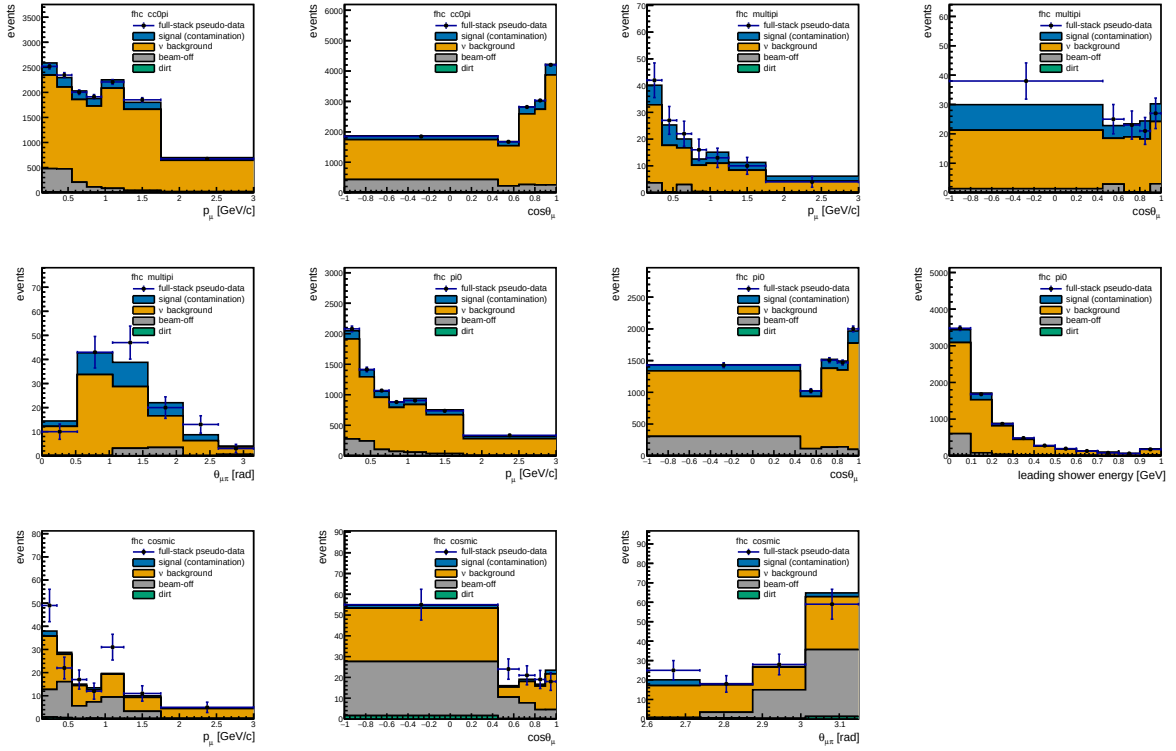


Figure 7: FHC control-region distributions at the frozen binning (the frozen plotting list): by row, CC0 π , multi- π , π^0 and cosmic; p_μ , $\cos\theta_\mu$ and, where defined, the muon-pion opening angle or the leading shower energy. Stack: signal (contamination), neutrino background, beam-off (per-run scaled) and dirt; points: full-stack pseudo-data (neutrino-MC throw plus a Poisson throw of the beam-off and dirt expectations). The beam-on comparison will use these plots with the beam-on points in place of the pseudo-data; overflow is in the last bin.

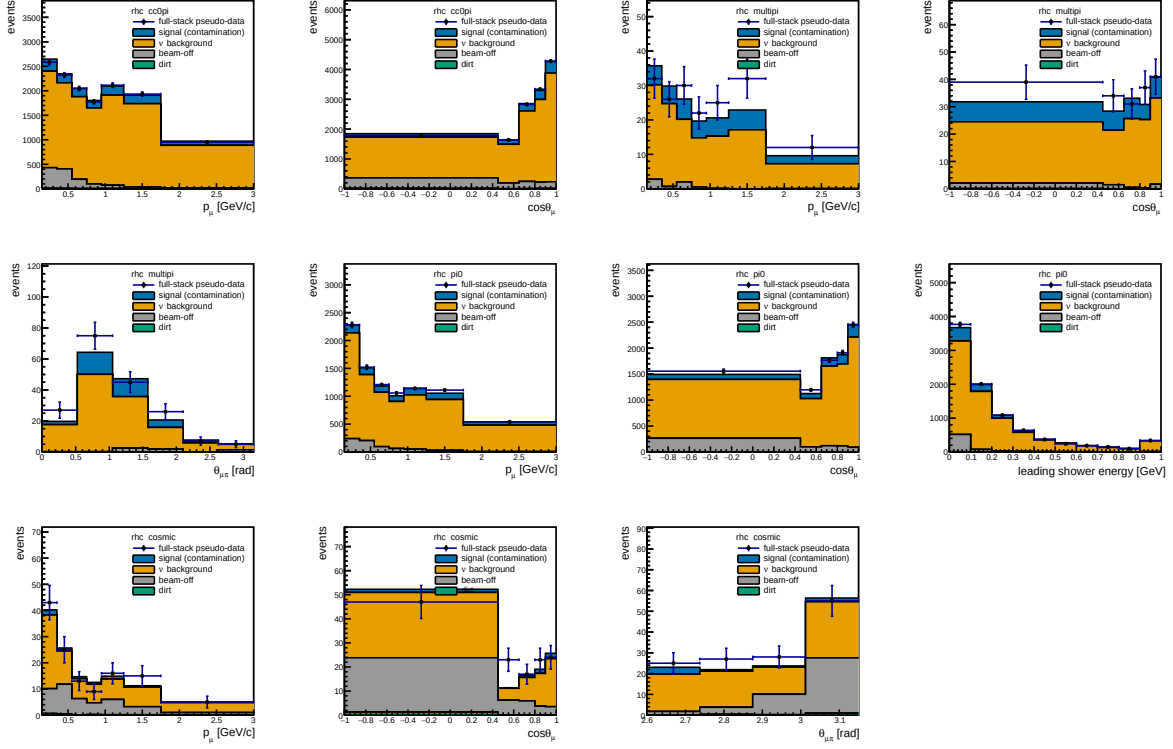


Figure 8: As Fig. 7, for RHC.

21.3 Transfer factors and control-region uncertainties

Whether a control-region discrepancy can be propagated into the signal region depends on the transfer factor $T_{b,c} = N_b^{\text{SR}}/N_b^{\text{CR}_c}$ for each background class b : a region constrains a class only where T is small and the ratio is stable against the model variations. Table 28 gives the composition of each region by background class and the transfer factors.

Table 28: Composition of the signal region (SR) and of each control region by background class (CV-weighted neutrino MC, per-run POT-scaled; beam-off and dirt excluded), and the transfer factors $T = N^{\text{SR}}/N^{\text{CR}}$ of each class. Classes are built from the selection’s truth counters: $\nu_\mu\text{CC } 0\pi$ (no charged or neutral pion above threshold), $1\pi^\pm$ outside the measured phase space, any π^0 , $\geq 2\pi^\pm$ without π^0 , out-of-fiducial-volume, and NC plus $\nu_e\text{CC}$. A region constrains a class only where T is small: CC0 π constrains the 0 π class ($T \simeq 0.03$) and π^0 the π^0 class ($T \simeq 0.02$; the adopted target class CC+NC π^0 , last row of each block, is 50–51% of the region’s neutrino MC); the multi- π region contains fewer multi-pion events than the signal region ($T \simeq 2.5$) and is a weak constraint; the cosmic region constrains only the beam-off component (not listed: 40 EXT events in the region against 23 in the signal region, $T = 0.58$, FHC).

Class	SR	CC0 π	multi- π	π^0	cosmic	$T_{\text{CC0}\pi}$	T_{multi}	T_{π^0}	T_{cosmic}
<i>FHC</i>									
signal (contamination)	833	987	26.9	582	5.1	0.84	31	1.43	164
$\nu_\mu\text{CC } 0\pi$ (proton or other track as pion)	215	7 694	2.2	887	20.0	0.03	96	0.24	11
$\nu_\mu\text{CC } 1\pi^\pm$ outside the measured phase space	46.5	554	2.4	318	13.0	0.08	19	0.15	3.58
$\nu_\mu\text{CC}$ with a π^0	51.4	359	17.9	2 702	2.1	0.14	2.88	0.02	24
$\nu_\mu\text{CC} \geq 2\pi^\pm$, no π^0	128	195	51.6	582	3.5	0.65	2.47	0.22	37
out of fiducial volume	41.3	419	2.9	222	16.0	0.10	14	0.19	2.58
NC and $\nu_e\text{CC}$	79.0	482	6.3	555	5.6	0.16	12	0.14	14
total ν -MC	1 393	10 690	110	5 847	65.2				
CC+NC with a π^0 (adopted π^0 target; overlaps two rows above)	57.2	391	19.3	2 931	2.7	0.15	2.96	0.020	21
<i>RHC</i>									
signal (contamination)	1 015	1 206	34.9	780	6.4	0.84	29	1.30	158
$\nu_\mu\text{CC } 0\pi$ (proton or other track as pion)	257	8 981	2.4	1 117	25.7	0.03	108	0.23	9.98
$\nu_\mu\text{CC } 1\pi^\pm$ outside the measured phase space	49.1	678	3.4	384	16.7	0.07	14	0.13	2.93
$\nu_\mu\text{CC}$ with a π^0	75.5	445	29.0	3 756	2.7	0.17	2.61	0.02	28
$\nu_\mu\text{CC} \geq 2\pi^\pm$, no π^0	176	238	71.8	885	4.1	0.74	2.45	0.20	43
out of fiducial volume	54.5	511	4.9	302	17.2	0.11	11	0.18	3.17
NC and $\nu_e\text{CC}$	108	599	12.5	857	8.9	0.18	8.59	0.13	12
total ν -MC	1 735	12 658	159	8 080	81.8				
CC+NC with a π^0 (adopted π^0 target; overlaps two rows above)	81.0	491	31.1	4 118	3.6	0.17	2.61	0.020	22

The neutrino-MC prediction in each region carries the flux, cross-section and hadron re-interaction uncertainties evaluated directly from the multisim weights on the files instrumented with the control-region flags, with the same weight conventions as the extraction and the detector-variation term from the Run-4 detector-variation samples, which carry the region flags. Table 30 gives the uncertainty on the transfer factor of each region’s target class, computed as a ratio in every universe and variation.

Table 29: Normalisation uncertainty (%) of the neutrino-MC prediction in the signal region and in each control region: flux (PPFX), cross-section (GENIE multisim) and hadron re-interaction multisims evaluated on the files instrumented with the control-region flags, with the extraction’s weight conventions, and the detector term from the Run-4 detector-variation samples, which carry the region flags (eight variations against the detector-variation central value, summed in quadrature as in the extraction). “Total” is the quadrature sum. The last column is the correlation, across the flux universes, between the region’s yield and the signal-region yield: the flux term is a common normalisation and cancels in a control-region-to-signal-region transfer (Table 30). The ν -MC yields apply the safe-weight rule of Eq. 4 of the analysis note, as do Tables 11 of the analysis note and 13 and the pseudo-data throws; the yields agree between the tables.

Beam	Region	ν -MC yield	flux	cross section	re-interaction	detector	total	corr. with SR (flux)
FHC	signal region	1 581	18.0	16.9	2.7	20.7	32.3	1.00
FHC	CC0 π	12 192	16.7	11.9	1.1	21.8	30.0	1.00
FHC	multi- π	124	20.0	20.5	5.5	18.5	34.6	0.99
FHC	π^0	6 646	18.8	15.8	2.4	21.8	32.9	1.00
FHC	cosmic	75	15.9	13.2	1.8	24.3	31.9	0.98
RHC	signal region	1 735	16.2	16.2	2.7	14.2	27.1	1.00
RHC	CC0 π	12 658	15.1	11.8	1.1	8.7	21.1	0.99
RHC	multi- π	159	17.0	19.3	6.1	19.4	32.8	0.99
RHC	π^0	8 080	16.3	15.4	2.8	7.3	23.7	1.00
RHC	cosmic	82	15.2	12.8	2.5	11.9	23.3	0.99

Table 30: Transfer factor $T = N^{\text{SR}}/N^{\text{CR}}$ of each control region’s target class and its fractional uncertainty (%), with T recomputed as a ratio in every universe so that the numerator–denominator covariance is included. The flux term, 15–20% on either yield alone (Table 29), cancels to 2–3% in the ratio, as the yield correlations anticipated; the cross-section term does not cancel, because it changes the composition of the two regions differently, and the detector term is the largest. The detector entries carry the finite statistics of the detector-variation samples (a few thousand events per region), so those for the smaller classes are upper limits rather than measurements; the multi- π transfer’s detector entry is below its own Monte-Carlo-statistical floor (§21.3). The “all neutrino MC” row uses the total neutrino MC of the CC0 π region as a reference. The Monte-Carlo-statistical floor of the detector entries, estimated from two disjoint halves of the variation samples, is given in the text; for the adopted CC+NC π^0 class it is quoted in parentheses after the detector entry and, where it exceeds the entry, it replaces it in the total.

Beam	Region	Target class	T	flux	cross section	re-int.	MC stat	detector	total
FHC	CC0 π	ν_μ CC 0 π	0.0278	3.3	11.7	6.7	2.5	27.1	30.5
FHC	multi- π	ν_μ CC $\geq 2\pi^\pm$	2.5094	3.0	4.7	2.6	5.1	18.5	20.2
FHC	π^0	ν_μ CC with π^0	0.0189	2.6	6.1	1.8	4.4	15.1	17.1
FHC	CC0 π	all neutrino MC	0.1296	1.7	13.2	2.6	1.0	9.7	16.7
FHC	π^0	CC+NC with π^0 (adopted)	0.0194	2.2	5.8	1.9	4.2	14.9 (12.1)	16.8
RHC	CC0 π	ν_μ CC 0 π	0.0286	3.0	11.5	6.6	2.0	30.6	33.6
RHC	multi- π	ν_μ CC $\geq 2\pi^\pm$	2.4527	1.9	4.0	2.8	3.9	21.3	22.3
RHC	π^0	ν_μ CC with π^0	0.0201	1.8	6.3	1.8	3.3	11.5	13.7
RHC	CC0 π	all neutrino MC	0.1371	2.0	12.5	2.6	0.8	9.8	16.3
RHC	π^0	CC+NC with π^0 (adopted)	0.0197	1.7	6.1	1.6	3.1	11.8 (18.0)	19.4

Monte-Carlo-statistical floor of the detector term. The detector entries of Tables 29 and 30 are quadrature sums of eight differences between finite samples, so part of each is the statistics of the variation samples rather than a detector effect. The floor is estimated by evaluating every shift on two disjoint halves of the generated events (split on the true neutrino

energy, which is identical across the central-value and variation samples): half the difference of the two half-sample shifts estimates the statistical noise of the full-sample shift, and its quadrature over the eight knobs is the floor. For the region yields the floors are small: 1.4% (FHC) and 1.7% (RHC) against signal-region detector terms of 20.7% and 14.2%, and 1–2% for the $\text{CC}0\pi$ and π^0 regions. For the transfer factors they are not: the $\text{CC}0\pi$ target-class term (27% FHC, 31% RHC) sits well above its floor (9%, 6%) and is a genuine detector sensitivity, but the π^0 target-class term (15% FHC, 11% RHC) is comparable to or smaller than its floor (13%, 17%), so it is an upper limit set by the variation-sample statistics rather than a measurement (the same holds for the adopted $\text{CC}+\text{NC}$ π^0 class: 15% against a 12% floor in FHC, 12% against 18% in RHC), and the multi- π target-class term (18–21%) sits below its floor (25% FHC, 23% RHC); the half-sample estimator itself becomes unstable for the smallest classes, where floors of several hundred per cent appear. Any fitted correction under item 4(ii) of the procedure must therefore carry the larger of the quoted transfer uncertainty and its floor; for the π^0 class that is the floor.

Two limits of the control-region set. First, about 30% of the FHC neutrino background has no region: NC and $\nu_e\text{CC}$ without a π^0 (79 events), out-of-fiducial-volume (41) and $\text{CC}1\pi^\pm$ outside the measured phase space (47), the last subtracted from the same generator with a reconstruction bias of +30–66% for below-threshold pions (Table 8 of the analysis note). Second, the $\text{CC}0\pi$ region constrains the $\text{CC}0\pi$ rate, not the proton-as-pion misidentification that carries it into the signal region ($T = 0.03$, detector-dominated); an inverted pion-PID, proton-enriched region would test the misidentification without touching the signal region and has not been defined.

21.4 The π^0 region with and without the final multiplicity cut

The frozen π^0 definition drops the final multiplicity cut (Sec. 20). The comparison with the definition that retains it uses a reprocess of the Monte Carlo, the dirt and the per-run beam-off samples into a separate set of processed files that carry the alternative region flag and the two multiplicities, the Run-4 detector-variation samples included (nineteen reprocessed, the eighteen the detector term uses); no beam-on file was touched, and the frozen flag re-evaluated on the reprocessed files reproduces the yields of Table 13 of the analysis note and the uncertainties of §21.3. Table 31 gives the two definitions side by side.

Table 31: The π^0 control region without (frozen) and with the final multiplicity cut, full exposure, CV-weighted and per-run scaled. Purity and contamination are shares of the neutrino-MC prediction; the target share is of the full stack. The transfer-factor uncertainty is the multisim (flux, cross section, reinteraction) and MC-statistical total of §21.3; the detector term is the full-sample quadrature over the eight knobs, as in Table 30, with its MC-statistical floor from the split-sample method, and the total takes the larger of the two as there. The non-target row is all neutrino background other than the target class, the leverage denominator of Sec. 20; the CC0 π line is part of it.

	FHC		RHC	
	frozen	with cut	frozen	with cut
neutrino MC	6646	5889	8080	6942
target CC+NC π^0	3331	2974	4118	3567
signal	662	598	780	698
non-target neutrino background	2654	2318	3183	2678
of which ν_μ CC 0 π	1011	953	1117	1041
beam-off	778	733	678	632
dirt	12	11	9	9
target purity (of ν -MC)	50.1%	50.5%	51.0%	51.4%
target share (of full stack)	44.8%	44.8%	47.0%	47.0%
signal contamination	10.0%	10.1%	9.6%	10.1%
beam-off share (of full stack)	10.5%	11.1%	7.7%	8.3%
overlap with multi- π (ν -MC events)	46.9	46.9	68.1	68.1
leverage: non-target / target	0.80	0.78	0.77	0.75
T (target)	0.0194	0.0217	0.0197	0.0227
T uncertainty, multisim+MC-stat	7.8%	7.6%	7.3%	7.1%
T detector term / MC-stat floor	14.9/12.1%	15.7/12.5%	11.8/18.0%	11.4/17.4%
T uncertainty, total (larger of term and floor added in quadrature)	16.8%	17.4%	19.4%	18.8%
E_{shr} shape, events kept vs removed by the cut, unit-area χ^2 /bins	10.4/10		121/10	
largest per-bin change of the E_{shr} shape, frozen vs with cut	0.2%		0.7%	

The cut removes 11% (FHC) and 14% (RHC) of the region, and it removes every component in nearly the same proportion: the target class keeps 89/87%, the signal 90/90%, the beam-off 94/93%. The composition is therefore unchanged to within half a percent in purity and identical in the target's share of the full stack, and the overlap with multi- π is identical because multi- π requires the cut anyway. The leading-shower-energy distribution the comparison will be made on is tested between the events the cut keeps and the events it removes, which are disjoint samples (the two definitions themselves are nested, so a test between them would understate the difference): in FHC the removed events have the same shape (χ^2 10.4/10); in RHC they do not (121/10; the removed 14% has a softer shower spectrum, 36% against 41% in the first bin). Because the removed fraction is small, the frozen distribution differs from the with-cut one by at most 0.7% of unit area in any bin (RHC; 0.2% FHC), well inside the shape test's resolution. The transfer factor rises by 12/15% with the smaller denominator, and its uncertainties are unchanged within their precision. What the cut changes is the multiplicity distributions themselves (Fig. 9): the frozen region extends to five and more primary tracks and four and more non-proton-like tracks, the topologies the multi-shower argument of Sec. 20 anticipated, and they are 11–14% of the region, not a dominant population. One limitation of the frozen definition follows from it: the 11–14% of the region that fails the signal-region multiplicity cut cannot populate the signal region, so a data/MC discrepancy confined to those events would appear in the region total and be attributed to the target class under item 4 of §21.5. That is accepted for the larger sample; the stored multiplicities allow the reconstruction-failure investigation of item 4(i) to locate such

a discrepancy if one appears.

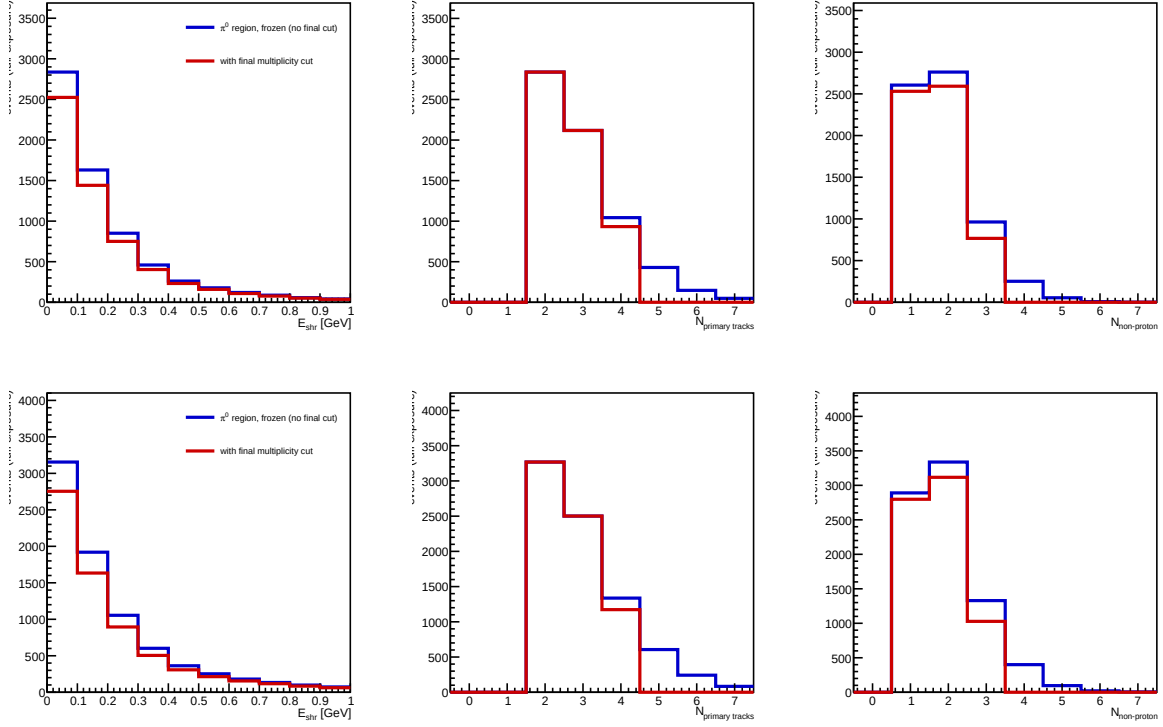


Figure 9: The π^0 region without (blue) and with (red) the final multiplicity cut, FHC (top) and RHC (bottom), neutrino MC at full exposure: the leading shower energy, the number of primary tracks and the number of non-proton-like tracks.

The frozen definition is kept: it is the larger and more inclusive sample, and the study shows that the choice is immaterial to the region's composition, purity and transfer uncertainty, and changes the frozen shape by less than a percent, so nothing about the eligibility question of §21.5 item 4(ii) turns on it. The premise that retaining the cut would sculpt the region towards single-shower events is not borne out at the level of the composition; it holds only for the multiplicity distributions, which are not compared.

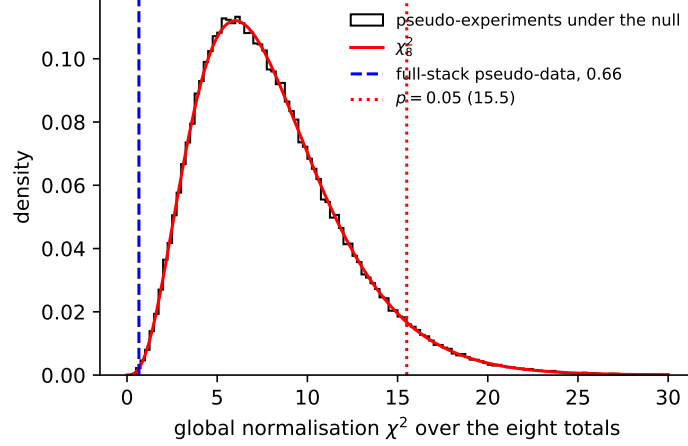


Figure 10: Null calibration of the global normalisation χ^2 over the eight (region, mode) totals: 2×10^5 pseudo-experiments with the prediction drawn from its pre-fit covariance and Poisson-fluctuated data, against the χ^2_8 density. The full-stack pseudo-data value 0.66 is marked; it is low because the pseudo-data is a Poisson throw of the same prediction and carries none of the prediction’s systematic variation.

21.5 Frozen procedure for the comparison with beam-on data

The procedure below is fixed before beam-on data in the control regions are examined.

1. **What is examined.** For each of the four regions: the event total, and the reconstructed distributions of p_μ , $\cos\theta_\mu$ and the muon–pion opening angle, and for the π^0 region the leading shower energy (the primary-track and non-proton multiplicities are stored by the frozen skim, §21.4, but are not in the pre-declared list and are not tested; they are available to the reconstruction-failure investigation of item 4(i) only), at the binning of the released observables where one exists and otherwise at the binning fixed in the frozen plotting list and shown, with full-stack pseudo-data, in Figs. 7 and 8. FHC and RHC are examined separately; the combined configuration is a cross-check only, and within each mode the event total is also compared per run period (FHC Runs 1, 2, 4, 5; RHC Runs 1, 2, 3, 4), so that a single-run excursion is detectable. Data are compared with the complete prediction, neutrino MC + beam-off + dirt, with the run-period beam-off scaling of Sec. 5 of the analysis note.
2. **Statistics.** Three quantities are computed for each region and beam mode, all against the pre-fit prediction. (a) The normalisation pull, $(N_{\text{data}} - N_{\text{pred}})/\sigma$, with $\sigma^2 = N_{\text{data}} + \mathbf{1}^\top V_{\text{pred}} \mathbf{1}$ and V_{pred} the summed covariance of the complete prediction (neutrino MC, beam-off and dirt: their finite statistics, flux, cross-section, re-interaction and detector terms, with all correlations among the prediction components). Because V_{pred} carries the 21–35% prediction systematics of Table 29, this pull cannot fail a discrepancy smaller than about 40–60%; it is therefore reported together with the pull against the statistical and detector terms alone, whose resolution (9–24%) is set by the detector term. (b) A shape χ^2 in the pre-declared distributions with the normalisation marginalised: the prediction is scaled to the data total and the covariance is projected onto shape (the normalisation mode removed), so that the common-mode flux and cross-section terms do not make the test powerless, as the closure χ^2 of Sec. 14.5 of the analysis note shows they otherwise would. (c) A global normalisation χ^2 over the eight (region, mode) totals with the full 8×8 pre-fit covariance, which is the single pass/fail statistic. All three are implemented in the protocol code (flux, cross-section and re-interaction multisims index-matched across the two modes,

the eight detector knobs as correlated unisims, MC, beam-off, dirt and data statistics) and were exercised on the full-stack pseudo-data of §21.2: the global χ^2 is 0.66/8 ($p = 1.00$), the sixteen pulls lie within ± 0.5 and all twenty-two shape tests have $p > 0.47$. The same run gives the power of the test: a discrepancy confined to one region fails the set at 95% confidence beyond 12% of the CC0 π or π^0 total and beyond 50–62% of the multi- π or cosmic total, while a common normalisation shift of all eight totals, which the 40–91% correlated flux, cross-section and detector terms absorb, fails it only beyond 58%. This is the blind spot of the test: it tests the background model region by region, not the overall normalisation, which the signal-region measurement carries as its flux and cross-section uncertainty.

3. **Acceptable agreement.** The control regions pass as a set if the global χ^2 of item 2(c) has $p > 0.05$; the expected false-failure rate is then 5% by construction. In the calibration, 2×10^5 pseudo-experiments under the null (prediction drawn from its pre-fit covariance, data Poisson-fluctuated, χ^2 with the nominal covariance) fail the nominal $p < 0.05$ threshold in 5.1% of cases and $p < 0.01$ in 1.1% (Fig. 10). The calibration treats the covariance as known; the finite-universe and detector-unisim sampling noise is not re-thrown, which is the residual approximation in the declared threshold. The per-region pulls and shape tests of items 2(a) and 2(b) are diagnostic: a region with $|\text{pull}| > 2$ (complete prediction covariance) or shape $p < 0.05$ is flagged for the response tree of item 4 even when the set passes, but the eight independent flags are not themselves pass/fail criteria (with sixteen pulls and twenty-two shape tests at these thresholds about two false flags are expected, and at least one with 85% probability). A single-bin excursion beyond 3σ with a passing total is recorded and examined for a reconstruction pathology but does not by itself flag the region.
4. **Permitted responses and the rule for choosing between them.** The response tree is entered when the global test of item 3 fails or a region is flagged; a flagged region within a passing set may receive at most response (b) below, never a fitted correction. Regions are treated in the order CC0 π , π^0 , then the two diagnostic regions, because the CC0 π class is 15% of the π^0 region and a CC0 π treatment changes the π^0 prediction before it is judged. (i) *Localised reconstruction failure.* If one run period's total in the region pulls beyond 3σ (statistical+detector) while the other periods pass, or one bin of a pre-declared distribution exceeds 3σ while the rest of the distribution passes, the excursion triggers a reconstruction-quality investigation, not a model correction and not an automatic exclusion: a 3σ excursion is expected to occur by chance somewhere among the periods, regions and bins examined, and a control-region discrepancy alone does not show that the signal region of that period is unusable. The period is excluded only if an independent detector or processing defect is established (detector-quality monitors, an independent inclusive sample, reproduction in more than one pre-declared variable, or a known production or calibration boundary); the exclusion then removes the period from both the control-region comparison and the measurement with its exposure removed from the normalisation, changes the flux weighting, the detector covariance and the response, and therefore makes a new frozen release. If no defect is established the period stays in the analysis and the discrepancy is treated through the model responses below. No cut or binning may be changed and a single bin is never removed. (ii) *Fitted normalisation correction (c).* Permitted only for CC0 π and π^0 (target-class purity above 50% of the neutrino-MC prediction; target classes ν_μ CC 0 π and CC+NC π^0) and only if the region's shape test 2(b) passes, so that the discrepancy is consistent with a pure normalisation shift: a one-parameter fit of the target class, to FHC and RHC simultaneously, applied if it improves the fit by a likelihood-ratio test at $p < 0.01$. The π^0 region meets the purity rule narrowly (50.1/51.0%) and with a class and denominator settled after the purities were known (Sec. 20), so a fitted π^0 correction is considered only after the pre-fit control-region

result is documented, together with the fit model, the leverage of the fixed components, the transfer-factor uncertainty and its Monte-Carlo-statistical floor, before any correction is propagated to the signal region. $\text{CC0}\pi$ (72%) is eligible under the same rule. The two fitted regions are disjoint by construction ($\text{CC0}\pi$ requires the shower veto to pass, π^0 requires it to fail), so a simultaneous fit counts no event twice; the multi- π/π^0 overlap (47/124 FHC, 68/159 RHC events) concerns a diagnostic region that is never fitted, and the global test of item 2(c) carries it as a covariance between totals. The fit updates the normalisation of the target class only, re-evaluated in every systematic universe so that its correlation with the flux and cross-section terms of the signal-region prediction is unchanged (the $1 + B/S$ coherence of Sec. 12.2 of the analysis note is retained, with B rescaled); the fitted uncertainty below replaces the class's prior normalisation uncertainty and enters as an independent term; the signal/background joint covariance is not otherwise modified. Its uncertainty is the transfer-factor uncertainty of Table 30 or its Monte-Carlo-statistical floor, whichever is larger (§21.3), added in quadrature to the leverage of the components held fixed in the fit: the prediction uncertainties of the non-target neutrino background, the beam-off and the signal contamination, each multiplied by its share of the target class (0.80, 0.23 and 0.20 in the π^0 region; 0.26, 0.15 and 0.13 in $\text{CC0}\pi$; Sec. 20). (iii) *Added uncertainty (b)*. Otherwise the central prediction is left unchanged and the full observed discrepancy, as a fraction of the target class in the region, is assigned as a one-standard-deviation normalisation uncertainty on that class in the signal region through its transfer factor. (iv) *Model revision (d)*. If the post-treatment global χ^2 , computed with the fitted normalisation or the added uncertainty in place and reported separately from the pre-fit value of item 3, still has $p < 0.05$, or a region remains flagged, the background model is revised before the signal region is opened. The pass/fail test of item 3 is always the pre-fit one; the post-treatment value is labelled as such wherever it is quoted. The response letters used here and in Sec. 20 are (a) no action, (b) added uncertainty, (c) fitted normalisation correction, (d) model revision. FHC and RHC share one background model: a class that passes in one mode and fails in the other is treated under (b), never corrected in one mode alone. The signal contamination of each region (987 events in $\text{CC0}\pi$ and 582 in π^0 for FHC, Table 28, more than the 833 signal events of the signal region) is held at the central-value tune in any fit, which keeps the regions blind to the signal: a 30% signal excess would move the $\text{CC0}\pi$ total by 2.8%, well inside the 9–24% detector-limited resolution of every pull. If more than one region is flagged, no independent corrections are combined without accounting for shared events and correlated systematic variations: in particular, multi- π and π^0 share events (§21.1) and are treated jointly, or multi- π is designated diagnostic only, as its transfer factor indicates.

5. **Prohibited responses.** Changing any signal-selection cut; changing the binning of any released observable; choosing a generator or tune after inspecting the regions; bin-by-bin reweighting; and any use of the beam-on signal-region sample.

21.6 Scope: inclusive regions only

The procedure of Sec. 21.5 covers the four inclusive regions. The proton-tagged regions of §21.7 apply the proton tag rather than inverting it and therefore do not constrain the $\approx 13\%$ proton mis-tag background that most affects the hadronic observables; the inverted-proton-PID region that would do so has not been implemented. The inherited proton-tagged regions are useful diagnostically but do not validate the proton-tagged background model.

21.7 Control regions for the proton-tagged sample

[PROTON-TAGGED]

The proton-tagged selection inherits its cut logic from the inclusive selection, and with it the four control-region definitions of Sec. 10.1 of the analysis note: the reprocessed proton-tagged files

already carry the four region flags. No new definitions or reprocessing are required to exercise the background model of the proton-tagged extractions; the same four regions can be evaluated either as inherited (the inclusive phase space) or with a reconstructed tagged proton also required, which places them in the same phase space as the measurement itself. Table 32 gives both.

All four regions are more signal-depleted for the proton-tagged signal than they were for the inclusive one: 5.8–7.2% signal contamination in $\text{CC}0\pi$ and π^0 , against 9.2–10.0% for the same regions in the inclusive phase space and a 52% purity in the signal region, because requiring a proton does not make a pion-less or π^0 event look like signal. The proton requirement costs about a third of the statistics ($\text{CC}0\pi$: 12 181 $\rightarrow$ 8 386 pseudo-data events for FHC) but suppresses the beam-off contribution by roughly a factor six (1 358 $\rightarrow$ 225, FHC $\text{CC}0\pi$, Table 32): cosmic-induced events rarely contain a reconstructed proton above threshold, so the proton tag is itself a strong cosmic rejector, and the EXT normalisation matters far less in the proton-tagged extractions than in the inclusive ones. The pseudo-data to neutrino-MC ratios of the two large regions stay at 0.987–1.007 with the tag, so the per-run bookkeeping holds in the proton-tagged phase space as well; the small regions scatter more widely (0.85–1.52), which on 37–62 pseudo-data events is Poisson scatter of one to two standard deviations, not a normalisation effect.

Limits of the proton-tagged regions. All four apply the proton tag rather than invert it, so none of them constrains the proton mis-tag background: the $\sim 13\%$ of tagged protons that are really a pion or another track, which smears $W_{\pi p}$ and the TKI variables directly. Testing that requires a region defined by inverting the proton PID (selecting candidates that pass the momentum and containment requirements but fail the LLR cut), which does need a new flag in the proton-tagged selection and a reprocess. Slicing a measured W/TKI observable is not an alternative: because the signal definition is inclusive in those variables, cutting on them splits the signal region rather than depleting it. On the Run-1 FHC overlay the signal region has 53.7% purity and the slices $W_{\pi p} > 1.5$ GeV, $W_{\pi p} < 1.15$ GeV, $p_n > 0.8$ GeV/ c and $\delta p_T > 0.7$ GeV/ c give 50.0%, 48.9%, 47.3% and 52.0% respectively, so none of the slices is depleted. Even the kinematically forbidden region $W_{\text{had}} < m_\pi + m_p$ retains 53.4% signal, because the calorimetric mass is systematically under-reconstructed and true signal populates it through resolution.

Table 32: Control-region yields for the proton-tagged sample at full exposure, evaluated with the inherited definitions both as-is and with a tagged proton also required. “Sig.” is the proton-tagged signal contamination of the neutrino MC. $N_{\text{data}}/N_{\nu\text{MC}}$ compares the Poisson-thrown pseudo-data with the CV-weighted neutrino MC; EXT and dirt are listed for scale.

Beam	Region	N_{data}	Signal	ν bkg	EXT	Sig. frac.	$N_{\text{data}}/N_{\nu\text{MC}}$
<i>inherited definitions (inclusive phase space)</i>							
FHC	CC0 π	12 181	705	11 487	1 358	5.8%	0.999
FHC	multi- π	125	30	94	7.2	23.9%	1.012
FHC	π^0	6 628	466	6 181	778	7.0%	0.997
FHC	cosmic	82	3.5	72	53	4.6%	1.089
RHC	CC0 π	12 590	731	11 927	1 214	5.8%	0.995
RHC	multi- π	175	34	125	5.9	21.5%	1.101
RHC	π^0	8 176	524	7 556	678	6.5%	1.012
RHC	cosmic	81	3.5	78	42	4.3%	0.990
Comb	CC0 π	24 771	1 436	23 414	2 572	5.8%	0.997
Comb	multi- π	300	64	219	13	22.6%	1.062
Comb	π^0	14 804	990	13 736	1 455	6.7%	1.005
Comb	cosmic	163	6.9	150	94	4.4%	1.037
<i>with a tagged proton required (measurement phase space)</i>							
FHC	CC0 π	8 386	486	7 886	225	5.8%	1.002
FHC	multi- π	59	9.8	46	0.2	17.7%	1.064
FHC	π^0	4 011	293	3 769	189	7.2%	0.987
FHC	cosmic	39	1.7	24	0.1	6.5%	1.517
RHC	CC0 π	8 621	500	8 127	197	5.8%	0.999
RHC	multi- π	62	14	58	1.3	19.8%	0.852
RHC	π^0	4 902	333	4 534	162	6.8%	1.007
RHC	cosmic	37	2.3	29	0.5	7.5%	1.187
Comb	CC0 π	17 007	985	16 013	422	5.8%	1.000
Comb	multi- π	121	24	104	1.5	18.9%	0.944
Comb	π^0	8 913	626	8 303	351	7.0%	0.998
Comb	cosmic	76	4.0	53	0.6	7.1%	1.336

21.8 Validation of the background fit

The background fit, the normalisation fit of the control-region target classes to the control-region totals and distributions, is validated on simulation (Fig. 11).

- **Toys.** Sets of 300 toys give pull means of 0.02–0.07, widths of 0.89–1.03 and coverage of 67–75%; ensembles of up to 10^5 toys per model are described in §21.8.
- **Injections** (Asimov data, two-parameter fit of the totals). Shifts of a class are recovered exactly; a 30% signal excess moves the normalisations by 0.13σ , an unfitted class at +30% by -0.07σ , a detector variation by at most 0.14σ , and flux and GENIE universes by -0.97 to $+0.84\sigma$, within the uncertainty they are part of.
- **FHC against RHC.** A likelihood-ratio test of common against beam-specific normalisations inside the joint covariance (a parameter goodness-of-fit test is not valid, since the modes share universes): a +15% shift of the RHC neutrino simulation alone gives $\Delta\chi^2 = 12.9/2$ ($p = 0.002$, totals) and $20.3/2$ ($p = 4 \times 10^{-5}$, two-dimensional).
- **Markov chain.** An adaptive Metropolis sampler with four chains and flat priors gives posterior means within 0.02 and widths within 2% of MIGRAD/MINOS on the Asimov dataset and on a CC0 $\pi \times 1.25$ injection, with $\hat{R} \leq 1.004$. With a few parameters and a nearly quadratic likelihood the profile likelihood suffices, and the sampler is kept as a cross-check.

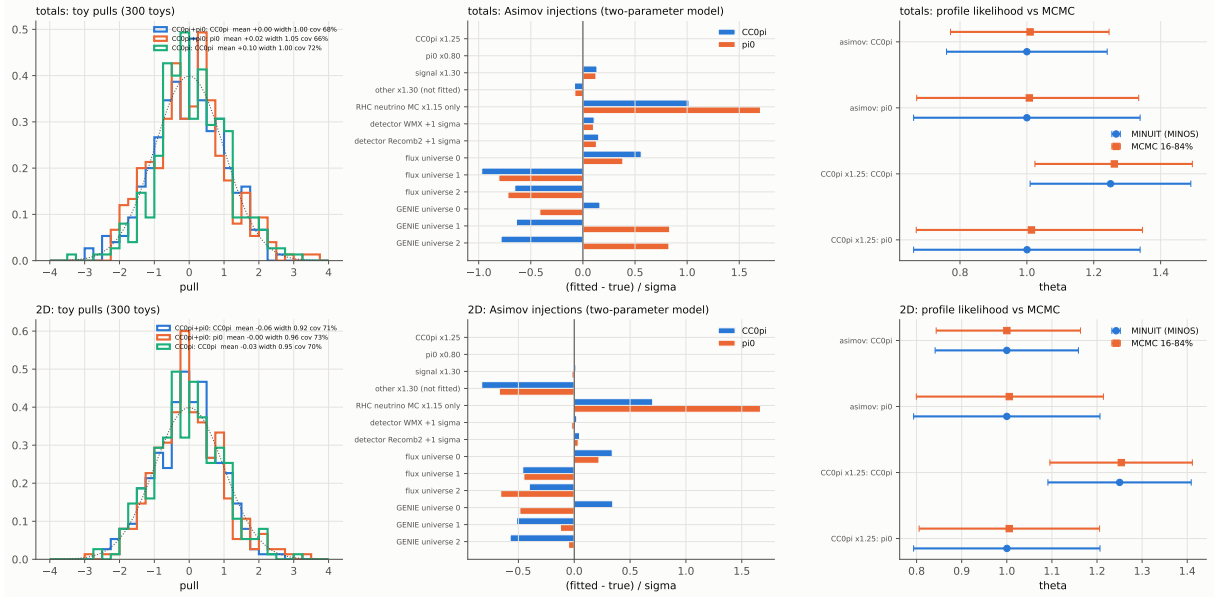


Figure 11: Validation of the background fit on simulation, for the totals-only fit (top) and the two-dimensional fit (bottom): pull distributions from toy ensembles, recovery of injected normalisation shifts, and the profile-likelihood interval against the Markov-chain posterior.

Convergence of the pull means. With 300 toys the uncertainty of a pull mean is 0.058, so a mean of +0.07 cannot be distinguished from zero. Ensembles of 10^5 toys per model (uncertainty 0.003) and three diagnostic variants of 5×10^4 toys each resolve this (Table 33, Fig. 12). With the combined Neyman–Pearson data variance, the pull mean of the two-dimensional fits does not converge to zero: +0.019 remains for $CC0\pi$ and +0.034 and +0.031 for $CC0\pi + \pi^0$, six to ten standard deviations of the ensemble. The offset persists with Gaussian instead of Poisson fluctuations (+0.031), so it does not come from Poisson counting in sparse bins. It disappears when the data variance is the prediction at the nominal parameters (+0.000, −0.004, −0.001): the combined Neyman–Pearson variance depends on the fluctuating data, which correlates the weight of a bin with its own fluctuation. The bias is small, 0.3–0.7% on the normalisation against a 16–21% uncertainty, but avoidable, so the fit uses the Pearson form, which every data result uses. The pull widths are 1.00 and the coverage 68% in every variant.

Table 33: Pull means $\pm$ their ensemble error. “standard”: systematic throw from the covariance, Poisson, CNP data variance (10^5 toys); “Gaussian”: Gaussian instead of Poisson fluctuation; “Pearson”: standard toys fitted with the prediction as data variance (5×10^4 each).

Bins	Model (parameter)	standard, CNP	Gaussian, CNP	standard, Pearson
totals	$CC0\pi$	$+0.003 \pm 0.003$	$+0.005 \pm 0.005$	-0.011 ± 0.005
totals	$CC0\pi + \pi^0$ ($CC0\pi$)	$+0.012 \pm 0.003$	$+0.014 \pm 0.005$	$+0.003 \pm 0.005$
totals	$CC0\pi + \pi^0$ (π^0)	$+0.016 \pm 0.003$	$+0.014 \pm 0.005$	$+0.008 \pm 0.005$
2D	$CC0\pi$	$+0.019 \pm 0.003$	$+0.031 \pm 0.005$	$+0.000 \pm 0.005$
2D	$CC0\pi + \pi^0$ ($CC0\pi$)	$+0.034 \pm 0.003$	$+0.031 \pm 0.005$	-0.004 ± 0.005
2D	$CC0\pi + \pi^0$ (π^0)	$+0.031 \pm 0.003$	$+0.029 \pm 0.005$	-0.001 ± 0.005

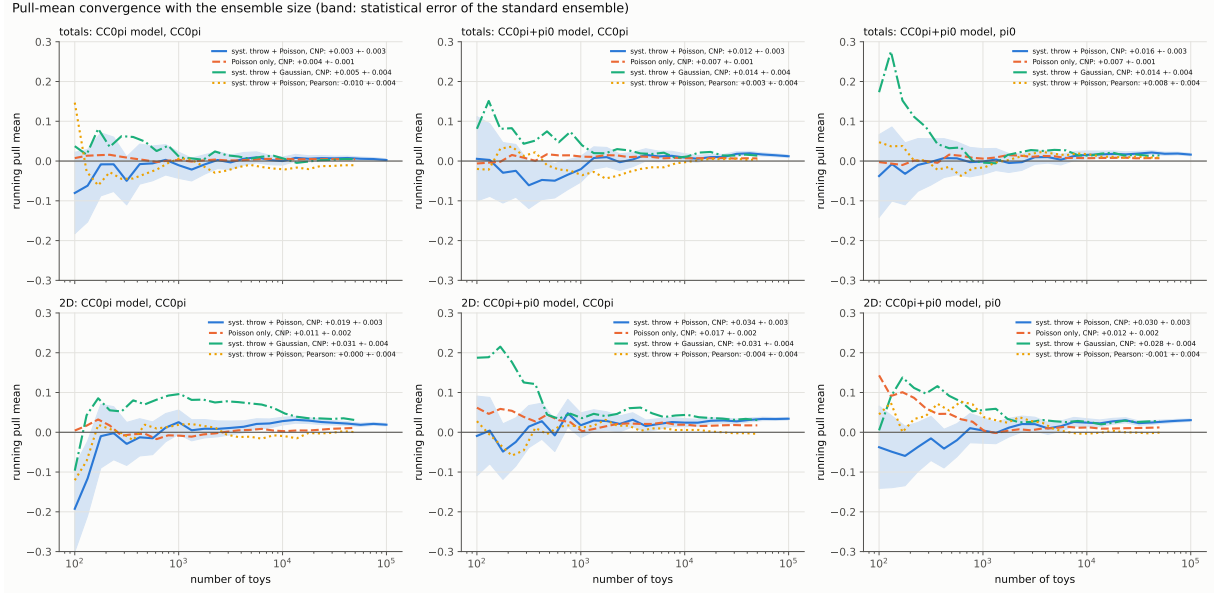


Figure 12: Running pull mean against the ensemble size for each model and parameter; band: statistical error of the standard ensemble. The ensemble thrown with statistical fluctuations only, and no systematic variation, has narrower pulls as expected.

22 Selection studies

[INCLUSIVE]

22.1 Selection-optimisation study

Choice of phase space and opening-angle cut. The measured phase space and the $\theta_{\mu\pi} < 2.6$ rad cut were chosen on the Run-1 FHC sample, starting from a reference phase space of $p_\mu, p_\pi > 0.1$ GeV/c. The numbers below and in Table 34 come from that study. The full-exposure yields, efficiency and purity of the released measurement are those of Table 11 of the analysis note.

- Signal MC (POT-weighted): 324.7 events
- Total background: 243.6 (≈ 199.7 ν -bkg + 40.1 EXT + 3.8 dirt)
- True signal generated (efficiency denominator): 2577–2684 events (observable-dependent)
- Signal purity: 57.1%
- Signal efficiency: 12.1–12.6%
- Measured phase space (reported result): efficiency 15.6%, purity 54.9% (reference phase-space study: selected signal 311.9, generated 1999, background 256.4; Table 34)

Per-run scaling. The yields above and in Table 11 of the analysis note use the per-run POT factors of Table 5 of the analysis note (each run scaled by its own $D_{\text{run}}/\text{MC}_{\text{run}}$), and the EXT by the per-run trigger ratios (per-run gate ratios, Sec. 5 of the analysis note). A single per-mode factor would differ from the per-run scheme by about 2% in the FHC final-cut signal, a small and nearly uniform re-weighting of the run mixture that moves efficiency and purity by $< 0.5\%$. The per-run scheme is used because real beam-on data require it. The pseudo-data are thrown from the same per-run-scaled prediction, so the closure tests the bookkeeping of the scheme, not its correctness against data.

Measured phase space and efficiency turn-on. The reference selection has almost no efficiency at low momentum and at large opening angle. In the fine truth distributions the per-bin efficiency is only $\sim 2\text{--}5\%$ for $p_\mu, p_\pi < 0.15$ GeV/c (the 10/20 cm track-length cuts), and the region above the reco $\theta_{\mu\pi} = 2.6$ rad cut is dominated by cosmics ($\sim 7\%$ purity). The measured phase

space $p_\mu > 0.15$, $p_\pi > 0.175$ GeV/c, $\theta_{\mu\pi} < 2.6$ rad (Sec. 6 of the analysis note) raises and flattens the efficiency to ~ 13 – 20% (Fig. 21 of the analysis note). Table 34 compares the selection across the iterations, and it fixes the two opening-angle choices.

- (i) The 2.6 rad upper cut is kept. On the full exposure the events it removes are exactly those of the cosmic control region (Table 13 of the analysis note): in FHC, 5.1 signal, 60 neutrino background, 40 beam-off and 1.9 dirt, a signal purity of 4.7%. Dropping the cut would lower the signal-region purity from 58.8% to 55.0%, raise its beam-off content from 23 to 64 events and recover only 2.7% of the generated signal ($2055 \rightarrow 1999$ in the table). The removed region serves instead as the beam-off control region.
- (ii) No lower opening-angle cut is used. The small-angle region has high purity ($\sim 58\%$), so a lower cut reduces the selected signal without improving purity.

Table 34: Selection-optimisation study across the phase-space iterations, on the Run-1 FHC sample. Every entry is Monte-Carlo derived (GENIE overlay signal and background, EXT, dirt); the thresholds chosen here are those of the full-exposure selection of Table 11 of the analysis note. The beam-off column uses the pooled beam-off scaling of that study, so its absolute values are not those of the full exposure; the ranking of the options is unaffected. “no upper cut” drops the $\theta_{\mu\pi} < 2.6$ cut; “[0.314, 2.6]” adds a lower OA cut. The final choice keeps the 2.6 rad upper cut with no lower cut.

	Reference ($p > 0.1$)	no upper cut ($p > .15/.175$)	[0.314, 2.6] (lower cut)	Final ($\theta < 2.6$)
Generated signal	2684	2055	1916	1999
Selected signal	324.7	320.3	301.8	311.9
Efficiency	12.1%	15.6%	15.8%	15.6%
EXT (cosmic)	40.1	136.2	38.0	40.1
Total bkg	243.6	385.4	252.3	256.4
Purity	57.1%	45.4%	54.5%	54.9%
sig $\times$ purity	185.6	145.4	164.4	171.2

22.2 Momentum-threshold validation

The two signal-definition momentum thresholds sit at the selection efficiency turn-on, measured against a threshold-free true signal (the CC ν_μ $1\mu 1\pi^\pm$ topology with the momentum/angle cuts removed):

Threshold	below	above	turn-on
$p_\mu > 0.15$ GeV/c	~ 0 – 1.5%	~ 9 – 12%	$0.145 \rightarrow 0.155$
$p_\pi > 0.175$ GeV/c	~ 0 – 3%	~ 12 – 16%	$0.165 \rightarrow 0.185$

Both are correctly placed: lowering them would add near-zero-efficiency phase space, raising them would discard measurable signal. The reconstructed pion momentum (muon-hypothesis range) is biased low and increasingly so with momentum (mean reco–true ≈ 0 at threshold, -0.15 GeV/c at 0.40) from the μ -vs- π mass hypothesis plus pion re-interaction; a pion-hypothesis mass correction is applied to the p_π reco bins.

22.3 Combined-configuration reconstruction-level spectra

The two figures below show the selected and background-subtracted spectra of the combined configuration; the FHC and RHC spectra are in the analysis note.

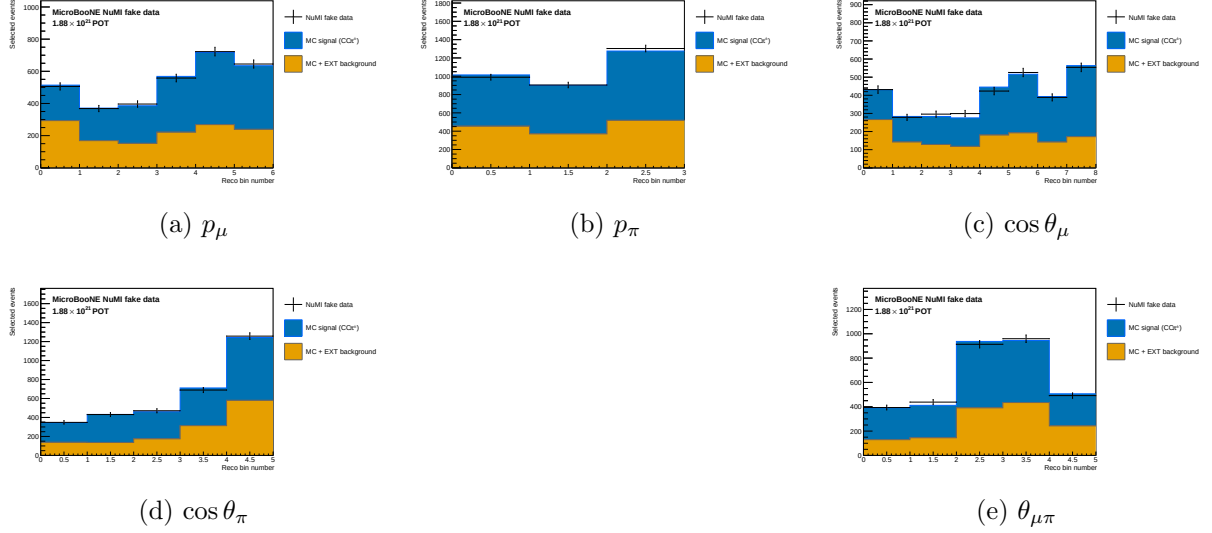


Figure 13: Stage 1, combined: selected reconstructed spectra in the combined configuration.

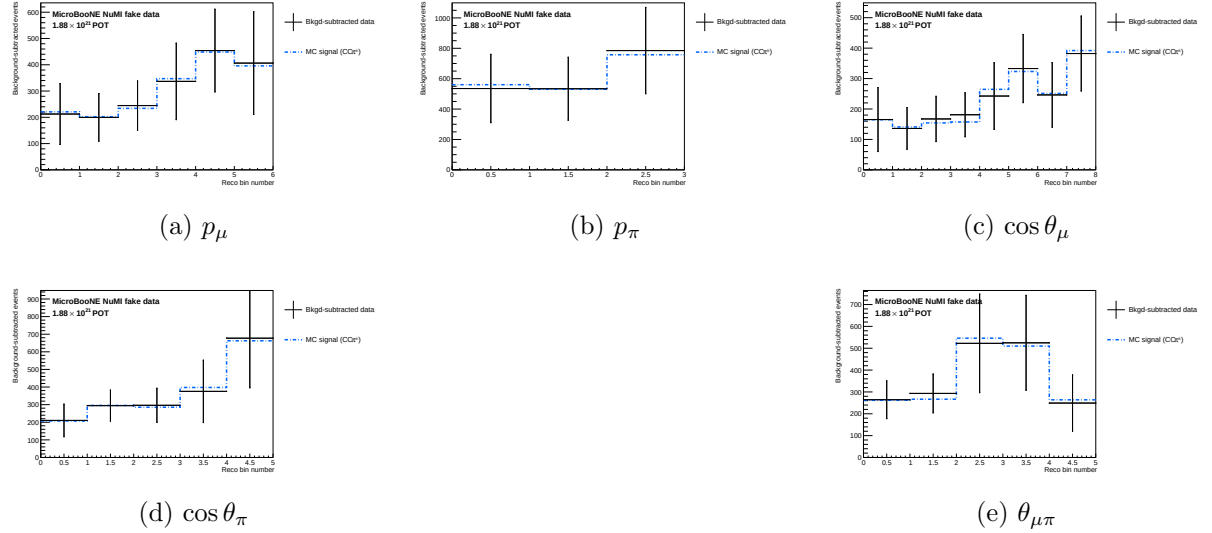


Figure 14: Stage 2, combined: background-subtracted data compared with the central-value MC signal.

23 Analysis pipeline and automated guards [SHARED]

Every released number is produced by the XSECANALYZER chain in four stages that are re-runnable from the raw PeLEE ntuples:

1. the event processing applies the selection(s) to each PeLEE ntuple and writes the processed event files, for the neutrino overlays, the dirt overlay and the per-run beam-off samples;
2. the universe builder constructs, per observable and configuration, the true and reconstructed histograms of every systematic universe from the files listed in that configuration;
3. the unfoldr sums the universes with the per-run exposure scaling, builds the covariance, unfolds, and writes the closure sidecar, the systematics dump and the matrix tables;
4. the export step writes the data release (curves, A_C matrices, covariance decompositions); the figure macros draw from the closure sidecars; the one-bin total uses the same chain

with the single-bin configurations; the count-level tables (cut flow, cut-and-count check, control-region yields) are produced with the same weights and scalings.

Five automated guards protect the normalisation, the currency of released artefacts, the result tables and the blinding:

- A normalisation manifest reads the authoritative configuration files and asserts that exposure, flux, target count and beam-off scaling agree across all configurations, and that every declared systematic is backed by a file.
- A coupling assertion in the unfolders refuses a configuration whose declared detector variations are not present in its file list.
- A staleness check refuses to ship any released artefact or result figure older than the extraction it depicts, and compares the integrals printed on the p_π figures with their sidecars.
- A table check parses every result table of the analysis notes and this supplement and fails if a value disagrees with the released results table, if a table does not sit under its own label, or if a row's column count differs from its header.
- A blinding guard treats a beam-on file without MC truth as real data and refuses it unless an explicit unblinding switch is set; the same detection selects the beam-off treatment (added to pseudo-data, subtracted from real data), so no manual switch exists to forget.

The release was built from the processed ntuples in a single pass with every cached universe deleted, so no intermediate result is carried over. A clean-cache rebuild of the same kind reproduced a previous release bin by bin to 10^{-9} (recorded in the change log); that comparison has not been repeated on the current release.

Part III

Rejected alternatives and development history

24 The first binning optimisation

[INCLUSIVE]

Historical configuration, not used in the submitted results. The change log gives the chronology.

The first binning optimisation gave p_π five bins, $\theta_{\mu\pi}$ seven and $\cos\theta_\pi$ five. The response and A_C tests of §7.2 and §24.2 coarsened or re-binned these, and the binning review of §8 replaced them; the binnings of the results are those of §3.

24.1 Binning optimisation

Bins were designed to be at least the local RMS detector resolution wide (so the response stays well conditioned and the Wiener-SVD does not amplify the correlated flux/cross-section uncertainty) and to hold ≥ 45 selected-signal events (statistical floor). This lowers the total fractional uncertainty in every observable, chiefly through the statistical and unfolding-amplified terms; the flux/beamline term is correlated across bins and sets a floor:

Observable	bins	Total before	Total after
p_μ	22→7	33.2%	25.0%
p_π	5→5	27.7%	27.0%
$\cos\theta_\mu$	12→5	31.3%	26.7%
$\cos\theta_\pi$	12→5	48.5%	33.9%
$\theta_{\mu\pi}$	9→7	28.4%	23.5%

24.2 Re-check of the optimised binnings

The binnings of the optimisation stage were coarsened where A_C smearing left no resolved information ($\cos\theta_\pi$ 5 $\rightarrow$ 4, $\theta_{\mu\pi}$ 7 $\rightarrow$ 5). The resulting binnings, called released below, were then re-tested with the final normalisation, weights, beam-off treatment, pseudo-data throw and detector variations, all of which condition the response, to find whether finer binnings are admissible. The test had two stages. The first is a pre-screen on the processed files (`macros/binning_screen.C`): for the released binning and finer candidates of each observable, the column-normalised migration diagonal per true bin, the efficiency per bin and the selected events per reconstructed bin, with the per-run scales and the central-value weight, and the 0.68 diagonal criterion applied to every bin. In the second, the candidates that passed were built and unfolded in full (universes, covariance, closure) and compared with the released extraction on the A_C row sums, the A_C conditioning ($s_{\min}/s_1$), the data-statistical term, the per-bin total uncertainties and the closure ratios.

Table 35: Binning re-check; “released” denotes the binning under test, and the forward-split five-bin $\cos\theta_\pi$ row is the reported binning. Pre-screen: smallest column-normalised migration diagonal, on the analysis sample, over the true bins of the released binning and of each finer candidate, FHC / RHC. The two-bin p_π row is evaluated here with a lower edge on the first reconstructed bin, which the released configuration does not have (the framework’s own diagonals are 0.91 and 0.76); the finer p_π candidates fail in their upper bins regardless.

Observable	Binning	min. diagonal FHC / RHC	criterion
p_μ	released 7	0.33 / 0.31	fails already (MCS resolution; §24)
	8, 9 bins	0.17 / 0.18, 0.22 / 0.22	fail
$\cos\theta_\mu$	released 5	0.68 / 0.68	at the limit
	6, 7 bins	0.64 / 0.63, 0.61 / 0.61	fail
θ_μ	released 5	0.67 / 0.65	at the limit
	7 bins	0.63 / 0.62	fail
$\cos\theta_\pi$	released 4	0.81 / 0.83	pass
	5 (backward split)	0.77 / 0.77	pass
	5 (forward split)	0.74 / 0.76	pass
	6 bins	0.73 / 0.75	pass
$\theta_{\mu\pi}$	released 5	0.79 / 0.79	pass
	6, 7 bins	0.74 / 0.73, 0.75 / 0.73	pass
p_π	released 2	0.58 / 0.56 (see caption)	–
	3, 4 bins	0.39 / 0.40, 0.29 / 0.29	fail

Table 36: Binning re-check; “released” denotes the binning under test, and the forward-split five-bin $\cos\theta_\pi$ row is the reported binning. Full extraction of the candidates that passed the pre-screen, FHC / RHC: range of the A_C row sums, A_C conditioning $s_{\min}/s_1$, bin-averaged data-statistical term, range of the per-bin total uncertainty, and range of the per-bin closure ratio to the A_C -smeared truth. Backward split: edges $-1, -0.3, 0.2, 0.5, 0.75, 1$; forward split: $-1, -0.1, 0.35, 0.6, 0.8, 1$; six bins: $-1, -0.4, 0, 0.3, 0.55, 0.75, 1$; $\theta_{\mu\pi}$ six bins: $0, 0.55, 0.75, 0.95, 1.3, 1.85, 2.6$ rad; seven bins: $0, 0.5, 0.7, 0.9, 1.15, 1.45, 1.9, 2.6$ rad.

Binning	A_C row sums	$s_{\min}/s_1$	data stat	per-bin unc.	closure
$\cos\theta_\pi$ rel. 4	0.71–0.82 / 0.79–0.86	0.27 / 0.24	12.0 / 11.5%	30–49 / 32–55%	0.95–1.09 / 0.94–1.07
backward split 5	0.65–0.76 / 0.63–0.80	0.20 / 0.27	16.0 / 15.2%	29–73 / 34–91%	0.83–1.08 / 0.87–1.03
forward split 5	0.68–0.77 / 0.70–0.84	0.46 / 0.38	12.7 / 12.3%	30–53 / 33–48%	0.89–1.07 / 0.90–1.05
6 bins	0.73–0.84 / 0.77–0.89	0.18 / 0.15	19.0 / 16.3%	32–91 / 32–54%	0.66–1.13 / 0.81–1.23
$\theta_{\mu\pi}$ rel. 5	0.52–1.03 / 0.58–1.15	0.008 / 0.002	8.2 / 8.4%	35–40 / 30–39%	0.95–1.06 / 1.03–1.08
6 bins	0.56–1.29 / 0.58–1.36	0.002 / 0.000	8.4 / 10.2%	34–40 / 30–41%	0.95–1.06 / 1.03–1.11
7 bins	0.63–1.41 / 0.65–1.06	0.000 / 0.000	11.5 / 11.2%	37–40 / 32–46%	0.85–1.04 / 0.87–1.14

p_μ , $\cos\theta_\mu$, θ_μ and p_π cannot go finer: their released binnings already sit at or below the diagonal

criterion, set by the MCS momentum resolution and the angular resolution against the neutrino direction, and every finer candidate fails it. $\theta_{\mu\pi}$ admits six or seven bins by the diagonal and closes acceptably, but the additional bins carry no resolved information: the smallest singular value of A_C , already 0.008 of the largest at five bins, goes to zero, and the row sums overshoot to 1.3–1.4 in the bins the regularisation has to fill in, so the released five bins stand. $\cos\theta_\pi$ is the one observable with room, and only in the forward hemisphere where the events are: splitting the backward region produces a bin of 73–91% uncertainty that closes at 0.83–0.87, and six bins do the same, whereas the forward split (edges $-1, -0.1, 0.35, 0.6, 0.8, 1$) keeps every diagonal above 0.74, keeps the per-bin uncertainties in the released range, raises the data-statistical term by only 0.7% and improves the conditioning of A_C ($s_{\min}/s_1$ from 0.27 to 0.46 in FHC, 0.24 to 0.38 in RHC) while closing to 0.89–1.07. This is the only change the re-check supports, and it is the reported binning. The generator predictions are computed at these edges from the event-level files (the unchanged observables reproduce the prediction files of the four-bin scheme to machine precision, which also verifies the generator chain end to end), and the D’Agostini cross-check is run on it. No control-region distribution uses $\cos\theta_\pi$, so the frozen control-region binning is unaffected. The proton-tagged $\cos\theta_\pi$ keeps its four bins.

The released binnings of the proton-tagged observables were re-tested on the maximin scan against a relaxed 0.50 diagonal criterion in addition to the nominal 0.68. Twenty-six candidate binnings pass the relaxed diagonal, and every one of them fails on the conditioning of A_C . The single exception is a two-bin W_{had} (0.58/0.54), which is better conditioned than the released six-bin scheme but was not adopted at this stage, since two bins buy that conditioning by discarding the shape W_{had} is measured for. The re-check therefore changes no released proton-tagged binning. The one proton-tagged binning set at this stage is that of θ_p , whose edges were taken from the RHC chain rather than the FHC one after the scan was run on both: at the same four bins they condition three to four times better in both usable configurations.

Revisions from the binning review. The review of §8, on the corrected exposure and with the edges optimised at every bin count rather than chosen by hand, revises three of the conclusions above. First, the released five-bin $\cos\theta_\mu$ and θ_μ fail the 0.68 criterion (worst diagonals 0.67 and 0.65), optimised edges at the same five bins pass it, and both variables admit finer binnings that retain more shape. Second, the proton-tagged candidates were rejected on a conditioning threshold that the released binnings themselves do not meet; the review measures the retained shape by the Wiener filter factors of A_C instead. Third, the six-bin W_{had} fails 0.68 in all six bins and 0.50 in five, and the RHC-derived θ_p edges leave 21 selected events in the last bin. The binnings adopted from the review replace those of this section.

25 The muon angle: $\cos\theta_\mu$ versus θ_μ

[INCLUSIVE]

Historical configuration, not used in the submitted results. The change log gives the chronology.

25.1 Angular variable: $\cos\theta_\mu$ versus θ_μ

The muon angle is reported only as $\cos\theta_\mu$. The binning review (§8) shows that the same partition of the angle retains up to one degree of freedom more as $\cos\theta_\mu$ than as θ_μ , because the regulariser acts on the density, so a θ_μ result adds no information to a $\cos\theta_\mu$ result with the same bins. The comparison below uses the five-bin $\cos\theta_\mu$ and θ_μ binnings that preceded the review.

In those binnings the two variables spanned the same full range but were different partitions of it: the θ_μ edges (0.43, 0.62, 0.84, 1.17 rad) are not the arccos images of the $\cos\theta_\mu$ edges (0.45, 0.64, 0.86, 1.10 rad). The two behave very differently under the additional-smearing matrix A_C of the Wiener–SVD unfolding, which affects how the measurement compares to models.

In $\cos\theta_\mu$ the signal is sharply forward-peaked: the last two bins ($\cos\theta_\mu > 0.8$) hold most of the rate, and A_C suppresses the integral of any truth-level prediction strongly (row sums 0.17–0.39 in FHC at that binning, against 0.66–0.91 for p_μ). Because the same A_C is applied to the data, the tune and every generator, they are compared on equal footing.

The suppression is not a property of the shape of the truth distribution. Across the observables the A_C integral factor shows no correlation with peakedness: $W_{\pi p}$ is the sharpest distribution in the analysis (peak-to-minimum ratio 36) yet its factor is 0.75, δp_T is comparatively flat (ratio 10) at 0.70, and the six-bin W_{had} then under study, neither sharp nor flat, was the most suppressed at 0.61; the inclusive $\cos\theta_\mu$ factors sat far below all of these (0.17–0.39 against 0.66–0.91 for p_μ) because its response is the worst conditioned, not because its peak is the sharpest. Within $\cos\theta_\mu$ itself NEUT and NuWro are the flattest of the five truth-level curves (peak-to-minimum 3.8 and 4.3, against 16.3 for the tune) and are smeared no differently from the others: no smeared generator bin is negative and both sit above the tune in every bin. A_C is the Wiener filter acting on the SVD modes of the response: it removes whatever the data cannot statistically constrain, which is set by the conditioning of the response for that observable and binning, not by the shape of the truth distribution, and it is not norm-preserving, so it can suppress or enhance.

θ_μ with the width-aware operator. The pseudo-data are Poisson throws from the Micro-BooNE tune, so the tune is the model that should fit best, and it does (FHC θ_μ , χ^2 on five bins: tune 2.14, $p = 0.83$; GENIE 5.55; GiBUU 6.07; NEUT 8.46; NuWro 9.10; closure 1.52, $p = 0.91$). Under the uniform stencil (§12) every model gave 2.8–4.7, indistinguishable, because every model had been mapped onto the same curve. At those binnings the θ_μ χ^2 were systematically larger than the $\cos\theta_\mu$ ones for every model, the truth included.

26 Extending the pion threshold below 0.175 GeV/c (not adopted) [INCLUSIVE]

Rejected alternative, not used in the submitted results. The change log gives the chronology.

The extension described here was tested and not adopted; the measurement threshold is $p_\pi > 0.175$ GeV/c.

Test with a low-momentum bin. For the test, a [0.113, 0.175] GeV/c bin is prepended to the pion-momentum binning, lowering the measured phase space to the ν_e threshold. The truth-level signal below 0.175 GeV/c is recovered from the stored signal-component branches (the reconstruction reproduces MC_Signal above 0.175 GeV/c to 0.4%, the residual being the unstored kaon veto); 3323 true signal events populate the new bin. Its selection efficiency is 2.2%, an order of magnitude below the 15–17% of the standard bins (Table 37), as expected from the reconstruction turn-on. The Wiener–SVD extraction nevertheless remains well behaved with the bin included: the M_A^{RES} pseudo-data closure gives $\chi^2/\text{ndf} = 1.41/6$ ($p = 0.97$) and the flux term is unchanged (17% reco, 18.4% unfolded, amplification 1.08 $\times$). In this test the bin therefore adds information, at the price of a large per-bin uncertainty driven by the low efficiency.

For the test the generator predictions were regenerated on the six-bin scheme (the ppi histogram uses $p_\pi > 0.113$ while the other observables retain $p_\pi > 0.175$, mirroring the analysis). GENIE, NuWro and NEUT all provide the low bin; GiBUU could not be re-binned as its generator-level event record was not retained, so it is omitted from the extended p_π comparison. Against the pseudo-data the generators give $\chi^2/\text{ndf} = 11.1/6$ (GENIE, $p = 0.09$), 7.0/6 (NuWro, $p = 0.33$) and 6.3/6 (NEUT, $p = 0.39$): NEUT and NuWro describe the extended spectrum well, while GENIE is mildly disfavoured, consistent with the known GENIE over-prediction of low-momentum $\text{CC}\pi^\pm$ production.

Table 37: Selection efficiency per true p_π bin with the low-momentum bin added. The new $[0.113, 0.175]$ GeV/c bin sits at 2.2%, an order of magnitude below the standard bins.

True p_π [GeV/c]	N_{true}	N_{reco}	Efficiency
0.113–0.175 (new)	3323	74	2.2%
0.175–0.250	3368	526	15.6%
0.250–0.320	2190	367	16.8%
0.320–0.420	2831	479	16.9%
0.420–0.550	2499	410	16.4%
0.550–1.000	2933	429	14.6%

27 The rejected five-bin p_π response

[INCLUSIVE]

Rejected configuration, not used in the submitted results. The change log gives the chronology.

The migration diagonals of this response and the reasons for rejecting it are in §7.2.

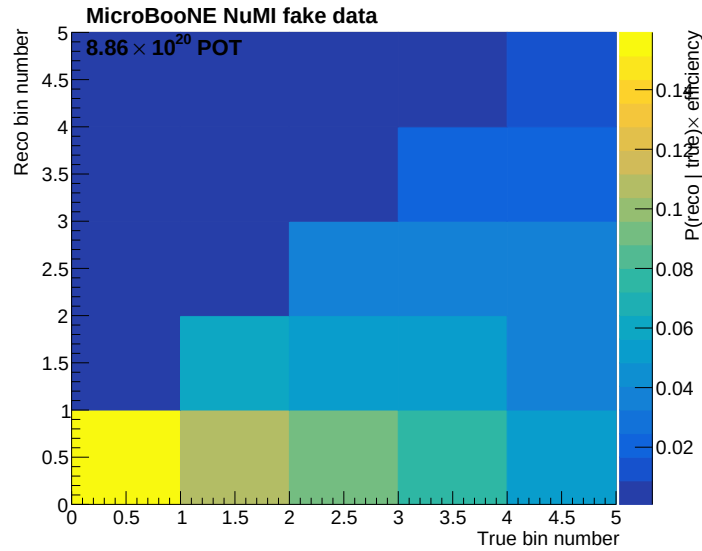


Figure 15: Smearceptance matrix of the rejected five-bin p_π scheme (FHC, analysis sample): reconstructed events pile into the lowest bin from every true bin, the diagnostic that motivated the merged regions of §7.2.5; the three-region response of the results is in the analysis note.

28 Golden-pion classifier: details

[INCLUSIVE]

Rejected alternative, not used in the submitted results. The change log gives the chronology.

28.1 Available inputs

The BNB note supplies a tested recipe, and all four of its BDT inputs are available in this analysis. The Bragg likelihood ratios (`trk.bragg_pion`, `_mip`, `_mu`, `_p`) and the track-shower score are in the ntuples; the multi-pion pion-ID BDT in this analysis trains on exactly those, and the descendant counts are `pfp.trk_daughters_v` / `pfp.shr_daughters_v`. Wiggleness is available under a different name: the PeLEE ntuples carry `trk.avg_deflection_stdev_v` (with companions `_mean_v` and `_separation_mean_v`), which is the standard deviation of the angular deflection between adjacent track points. It is fully populated and is carried forward by `ProcessNTuples`.

28.2 Track wiggleness as a golden-pion tag

The variable was tested on truth-matched charged pions in the Run-1 FHC overlay, calling a pion “golden-like” when the mass-corrected range momentum recovers the true momentum to within 20%, and “interacting” when it recovers less than 60%. The two populations separate clearly, but golden pions are the wigglier ones, by about a factor two in the mean (0.0048 against 0.0027), the opposite of the naive expectation. This is not a momentum artefact: it persists when binned in track length, with the ratio staying at 0.51–0.63 from 5 cm to 400 cm.

The sign has a physical explanation. A pion that stops travels its full range and scatters strongly through its final slow centimetres, so the deflection spread averaged along the track is large. A pion that interacts hadronically is cut short while still fast and never reaches that wiggly end, so its track is straighter. High wiggleness is thus a signature of having stopped, which is the population for which a range-based momentum is meaningful.

As a single cut it is useful: on this sample the golden fraction is 39.9% before any cut, and requiring `trk_avg_deflection_stdev_v` > 0.001 keeps 86.8% of golden pions while rejecting 48.8% of interacting ones, raising the golden fraction to 53.0%. A tighter cut at 0.002 gives 57.8% efficiency and 55.7% golden. For comparison, the BNB four-variable BDT reaches 67.7% golden at roughly half the generic-selection efficiency, so one variable recovers a good part of the gain, and the remaining three variables, all available here, could recover more.

Three options follow, in increasing order of effort: rebin p_π against a migration criterion, which on the numbers above means one or two resolved intervals at threshold plus a wide open region; add a golden-pion cut for the p_π cross section built from the available variables, accepting the factor-two efficiency loss the BNB analysis accepts; or reproduce the BNB BDT in full. None changes the total or angular results, which do not use this estimator. The first option, applied with the 0.50 criterion of the analysis (§8), gives the adopted binning (§7.2.5); the classifier was trained and tested and does not change this conclusion (§7.2.6).

28.3 Variable ranking

The variable ranking of the classifier of §7.2.6 is dominated by the deflection variables (`wmean`, then `wstd`); the Bragg ratios contribute almost nothing, although Bragg was expected to carry the stopping signature.

28.4 Track-sample and analysis-sample diagonals

Table 7 is built on the track sample (§7.1). On it the five-bin scheme reads 99/24/13/6/6% and the two-bin scheme 98/62%, against 96/35/21/10/6% and 91/76% on the analysis sample. The track-sample figures are therefore not the analysis response, and the 62% does not mean that the inclusive p_π measurement fails the usable-bin criterion: on the analysis sample the upper bin of the two-bin scheme is at 76%. The comparisons made within the table, between rows of the same sample, are valid as they stand.

28.5 Multiple Coulomb scattering as estimator and classifier input

Range saturates because the pion interacts before stopping, which truncates the track. Multiple-Coulomb-scattering (MCS) momentum is inferred from the scattering along whatever track exists and is not truncated by an interaction, so it is the natural candidate to replace range. The CC0 π analysis uses this pairing, range for contained tracks and MCS for exiting ones. Measured on 122 878 truth-matched pion tracks (`trk_mcs_muon_mom_v`, filled for 99.9% of them):

true p_π [GeV/ c]	$\langle \text{len} \rangle$	$\langle \text{range/true} \rangle$	RMS	$\langle \text{MCS/true} \rangle$	RMS
0.175–0.250	25 cm	0.834	0.16	4.335	14.1
0.320–0.420	46 cm	0.629	0.23	2.438	7.2
0.550–0.800	71 cm	0.446	0.18	1.423	3.5
0.800–1.500	84 cm	0.319	0.16	1.011	2.1

In the top bin range is biased to 0.319 (the saturation), while MCS is unbiased at 1.011 within uncertainties; but MCS carries a 210% RMS there and 1400% in the lowest bin, which makes it unusable as a per-event estimator. The reason is structural and explains why the CC0 π recipe does not transfer: that analysis applies MCS to exiting tracks, which are long by construction, whereas the problematic pions here are contained but interacting, 25–84 cm long, and MCS needs of order a metre.

The CC0 π note also uses the disagreement between the two estimators as a track-quality BDT input rather than as an estimator. The motivation applies here: a pion that interacts has a truncated range but an untruncated MCS, so the fractional difference $(p_{\text{range}} - p_{\text{MCS}})/p_{\text{range}}$ should flag the events that degrade the response. Added to the golden-pion classifier, it ranks second of ten in importance, behind only the mean deflection, but it leaves the performance unchanged:

classifier inputs	ROC integral	diagonals at $\sim 44\%$ efficiency
baseline (nine variables)	0.758	99.1/33.1/24.5/15.4/9.1
+ range–MCS difference	0.757	99.0/34.8/24.8/14.8/9.1

The new variable is informative but provides no additional information: it is strongly correlated with the deflection variables already present, since both measure whether the track scattered heavily and ended abruptly. A high importance rank with an unchanged ROC integral is the signature of a redundant input. This independently supports the ceiling argument above: the classifier already extracts what the golden label can offer, and the limit is the physics of the label rather than the quality of the classifier.

A further variant was not pursued. Retraining on the label “range is accurate”, $|p_{\text{corr}}/p_{\text{true}} - 1| < 0.2$, would target the quantity of interest directly and would raise the diagonals, but it would select on the answer: keeping events where the estimator happens to be right sculpts the response by construction and inflates the diagonal without measuring anything, the same effect described for the tight BDT working points below. The golden label is used because it is a physical property of the pion, independent of how well it was reconstructed.

28.6 Tight working points

The > 0.15 row of Table 7 appears better than perfect truth selection (46.9, 42.2, 36.1, 19.5%), which is impossible for an unbiased classifier. This arises because a hard cut on the classifier preferentially keeps long tracks: inside the top true bin, the surviving tracks have a mean length of 120.8 cm against 78.2 cm for all pions, and a mean reconstructed momentum of 0.406 against 0.307 GeV/ c . The cut is thus partly a cut on the reconstructed momentum itself, which raises the diagonal by construction while sculpting the spectrum. The apparent gain is circular, and the tight working points are not used. This is the failure mode anticipated for track length as a direct input; it enters here through the deflection variables, which correlate with length, even though length itself was excluded from training.

28.7 Residual bias of golden pions

Golden pions still reconstruct at $\langle p^{\text{corr}}/p^{\text{true}} \rangle = 0.73$ on the track sample. Interaction is therefore not the only problem: the momentum conversion is itself biased. A single flat rescaling does

not repair it: applying the best global factor drops the first bin from 98.9 to 49.1% and leaves 0/5 bins passing, because the bias is momentum-dependent, as the saturation curve of Figure 1 shows.

Coarser binning helps but is not sufficient. Collapsing to two bins (0.175, 0.230, ∞) gives 98.7 and 58.3% with perfect golden selection, an improvement but still short of 0.68. Only a single bin, i.e. no differential measurement, passes trivially.

29 Pion-momentum estimators: detailed studies

[INCLUSIVE]

Rejected alternatives, not used in the submitted results. The change log gives the chronology.

Calorimetric energy. Deposited charge stops at the interaction point just as range does, so $\langle \text{calo}/\text{true} \rangle$ falls from 1.136 to 0.383 across the momentum range while the RMS is three to five times that of range (1.007 to 0.313, against 0.157 to 0.158). Calorimetry is subject to the same truncation as range, with more noise.

Golden-pion selection. Splitting the sample by the truth label shows that golden pions are not well reconstructed either, and that their advantage disappears at high momentum, where it would be needed:

true p_π [GeV/c]	golden fraction	golden $\langle \text{range}/\text{true} \rangle$	non-golden
0.175–0.250	77.8%	0.873	0.699
0.320–0.420	44.1%	0.705	0.570
0.550–0.800	21.4%	0.507	0.429
0.800–1.500	17.2%	0.327	0.317

Above 0.8 GeV/c the golden and non-golden values are identical to within one per cent. The reason is physical: a 1 GeV/c pion has a CSDA range of several metres in argon and cannot stop inside a 2.5 m TPC unless it has already lost most of its energy through processes that truncate or break the track. “Golden” at high momentum therefore selects a pathological subsample rather than a clean one. This is why the classifier of §7.2.6 reaches a ceiling: above ~ 0.5 GeV/c there is no clean subsample to find.

Multivariate regression. A BDTG regression on p_π^{true} , using range, MCS, calorimetry, length, the three deflection variables, the four Bragg likelihoods, track score and daughter count, was trained on Runs 1 and 4 and evaluated on the held-out Run 2. It largely removes the saturation in the mean, $\langle \text{pred}/\text{true} \rangle$ running 1.196 to 0.722 against 0.835 to 0.319 for range, at comparable RMS. Rebuilding $W_{\pi p}$ with it:

pion estimator	W bias	W RMS
range (analysis)	−0.127	0.144
BDTG regression	− 0.005	0.119
true p_π (ceiling)	+0.003	0.046

The bias falls by a factor of 25, which is relevant independently of binning: a $-127 \text{ MeV}/c^2$ shift distorts the whole W distribution. The resolution improves only from 0.144 to 0.119 against a ceiling of 0.046, so the regression recovers about a quarter of what is available.

Its effect on the binning is a redistribution of the migration rather than an overall improvement. The six-bin diagonals move from 98/30/19/19/13/6% to 63/38/33/43/52/56%: the top bin improves nearly tenfold, but the first bin degrades from 98% to 63%, and in the two-bin scheme the worst bin falls from 73% to 58%. That degradation is regression to the mean, the estimator

being pulled toward the sample centre, and it damages the low- W region where the resonance physics sits. The same pattern appears in p_π itself, where the first bin falls from 99.2% to 60.6% while the top bin rises from 0.5% to 33.2%.

Two considerations argue against adopting it. The estimator would be MC-trained and would sit upstream of every observable, so its model dependence would become a systematic on every result rather than on one cut. And the trade it offers moves measurability out of the region of interest. It is not adopted.

Remaining handles. The ceiling measurements above bound what is available: a perfect pion momentum would give $W_{\pi p}$ a resolution of 0.046, a factor of three better than the regression achieves and enough to support several bins. The information is carried away by the interaction products, so the approach that addresses the cause is to recover their energy: the pion and its daughters treated as a mini-calorimeter, using the per-PFP calorimetric energies and daughter links already present in the input files. For the proton-tagged sample there is a second, independent handle in transverse kinematic imbalance, where the muon and proton constrain the pion transverse momentum without using the pion’s own reconstruction at all, at the cost of Fermi-motion smearing of order 250 MeV/ c . The transverse-imbalance route has not been tried; the daughter-recovery route was tested next.

Recovery of the interaction products. The ceiling measurements above locate the missing information in the secondaries produced when the pion interacts, so the natural remedy is to treat the pion and its products as a single calorimeter, and this was tested. PeLEE carries no parent index, so association is geometric, by PFP start point relative to the pion track end, optionally restricted to the generation immediately below the pion in the Pandora hierarchy. The hierarchy restriction is needed: at a 40 cm radius the unrestricted sum collects 0.371 GeV in the lowest momentum bin, where only 0.034 GeV is missing, so it is dominated by unrelated activity.

With the hierarchy cut applied, on Run 1:

true p_π [GeV/ c]	$\langle \text{KE missing} \rangle$	$\langle \sum E \text{ of daughters within } R \rangle$ [GeV]			
		5 cm	10 cm	20 cm	40 cm
0.175–0.420	0.034	0.023	0.026	0.029	0.037
0.420–0.800	0.246	0.040	0.048	0.055	0.062
0.800–1.500	0.659	0.056	0.074	0.091	0.104

“KE missing” is the pion’s true kinetic energy less the calorimetric energy of its own track. In the top momentum bin 659 MeV is unaccounted for and the visible daughters supply 104 MeV, of which about 37 MeV is the baseline seen even in the lowest bin where almost nothing is missing. Genuine recovery is therefore of order 10% at high momentum and 25% in the middle bin.

The effect on the binning was tested directly rather than inferred from the bias, and the daughter sum does not change the binning assessment. Adding the daughter energy improves $\langle \text{reco}/\text{true} \rangle$ everywhere, the top bin moving from 0.373 to 0.470 and the first bin becoming nearly unbiased at 0.953, and it roughly triples the upper migration diagonals. Neither is sufficient: the best upper diagonal rises from 5.5% to 15.7%, against the 68% of the BNB criterion. In the two-bin scheme there is no net gain and arguably a loss, the worst bin improving from 61.4% to 71.0% while the first degrades from 97.7% to 82.4%, the same regression-to-the-mean behaviour seen with the multivariate estimator. Since both bins already clear the criterion with plain range, the daughter association would add a new systematic without improving the response. The coarse p_π binning is therefore robust against every estimator tested, which the binning study alone could not establish.

The recovered energy is far too small to change the measurement, and the reason is physical rather than algorithmic. Pion absorption on argon, $\pi A \rightarrow NN \dots$, deposits much of the energy in nucleons, roughly half of which are neutrons and therefore invisible in liquid argon; the

charged products are frequently short and fall below the threshold to be reconstructed as separate particles. The energy is therefore unassociated and largely undetected. Applying the daughter sum would move $\langle \text{reco}/\text{true} \rangle$ in the top bin from 0.32 to about 0.42, still less than half of truth.

30 Cosmic-direction cuts considered and rejected [INCLUSIVE]

Rejected alternative, not used in the submitted results. The change log gives the chronology.

Directional cut against residual cosmics. Part of the surviving beam-off sample is thought to consist of broken cosmic muon tracks, in which the break is reconstructed as a vertex and the track direction bears no relation to it. Such events carry a directional signature, which is seen in the data: among selected events the beam-off sample is enriched at backward reconstructed muon angles by nearly an order of magnitude relative to signal.

reco $\cos \theta_\mu$	EXT (beam-off)	MC signal	enrichment
< 0	31.8%	4.1%	$7.8\times$
< -0.5	9.1%	1.4%	$6.4\times$

Despite this enrichment, a cut does not pay off. At the final cut the FHC sample is 833 signal, 23 beam-off and 1 418 predicted events in total (Table 11 of the analysis note). A $\cos \theta_\mu < 0$ requirement would remove about 32% of the beam-off background, about 7 events, and about 4% of the signal, about 33 events. The beam-off background is subtracted from the data directly, and that subtraction is unbiased by construction, so removing part of it only reduces the statistical uncertainty of the subtraction. That uncertainty is 4.2 events for the 23, dominated by the 8 selected Run-1 beam-off events scaled by 1.23 for Run 1 and by 0.76 for Run 2; the cut would reduce it to about 3.5 events. Losing 33 signal events to reduce the uncertainty by less than one event is a net loss.

The same bound applies to any cosmic cut. The beam-off background is 1.6% of the selected sample and is subtracted without bias, so removing all of it with no signal loss would gain about 4 events of statistical uncertainty on a sample of 1 418. Any cut costing more than about 0.5% of the signal is a net loss regardless of how efficiently it tags cosmic rays. The residual beam-off contribution is therefore left in place and handled by subtraction.

A more selective variant has not been tested. The $\text{CC}0\pi$ analysis tags mis-directed tracks with a calorimetric direction-preference flag in coincidence with a poor forward-muon fit, $\chi_\mu^2 > 6$, which fires on tracks whose dE/dx profile disagrees with the assigned direction rather than on backward-going tracks as such. That would not cut genuine backward muons, which outnumber mis-reconstructed ones here by roughly 8:1. Both quantities exist in the input ntuples but are not carried into the processed files. Such a cut is bounded by the same ≈ 6 -event ceiling on the EXT statistical uncertainty, so it could not bring a significant improvement.

Part IV

Physics background

31 Physics background and prior measurements (extended)

[SHARED]

31.1 Production mechanisms

At neutrino energies of 0.5–5 GeV, charged-current single charged-pion production proceeds through three mechanisms.

Baryon resonance production dominates at $E_\nu \lesssim 2$ GeV. The neutrino excites the struck nucleon to a baryon resonance, most prominently the $\Delta(1232)$ isobar, which then decays to a nucleon and pion:

$$\nu_\mu n \rightarrow \mu^- \Delta^{++} \rightarrow \mu^- \pi^+ p, \quad \nu_\mu p \rightarrow \mu^- \Delta^+ \rightarrow \mu^- \pi^0 p \text{ (or } \pi^+ n). \quad (3)$$

The standard model is that of Rein and Sehgal [10], which uses the Feynman–Kislinger–Ravndal (FKR) quark model to compute form factors for all baryon resonances up to $W \lesssim 2$ GeV. The modern implementation used in GENIE for resonance (RES) production is the Kuzmin–Lyubushkin–Naumov Berger–Sehgal (KLN-BS) model [11, 12, 13, 14], which combines the lepton-mass-corrected vector and axial form factors of Kuzmin *et al.* [11] with the updated $\Delta(1232)$ electromagnetic form factors of Graczyk and Sobczyk [12] and the improved treatment of the Δ line shape and the lepton-mass effects of Berger and Sehgal [13]. Berger and Sehgal also discuss the interference between the resonant and non-resonant amplitudes; that interference term is not implemented in the GENIE v3 configuration underlying the MicroBooNE tune [2], where the non-resonant contribution is added incoherently. Higher resonances ($N(1440)P_{11}$, $N(1520)D_{13}$, etc.) are included up to $W \lesssim 2$ GeV and contribute at $E_\nu \gtrsim 1.5$ GeV.

Non-resonant and DIS backgrounds. A non-resonant continuum contributes at all W ; in a full amplitude-level treatment it interferes (constructively or destructively by isospin) with the resonant amplitude, although the generators used here add it incoherently. Deep-inelastic scattering (DIS) becomes significant above $W \approx 1.6$ GeV. Together these processes represent $\sim 20\%$ of the $\text{CC}\pi^\pm$ rate in the energy range of this analysis and are modelled in GENIE via the Bodek–Yang formalism.

Coherent pion production. The neutrino can scatter coherently off the entire nucleus [7, 8], $\nu_\mu A \rightarrow \mu^- \pi^+ A^*$, leaving the nucleus in its ground state. The cross section is small ($\lesssim 10\%$ of the total) below 5 GeV. The signal definition of Sec. 6 of the analysis note is a final-state topology (one muon, one charged pion, no other mesons) with no truth-level veto on the interaction mode, so coherent $\text{CC}\pi^\pm$ events that satisfy it are included in the signal as a small component, not treated as background.

31.2 Nuclear effects and final-state interactions

On a heavy nucleus such as argon ($A = 40$), two classes of nuclear correction modify the $\text{CC}\pi^\pm$ rate and kinematics significantly.

Initial-state effects. Fermi motion of the struck nucleon broadens kinematic distributions, and Pauli blocking suppresses transitions to low-momentum nucleon final states near threshold. Nuclear binding also shifts the effective nucleon mass relative to free-nucleon predictions.

Final-state interactions (FSI). The produced pion must traverse the nuclear medium before being detected. Intranuclear rescattering occurs via:

- *Pion absorption* ($\pi A \rightarrow NN \dots$): probability $\sim 25\text{--}30\%$ on Ar; converts $\text{CC}\pi^\pm$ events into $\text{CC}0\pi$ topology.
- *Charge exchange* ($\pi^+ n \rightarrow \pi^0 p$): produces a $\text{CC}\pi^0$ event.
- *Quasi-elastic scattering*: pion survives but changes direction; modifies p_π and θ_π distributions.
- *Inelastic production* ($\pi N \rightarrow \pi\pi N$): rare below $p_\pi \approx 400$ MeV/ c .

FSI are modelled in GENIE via the INTRANUKE (hA) cascade. Because absorption and charge exchange mix $\text{CC}\pi^\pm$, $\text{CC}\pi^0$, and $\text{CC}0\pi$ topologies at the $\sim 30\%$ level, the signal in any $\text{CC}\pi$ analysis is necessarily defined by the final-state particle content observed in the detector, not by the underlying production process (Sec. 6 of the analysis note).

31.3 Generator predictions and known tensions

Major neutrino event generators reproduce the inclusive CC cross section to 10–15% but show systematic spread in the $\text{CC}\pi^\pm$ rate, owing to differences in the resonance model, the treatment of the transition region between resonance and DIS, and the FSI cascade.

GENIE v3 [1, 3]. For resonance production GENIE implements the KLN-BS model [11, 12, 13, 14] for the full tower of 18 baryon resonances up to $W \lesssim 2$ GeV, including the $\Delta(1232)P_{33}$, $N(1440)P_{11}$ (Roper), $N(1520)D_{13}$, $N(1535)S_{11}$, $\Delta(1620)S_{31}$, $\Delta(1700)D_{33}$, $N(1680)F_{15}$, $\Delta(1950)F_{37}$, and their isospin partners. The hadronic invariant mass range $1.0 < W < 1.7$ GeV is bridged by a resonance–DIS transition using the AGKY (Andreopoulos–Gallagher–Kehayias–Yang) model, which applies KNO scaling in the transition region and the Bodek–Yang DIS model above $W \approx 1.7$ GeV. Non-resonant single-pion production is included through the low- W extension of the Bodek–Yang DIS model and is added incoherently to the resonant contribution; no resonant/non-resonant interference is implemented. Coherent π^+ production uses the Rein–Sehgal coherent model. FSI are handled by the INTRANUKE hA empirical model, in which each hadron undergoes at most one effective interaction whose probability and outcome are tuned to hadron–nucleus data. The alternative hN microscopic cascade, also available in GENIE, propagates hadrons step by step using free-nucleon πN and NN cross sections. GENIE has been reported to over-predict the $\text{CC}\pi^\pm$ rate by 20–35% relative to bubble-chamber and MiniBooNE data in several comparisons when normalised to the BNL dataset; the tension is reduced but not eliminated in the AR23 tune.

NuWro [4]. NuWro models resonance production via the $\Delta(1232)$ only; no higher baryon resonances are included. The Δ is treated with the Paschos–Yu formalism for the Δ excitation amplitude, retaining the full angular structure of the pion decay. A smooth analytic interpolation connects the single- Δ resonance region to the DIS regime at higher hadronic invariant mass ($W \gtrsim 1.4$ GeV), using Bodek–Yang structure functions for the deep inelastic contribution; the transition avoids a hard boundary between resonance and continuum. Non-resonant single-pion production is included as a separate amplitude. FSI are implemented using the microscopic Salcedo–Oset model, which treats pion propagation in nuclear matter through the pion self-energy in the random-phase approximation, naturally accounting for medium modifications such as Δ broadening and pion absorption via two-body processes ($\pi NN \rightarrow NN$) that are absent in purely cascade-based approaches. NuWro has historically shown better agreement with MiniBooNE and bubble-chamber data than GENIE, partly attributable to the absence of overestimated higher-resonance contributions.

NEUT [15, 6]. NEUT, the generator used by T2K and Super-Kamiokande, implements resonance production via a modified Rein–Sehgal model including 18 resonances up to $W \lesssim 2$ GeV, with Graczyk–Sobczyk [12] electromagnetic form factors for the $\Delta(1232)$. The axial mass M_A^{RES} is a tunable parameter; T2K applies a data-driven tune that reduces the overall $\text{CC}\pi^\pm$ normalisation by ~ 10 –20% relative to the default. The transition to DIS is handled similarly to GENIE, using GRV98 LO parton distribution functions with Bodek–Yang corrections above $W \approx 2$ GeV. Coherent pion production follows the Rein–Sehgal coherent model with a Berger–Sehgal-style correction for finite lepton mass. FSI are modelled by a custom NEUT intranuclear cascade that tracks each produced hadron through the nucleus using step-by-step interactions sampled from πN cross sections, with nuclear density and mean-free-path corrections derived from the optical potential.

GiBUU [5]. The Giessen Boltzmann–Uehling–Uhlenbeck (GiBUU) generator takes a different approach: instead of a probabilistic step-by-step cascade, it solves a coupled set of kinetic

transport equations for the full hadronic system inside the nuclear medium. Resonance production includes the $\Delta(1232)$ and higher resonances whose in-medium spectral functions are broadened by collisional damping; the Δ width in the nuclear medium is substantially larger than its free-space value of $\Gamma_0 \approx 118$ MeV, suppressing threshold production. Pion propagation after production is governed by a mean-field potential (the pion optical potential), which shifts pion momenta as they traverse the nucleus. This effect is absent in cascade models and is particularly important for low-momentum pions ($p_\pi \lesssim 200$ MeV/ c) near the pion production threshold. The transition to DIS uses PYTHIA for string fragmentation above $W \approx 1.6$ GeV. GiBUU predictions for pion angular distributions and for the backward-hemisphere $d\sigma/d\cos\theta_\pi$ can differ substantially from those of cascade-based generators, which makes GiBUU a useful cross-check of FSI systematics.

Because the present results are extracted on central-value pseudo-data, no data/MC deficit is measured here. The analysis does provide the spread of the standalone generators in the measured phase space, a factor of ~ 1.6 between GENIE and NEUT (Table 21), which brackets the injected MicroBooNE tune. The over-prediction seen for several generator configurations in parts of the relevant phase space in the measurements surveyed below is the context for that spread, not a result of this analysis.

31.4 Previous experimental measurements

Table 38 summarises published $\text{CC}\pi^+$ measurements on nuclear targets ordered chronologically.

Table 38: Selected $\text{CC}\pi^+$ cross-section measurements on nuclear targets. $\langle E_\nu \rangle$ is the approximate mean or peak neutrino energy. “Type” indicates whether a total integrated (T) or differential (D) cross section was reported. “Target” gives the nucleus primarily used for cross-section extraction.

Experiment	Target	$\langle E_\nu \rangle$ (GeV)	Type	Key result / reference
ANL 12-ft [16]	D	1.0	T	Bubble chamber; first σ vs E_ν
BNL 7-ft [17]	D	1.5	T	Higher statistics; 15% normalisation tension with ANL
K2K [18]	C	1.3	T	First accelerator $\text{CC}\pi^+$ on carbon near detector
MiniBooNE [19]	CH ₂	0.7	D	Q^2 , W , p_μ , θ_μ ; 20% excess over NUANCE
MINERvA [20]	CH	3.6	D	p_π , θ_π , Adler angles; NuMI medium-energy
T2K ND280 [23]	CH	0.85	D	p_μ , $\cos\theta_\mu$; off-axis beam similar to this work
ArgoNEUT [24]	Ar	9.6	T	First $\text{CC}\pi^+$ on Ar; 115 ν + 158 $\bar{\nu}$ events (1.25×10^{20} POT)
MicroBooNE BNB [26]	Ar	0.7	D	Full BNB dataset (1.11×10^{21} POT); $\sigma = (3.75 \pm 0.80) \times 10^{-38}$ cm ² /Ar
MicroBooNE NuMI ν_e [25]	Ar	0.73	D	First $\nu_e + \bar{\nu}_e$ $\text{CC}\pi^+$ on Ar; $\sigma = (0.93 \pm 0.28) \times 10^{-39}$ cm ² /nucleon

Deuterium bubble chambers: ANL and BNL (1979–1986)

The ANL 12-foot bubble chamber [16] and BNL 7-foot bubble chamber [17] provided the first high-statistics measurements of $\nu_\mu d \rightarrow \mu^- \pi^+ pp$ in the 1–3 GeV range. Because the deuterium target is close to a free nucleon, these measurements need smaller nuclear corrections than those on heavier targets. The ANL and BNL total cross sections disagree by $\sim 15\%$ in absolute

normalisation, a discrepancy long attributed to flux uncertainties; the BNL flux is believed to be the better constrained. This ANL–BNL tension propagated directly into the spread of generator predictions: models tuned to ANL tend to predict lower rates than those tuned to BNL. The MINERvA experiment [22] subsequently resolved much of this tension by simultaneously constraining the flux and the cross section.

K2K near detector (2008)

The K2K experiment [18] made the first accelerator-based total $\text{CC}\pi^+$ cross-section measurement on a carbon target at $\langle E_\nu \rangle = 1.3$ GeV using its fine-grained near detector. The result was consistent with NEUT predictions and established the nuclear-target correction from deuterium to carbon at intermediate energies.

MiniBooNE (ν_μ on CH_2 , BNB beam, 2011)

The MiniBooNE detector (818-tonne mineral-oil Cherenkov detector) published differential $\text{CC}\pi^+$ cross sections on CH_2 in the Booster Neutrino Beam ($E_\nu^{\text{peak}} \approx 0.7$ GeV) [19]. MiniBooNE was the first experiment to provide high-statistics differential distributions as a function of Q^2 , the invariant hadronic mass W , and the outgoing muon and pion kinematics. The flux-integrated total cross section was measured to be $\sigma \approx 3.4 \times 10^{-38}$ cm² per CH_2 molecule, or equivalently $\sim 2.0 \times 10^{-38}$ cm² per carbon nucleus at $\langle E_\nu \rangle \approx 1$ GeV [26]. The measured rate exceeded the NUANCE prediction by $\sim 20\%$, so that model under-predicted the data. The “generator over-prediction” discussed in the later literature is the opposite tension, in which GENIE-based predictions lie above subsequent measurements in parts of the phase space; it depends on the generator and the phase space and is not universal. The hydrogen content of CH_2 complicates the extraction of the carbon-only cross section, and the spherical Cherenkov geometry makes it difficult to measure low-energy pions or reconstruct proton recoils. These limitations led to the subsequent adoption of finer-grained tracking detectors and liquid-argon targets.

MINERvA (ν_μ on CH, NuMI medium energy, 2015–)

The MINERvA experiment [20] uses a finely-segmented scintillator-strip tracking calorimeter in the NuMI medium-energy beam ($\langle E_\nu \rangle \approx 3.6$ GeV). It has published the most kinematically complete $\text{CC}\pi^+$ measurements to date, reporting differential cross sections as a function of: pion momentum p_π , pion angle θ_π , muon momentum p_μ , muon angle θ_μ , and the full set of Adler angles describing the pion in the hadronic centre-of-mass frame. MINERvA also measured $\text{CC}\pi^+$ on iron and lead targets [21], providing direct constraints on nuclear A -dependence at the same beam energy. GENIE consistently over-predicts MINERvA by 20–35% across all kinematic variables in the $\Delta(1232)$ peak, the same qualitative tension seen at lower energies. The higher beam energy at MINERvA means that the $p_\pi > 0.5$ GeV/ c region is well populated, which complements the MicroBooNE NuMI result, statistically limited at high pion momentum.

T2K ND280 (ν_μ on CH, off-axis beam, 2020)

T2K measured $\text{CC}\pi^+$ at its near detector ND280 in the off-axis T2K beam (OAB), which peaks at $\langle E_\nu \rangle \approx 0.85$ GeV [23]. The T2K OAB spectrum closely overlaps with the MicroBooNE NuMI off-axis spectrum in the kinematically sensitive 0.3–1.5 GeV range, so this is the prior measurement most directly comparable to the present work. T2K reported differential cross sections in p_μ and $\cos\theta_\mu$, finding a data/MC deficit of $\sim 30\%$ relative to NEUT. The carbon target ($A = 12$) differs from argon ($A = 40$), but the overlapping beam spectrum enables direct comparison of nuclear-mass dependence at fixed neutrino energy.

ArgoNEUT ($\nu_\mu + \bar{\nu}_\mu$ on Ar, NuMI, 2018)

ArgoNEUT [24] was a 175-litre LArTPC placed upstream of the MINOS near detector in the NuMI low-energy beam ($\langle E_\nu \rangle \approx 9.6$ GeV). Using 1.25×10^{20} POT, ArgoNEUT selected 115 ν_μ and 158 $\bar{\nu}_\mu$ CC π^+ candidate events after background subtraction [24], publishing the first CC π^+ total cross section on argon as a function of neutrino energy. The result was consistent with GENIE predictions at these high energies, where sensitivity to the $\Delta(1232)$ resonance is suppressed. The LArTPC technology and argon target make ArgoNEUT the direct predecessor to MicroBooNE, but the factor ~ 10 higher beam energy means that extrapolation to the $E_\nu < 2$ GeV regime relevant for oscillation experiments requires generator assumptions and limits the constraining power on FSI models in the resonance region.

MicroBooNE BNB channel: inclusive precedent (2022)

The same MicroBooNE detector used in this analysis published the first measurement of energy-dependent inclusive ν_μ charged-current cross sections on argon in the Booster Neutrino Beam [9] ($E_\nu^{\text{peak}} \approx 0.7$ GeV), establishing the detector's absolute cross-section capability on argon together with the BNB flux modelling and reconstruction performance relevant to the present work. The dedicated exclusive CC π^+ measurement in the same beam followed on the full BNB dataset (below), and it is that exclusive result to which the present NuMI CC π^+ measurement is most directly compared.

MicroBooNE BNB channel: full dataset (2026)

Reference [26] reports CC π^+ differential cross sections using the complete MicroBooNE BNB dataset of 1.11×10^{21} POT [26], selecting 6816 events after background subtraction and measuring differential cross sections in muon momentum, muon angle, pion momentum and pion angle, four of the five observables of the present analysis. The flux-integrated total cross section is

$$\sigma = (3.75 \pm 0.07_{\text{stat}} \pm 0.80_{\text{syst}}) \times 10^{-38} \text{ cm}^2/\text{Ar}$$

at mean neutrino energy $\langle E_\nu \rangle \approx 0.8$ GeV. Most generators reproduce the data within uncertainties; an exception is the muon angle with respect to the beam, where all tested generators overestimate the cross section at very forward angles ($\cos \theta_\mu \rightarrow 1$) [26]. That pion momentum differential cross section was the first such measurement on an argon target.

The BNB and NuMI analyses probe complementary phase space: BNB is on-axis with a narrow ν_μ spectrum peaked at ~ 700 MeV and negligible $\bar{\nu}_\mu$ content, while NuMI FHC at 135 mrad off-axis provides a softer, broader spectrum with a $\sim 35\%$ $\bar{\nu}_\mu$ admixture and extends sensitivity to $E_\nu \sim 0.1\text{--}0.5$ GeV.

MicroBooNE NuMI $\nu_e + \bar{\nu}_e$ CC π^+ (2025)

Reference [25] presents the first measurement of $\nu_e + \bar{\nu}_e$ CC single charged-pion differential cross sections on argon, using the full MicroBooNE NuMI dataset (2.0×10^{21} POT, FHC and RHC combined). The observable set (electron energy E_e , electron angle θ_e , pion angle θ_π , and electron-pion opening angle $\theta_{e\pi}$) mirrors the muonic observables of the present analysis. The flux-integrated total cross section is measured to be

$$\sigma = (0.93 \pm 0.13_{\text{stat}} \pm 0.27_{\text{syst}}) \times 10^{-39} \text{ cm}^2/\text{nucleon}$$

at a mean ν_e energy of 730 MeV, in good agreement with all tested generator predictions (GENIE, NuWro, NEUT, GiBUU) within the combined uncertainties. Prior to this measurement, CC π^+ production by electron neutrinos on argon had not been measured; the result is an input to the DUNE ν_e -appearance analysis, where CC π^+ events are a background to the ν_e quasi-elastic signal.

Glossary

Signal definition The set of particles leaving the nucleus that counts as the process being measured, with momentum thresholds and a fiducial requirement. It makes no statement about the interaction mechanism, because final-state interactions mix mechanisms and only the final state is observable.

Fake data / pseudo-data A simulated dataset used in place of beam-on data while the analysis is blind. Here it is a Poisson throw: each simulated event is replicated a Poisson-distributed number of times with mean equal to its weight at data exposure, so the sample has the statistical fluctuations of a real dataset.

Blind The signal region has not been examined in beam-on data. Every plotted “data” point of the signal region is fake data.

Beam-on, beam-off (EXT), dirt Beam-on data are recorded in the beam spill; beam-off data are recorded between spills and measure cosmic-ray activity; “dirt” is simulated neutrino interactions outside the cryostat whose products enter the detector. The prediction is neutrino simulation + beam-off + dirt.

Gate ratio The factor scaling beam-off data to the beam-on exposure: the ratio of the number of beam-on to beam-off trigger gates in a running period.

Tune, central value (CV) The MicroBooNE tune is the collaboration’s adjusted GENIE model; the central value is the nominal prediction from it. “CV weight” is the per-event weight that turns the raw simulation into the tuned prediction.

Safe-weight rule Any event weight that is non-finite, negative or above 30 is replaced by 1. One rule, applied everywhere, so that the prediction and every pseudo-data throw agree.

Universe One alternative version of the prediction with a systematic parameter varied. A *multisim* is a set of hundreds of universes drawn randomly from the parameter uncertainties; a *unisim* is a single $\pm 1\sigma$ variation of one parameter. PPF universes vary the flux; GENIE universes the interaction model; reinteraction universes the hadron transport.

Detector variations (detVar) Dedicated simulation samples with one aspect of the detector response changed (light yield, recombination, space charge or wire response), used to estimate the detector uncertainty.

Migration matrix, efficiency, smearceptance The migration matrix M_{ij} is the probability that an event in true bin j is reconstructed in bin i ; the efficiency ε_j is the fraction of true bin j that is selected at all. Their product $S_{ij} = \varepsilon_j M_{ij}$ is the response the unfolding inverts; we call it the smearceptance matrix.

Migration diagonal The fraction of the selected events of a true bin that are reconstructed in the same bin (the diagonal of the column-normalised M). A binning is accepted only if every bin’s diagonal exceeds 0.68.

Unfolding, Wiener-SVD Solving for the true spectrum given the reconstructed one and the response. A plain inversion amplifies statistical noise; Wiener-SVD regularises it with a filter built from the simulation, trading a small bias for a large reduction in variance.

Additional-smearing matrix A_C The price of regularisation, made explicit: the unfolded result estimates $A_C T$ rather than the truth T . A prediction must be multiplied by A_C before it is compared with the result. A_C is released with every extraction.

Not normalisation-preserving The rows of A_C do not sum to one, so the integral of an unfolded differential result is not the total cross section and differs between observables of the same sample. The total is measured separately.

One-bin extraction The total cross section, extracted with a single true bin covering the whole phase space. Nothing is regularised, $A_C = 1$, and the result is the physical total.

Conditioning, $s_{\min}/s_1$, rank-deficient The ratio of the smallest to the largest singular value of the regularised response. Near zero, some direction in the true spectrum cannot be recovered from the data at all (rank-deficient); the result in that direction is the prior. A binning can pass the migration-diagonal criterion and still be unusable here.

Closure Unfolding pseudo-data and checking that the truth comes back. Closure against the A_C -smeared truth tests the implementation; it cannot test whether the model is right, because the same model is on both sides.

Covariance The matrix of uncertainties and their correlations between bins, one per systematic source, propagated through the unfolding. “Nominal configured covariance” means the sum over the sources implemented in the analysis; the sources known to be missing are listed in the limitations.

Flux-averaged cross section The cross section averaged over the neutrino energy spectrum of the beam, per argon nucleus, in the stated phase space. It depends on the flux, which is why each horn configuration is a separate measurement.

FHC, RHC Forward and reverse horn current: the two polarities of the NuMI focusing horns, giving a neutrino-dominated and an antineutrino-enriched beam.

Beam frame All angles are measured about the neutrino direction, which at MicroBooNE’s off-axis position is 28° from the detector z axis.

Control region, sideband A sample selected by inverting one requirement of the signal selection, so that it is disjoint from the signal region and enriched in one background. It can be inspected on beam-on data while the signal region stays blind.

Transverse kinematic imbalance (TKI) Variables built from the momenta transverse to the neutrino direction of the lepton and the hadronic system: δp_T , $\delta \alpha_T$, $\delta \phi_T$ and the inferred nucleon momentum p_n . They isolate nuclear effects from the unknown neutrino energy.

LLR particle identification The log-likelihood-ratio score built from track dE/dx that separates protons from muons and pions.

References

- [1] C. Andreopoulos *et al.*, “The GENIE Neutrino Monte Carlo Generator,” *Nucl. Instrum. Methods A* **614**, 87–104 (2010).
- [2] MicroBooNE Collaboration, P. Abratenko *et al.*, “New CC0 π GENIE model tune for MicroBooNE,” *Phys. Rev. D* **105**, 072001 (2022), arXiv:2110.14028 [hep-ex].
- [3] L. Alvarez-Ruso *et al.* (GENIE Collaboration), “Recent highlights from GENIE v3,” *Eur. Phys. J. ST* **230**, 4449–4467 (2021).
- [4] T. Golan, J. T. Sobczyk, and J. Żmuda, “NuWro: the Wrocław Monte Carlo generator of neutrino interactions,” *Nucl. Phys. B Proc. Suppl.* **229–232**, 499 (2012).
- [5] O. Buss *et al.*, “Transport-theoretical description of nuclear reactions,” *Phys. Rept.* **512**, 1–124 (2012); U. Mosel, “Neutrino event generators: foundation, status and future,” *J. Phys. G* **46**, 113001 (2019).

- [6] Y. Hayato and L. Pickering, “The NEUT neutrino interaction simulation program library,” *Eur. Phys. J. ST* **230**, 4469–4481 (2021).
- [7] D. Rein and L. M. Sehgal, “Coherent π^0 production in neutrino reactions,” *Nucl. Phys. B* **223**, 29–44 (1983).
- [8] C. Berger and L. M. Sehgal, “Partially conserved axial vector current and coherent pion production by low energy neutrinos,” *Phys. Rev. D* **79**, 053003 (2009).
- [9] MicroBooNE Collaboration, P. Abratenko *et al.*, “First Measurement of Energy-Dependent Inclusive Muon-Neutrino Charged-Current Cross Sections on Argon with the MicroBooNE Detector,” *Phys. Rev. Lett.* **128**, 151801 (2022), arXiv:2110.14023 [hep-ex].
- [10] D. Rein and L. M. Sehgal, “Neutrino Excitation of Baryon Resonances and Single Pion Production,” *Ann. Phys.* **133**, 79–153 (1981).
- [11] K. S. Kuzmin, V. V. Lyubushkin, and V. A. Naumov, “Lepton polarization in neutrino nucleon interactions,” *Mod. Phys. Lett. A* **19**, 2815–2829 (2004).
- [12] K. M. Graczyk and J. T. Sobczyk, “Form factors in the quark resonance model,” *Phys. Rev. D* **77**, 053001 (2008) [Erratum: *Phys. Rev. D* **79**, 079903 (2009)].
- [13] C. Berger and L. Sehgal, “Lepton mass effects in single pion production by neutrinos,” *Phys. Rev. D* **76**, 113004 (2007).
- [14] J. A. Nowak (MiniBooNE Collaboration), “Four-momentum transfer discrepancy in the charged current π^+ production in the MiniBooNE: Data vs. theory,” *AIP Conf. Proc.* **1189**, 243–248 (2009).
- [15] Y. Hayato, “A neutrino interaction simulation program library NEUT,” *Acta Phys. Polon. B* **40**, 2477–2489 (2009).
- [16] G. M. Radecky *et al.* (ANL Collaboration), “Study of single pion production by weak charged currents in low-energy νd interactions,” *Phys. Rev. D* **25**, 1161–1173 (1982).
- [17] T. Kitagaki *et al.* (BNL Collaboration), “Charged-current exclusive pion production in neutrino–deuterium interactions,” *Phys. Rev. D* **34**, 2554–2565 (1986).
- [18] A. Rodriguez *et al.* (K2K Collaboration), “Measurement of single charged pion production in the charged-current interactions of neutrinos in a 1.3 GeV wide band beam,” *Phys. Rev. D* **78**, 032003 (2008).
- [19] A. A. Aguilar-Arevalo *et al.* (MiniBooNE Collaboration), “Measurement of ν_μ -induced charged-current single positive pion production cross sections on mineral oil at $E_\nu \in 0.5$ –2.0 GeV,” *Phys. Rev. D* **83**, 052007 (2011).
- [20] B. Eberly *et al.* (MINERvA Collaboration), “Charged pion production in ν_μ interactions on hydrocarbon at $\langle E_\nu \rangle = 4.0$ GeV,” *Phys. Rev. D* **92**, 092008 (2015).
- [21] C. L. McGivern *et al.* (MINERvA Collaboration), “Cross sections for ν_μ and $\bar{\nu}_\mu$ induced pion production on hydrocarbon in the few-GeV region using MINERvA,” *Phys. Rev. D* **94**, 052005 (2016).
- [22] MINERvA Collaboration, J. Wolcott *et al.*, “Evidence for Quasielastic Charge Current Antineutrino Scattering Off Protons in Hydrocarbon at $\langle E_{\bar{\nu}} \rangle = 3.5$ GeV,” *Phys. Rev. Lett.* **116**, 081802 (2016).

- [23] K. Abe *et al.* (T2K Collaboration), “Measurement of the muon neutrino charged-current single π^+ production on hydrocarbon using the T2K off-axis near detector ND280,” *Phys. Rev. D* **101**, 012007 (2020), arXiv:1909.03936 [hep-ex].
- [24] R. Acciarri *et al.* (ArgoNEUT Collaboration), “First measurement of the cross section for ν_μ and $\bar{\nu}_\mu$ induced single charged pion production on argon using ArgoNEUT,” *Phys. Rev. D* **98**, 052002 (2018), arXiv:1804.10294 [hep-ex].
- [25] MicroBooNE Collaboration, P. Abratenko *et al.*, “First measurement of ν_e and $\bar{\nu}_e$ charged-current single charged-pion production differential cross sections on argon using the MicroBooNE detector,” *Phys. Rev. Lett.* **135**, 061802 (2025), arXiv:2503.23384 [hep-ex].
- [26] MicroBooNE Collaboration, P. Abratenko *et al.*, “Measurement of single charged pion production in charged-current ν_μ -Ar interactions with the MicroBooNE detector,” *Phys. Rev. D* **113**, 032007 (2026), arXiv:2509.03628 [hep-ex].
- [27] X.-G. Lu *et al.* (MINERvA Collaboration), “Measurement of final-state correlations in neutrino muon-proton mesonless production on hydrocarbon at $\langle E_\nu \rangle = 3$ GeV,” *Phys. Rev. Lett.* **121**, 022504 (2018).